Eastern Coalfields Limited

(A subsidiary of Coal India Limited)

Integrated Command and Control Centre (ICCC) in ECL

<u>BID SPECIFIC ATC</u>

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A. INTRODUCTION

Eastern Coalfields Limited (ECL) is a Subsidiary Company of Coal India Limited (CIL) which is under administrative control of Ministry of Coal. The Company has its registered office at Sanctoria, Asansol, West Bengal, and has its mining operation span across states of West Bengal and Jharkhand. The proposed system requirements will cover 14 locations, including 13 Areas and the ECL headquarters. The mining activities are divided into 13 Areas, each of which is further subdivided into multiple mines. There are entry and exit points, road weigh bridges and other important establishments at mines.

Objective of the Project

The objective of this project is divided into three parts: -

1. Part A: CCTV Surveillance System
2. Part B: Integrated solution for Access Control System and Weighbridge Automation System
3. Part C: Command & Control Centre at Areas & ECL HQ

B. Part A: CCTV Surveillance System

1. Objective:

The primary objective of this part is to implement and maintain a comprehensive CCTV Surveillance System consisting of 843 nos. of CCTV cameras. This system will include the necessary networking components for local monitoring at the mine level, as well as centralized monitoring at the area control room. The system will be responsible for monitoring and recording video footage from key locations such as coal heaps, road weighbridges, railway sidings, stores, magazines, workshops, attendance points, and other critical or vulnerable areas within Eastern Coalfields Limited (ECL).

It is to note that out of the total procurement of 1582 CCTV cameras, 843 cameras (comprising 355 Bullet cameras, 424 Dome cameras, and 64 PTZ cameras) will be installed as part the CCTV Surveillance System (this part). The remaining 739 CCTV cameras (including 580 Bullet cameras with ANPR licenses and 159 Dome cameras) will be installed at Entry - Exit Check Posts as part of the Access Control System.

The licenses for the 580 Bullet cameras for ANPR are separate and in addition to the ANPR licenses mentioned in the Video Analytics Application Software. The ANPR licenses as part of Video Analytics Application Software will be installed at other locations than the Entry-Exit Check posts. The bidder is required to provide the "5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR functionality" along with the cost of the associated ANPR license.

2. Major Items and their functionalities associated to the CCTV Surveillance System:

The following items mentioned with their major functionalities are required to be installed.

Item	Major Functionality	Location for installation
CCTV Cameras	Surveillance at end points, by capturing video footages	coal heaps, road weighbridges, railway sidings, stores, magazine, workshops, attendance points and other vulnerable locations
NVR and 32 inches Display	Recording and viewing CCTV cameras locally camera installation sites	Manager Office, Agent Office, Area Office, Weighbridges, Stores, Magazine etc.
Video Management Server with software, Video Storage/ NAS, Workstation PC, 55 inches Display, Enterprise CVMS license	Recording and viewing CCTV cameras at distant location Area Control Room	Area Control Room
Network Management Systems	For real-time monitoring of working/ not working of CCTV	Area Control Room

	Surveillance System components and network components	
Radios, Towers, Fibre Cable, CAT 6 cable, Switches	For network connectivity of CCTV cameras locally as well as at Area Offices	Various locations
UPS, SPD, CVT/ Voltage Stabilizer, MCB, Earthing	For power backup as well as power rectification	Various locations

3. Integration with other systems:

The CCTV Surveillance System must be integrated with the following systems

Video Analytics	The CCTV Surveillance System shall be integrated with the Video Analytics Server for running AI based Video Analytics
ICCC	Through this integration the system shall provide the real-time alerts and notifications to the ICCC for overall monitoring and surveillance with location details, timings, etc.
Upcoming Systems	The system must support integration with upcoming systems as required by ECL. This includes integrating APIs and SDKs from future systems and providing APIs and SDKs for future systems.
The bidder will be responsible for the development, customization, and implementation of these integrations during the contract period of six years without additional cost to ECL. The ECL will provide necessary support in obtaining the APIs from the relevant platforms to be integrated with this system.	

4. Implementation:

This project involves implementation of CCTV Surveillance System in following three parts:

Part	Purpose	Locations
CCTV Surveillance System for surveillance of Attendance Points	Surveillance of Attendance Room installed with Biometric Attendance Devices without network connectivity	**Complete ECL**
CCTV Surveillance System with network connectivity for overall surveillance	Overall Surveillance including coal heaps, road weighbridges, railway sidings, stores, magazine, workshops, Biometric Attendance Points, and other vulnerable locations with monitoring and network connectivity till Control Room	**SP Mines Area** **Sonepur Bazari Area** **ECL, HQ with associated units**
CCTV Surveillance System with network	Overall Surveillance including coal heaps, road weighbridges, railway sidings, stores, magazine, workshops, Biometric	**Rajmahal Area**

connectivity for overall surveillance and integration of existing IP based CCTV cameras	Attendance Points, and other vulnerable locations including network connectivity till Area Office and integration of existing IP based CCTV cameras and overall monitoring at Area Control Room	

a. **<u>Implementation CCTV Surveillance System for surveillance of Attendance Points in Complete ECL</u>**

   i. Out of the total 843 nos. of CCTV cameras, 330 CCTV cameras (all Dome type) will be installed at the sites installed with Biometric Attendance Devices for surveillance of Biometric Attendance points.

   ii. The CCTV cameras will be connected to the existing Network Video Recorder (NVR) wherever available. In cases where spare ports in the existing NVRs are unavailable, the cameras will be connected to the tendered 4-channel NVR in consultation with Area E&T Engineer.

   iii. The existing network connectivity and infrastructure up to the Area Control Room will be utilized for monitoring and integration. The bidder is required to utilized this existing infrastructure while installing the CCTV cameras. This includes the use of components such as cables, radios, UPS systems, racks, power conditioning equipment (including voltage stabilizers, power surge protectors, MCBs, and earthing), among others, to supply power and ensure connectivity of the CCTV cameras with the ECL network for both local and centralized viewing at the Area Control Room.

   iv. The bidder shall be responsible for the operation and maintenance of the CCTV cameras, including addressing any breakdowns caused by faults in the existing infrastructure, without any additional cost to ECL. The bidder must ensure that all systems remain fully operational in accordance with the Service Level Agreement (SLA) for their intended purpose and provide continuous support throughout the 6-year contract period.

   v. The CCTV cameras, excluding those at Rajmahal Area, SP Mines Area, Sonepur Bazari Area, ECL HQ with associated units, and Kalla Hospital, will be integrated with the existing Video Management Software (VMS) (Make: IDIS, Model: ISS - I NEX) at each respective Area Control Room. The supplied CCTV cameras must be compatible with the existing VMS, and the bidder is responsible for providing the necessary support for integration. The associated licenses for the VMS will be provided by ECL.

   vi. The CCTV cameras at Rajmahal Area, SP Mines Area, Sonepur Bazari Area, ECL HQ with associated units, and Kalla Hospital will be integrated with the tendered Video Management Software (VMS) with associated licenses to be provided by the bidder.

   vii. In locations where network connectivity is unavailable, monitors will be installed with NVRs for local viewing of the CCTV cameras. These cameras will be integrated with the existing Video Management Software (Make: IDIS, Model: ISS - I NEX) or any future VMS via ECL's own network or public internet, as provided by ECL. The bidder will be responsible for providing the necessary support for this integration.

viii. The CCTV cameras, NVRs, and monitor will draw power from the existing UPS and Switches wherever available.

ix. In this part the bidder will not be responsible for implementation of connectivity from mines to Area Offices or integration of the existing CCTV cameras.

x. Following are the locations for installation of CCTV Cameras at attendance points.

SN	Area Name	Group/Mine	Location	Quantity
1	Bankola Area	Area Office	Area Office	2
2		Bankola Colliery	Sankalap Khani	1
3			Bankola 7/8 no Pit	1
4			Bankola Agent Ofice	1
5		Kumardih A	Pit no 6	1
6			Agent Office	1
7		Kumardihi B/ Nakrakonda OCP	Agent Office	1
8			5/6 no pit	1
9			Nakrakonda OCP Workshop	1
10			CM Project	1
11		Tilaboni Colliery	Tilaboni pit 2/3	1
12			Old SSPUR	1
13			Agent Office	1
14		Shankarpur Colliery	Millenium	1
15			Shankarpur 3/4 Pit	1
16			Agent Office	1
17		S S Pur Colliery	Pit	2
18			Agent Office	1
19		Khandra Colliery	Khandra Nag	1
20			Khandra VK	1
21			Agent Office	1
22		Moira Colliery	Moira Colliery	1
23		Area Hospital	Area Hospital	1
24		Area Store	Area Store	1
25		Railway Siding 1	Railway Siding 1	1
26		Railway Siding 2	Railway Siding 2	1
27		POCP Siding	POCP Siding	1
28		Total for Bankola Area		29
29	Jhanjra Area	Area Office	Area Office	1
30		Area Store	Area Store/Workshop	1
31		Hospital	Hospital	1
32		MIC Unit, JPC	Caplamp Room	2

33			Agent Office	1
34			Glass House	1
35			Road Weighbridge	1
36		3 & 4 Incline unit, JPC	Caplamp Room	2
37			Road Weighbridge	1
38		POCP Siding 01	POCP Siding Office	1
39		Total for Jhanjra Area		12
40	**Kajora Area**	Area Office	Area Office	1
41		Area Store	Area Store	1
42		Nabajambad Group of Mines	Naba Kajora Agent/Manager Office	1
43			Naba Kajora Guest House	1
44			Madhabpur Manager Office	1
45			Madhabpur 2 no. pit	1
46			Jambad OCP	1
47			Jambad UG Agent Office	1
48			Jambad UG pit office	1
49		Khas Kajora Group of Mines	Central Kajora	1
50			Madhusudhanpur Agent Office	1
51			Madhusudhanpur 7 no. Pit	1
52			Khas Kajora Agent Office	1
53			Khas Kajora 5 & 6 no pit office	2
54			Khas Kajora 10 & 11 no pit office	1
55		Porascol Group of Mines	Parascol West pit office	1
56			Parascol Agent Office	1
57			Parascol East pit office	1
58		JK Ropeways	Area Office	1
59			Site 1/7	1
60		Total for Kajora Area		21
61	**Kenda Area**	Area Office	Area Office	2
62		Area store	Area store	1
63		Chora Hospital	Chora Hospital	1
64		Bahula Group of Mines	Bahula RJ Unit	1
65			Bahula NJ Unit	1

66			Bahula CHP	1
67			Bahula Agent Office	1
68			Lower Kenda Pit no 4	1
69		Siduli Group of Mines	Siduli Agent Office	1
70			Siduli JK Unit	2
71			CL Jambad Manager Office	1
72		Chora Group of Mines	Chora 10 pit Agent Office	1
73			Chora 10 Pit	1
74			Chora 7 & 9 pit Manager Office	1
75		New Kenda Group of Mines	New Kenda Agent Office	1
76			New Kenda OCP	1
77		CBI Group of Mines	Shankarpur Agent Office	1
78			CB Incline Manager Office	1
79			Bonbahal OCP	1
80		Total for Kenda Area		21
81	**Kunustoria Area**	Amritnagar Colliery	Pit 3	2
82			Agent Office	1
83		Mahabir Colliery	Manager Office	1
84		Narayankuri HW project	Project Office	1
85		Belbaid RLY siding	Siding office	1
86		Bansra Colliery	C pit	2
87			Agent Office	1
88		Bansra OCP	Rly siding	1
89		Kunustoria Colliery	Pit 2	1
90			Pit 3	1
91			Agent Office	1
92		N S OCP	Manager Office	1
93			Project Office	1
94		Belbaid Colliery	Pit office	1
95			Manager Office	1
96			Personnel Office	1
97		Parasia Colliery	Pit office	2
98			Agent Office	1
99		Area Office	Main Gate	1
100			First Floor	1

101			Area Store	1
102			System Dept	1
103		Bansra Hospital	CMO office	1
104		Total for Kunustoria Area		26
105	**Mugma Area**	Area Office	Area Office	2
106			Regional Hospital	1
107		Kapasara	Manager Office	2
108			Gopinathpur Ext office	1
109		Rajpura	Attandance room	2
110			Personnel Office	1
111		Kumardubi	BL-incline	3
112			Kumardubi_manager	1
113		Barmuri	Barmuri Office	1
114			Barmuri_Project_office	2
115		Lakhimata	Attendance Point	3
116			Manager Office	1
117			CP Siding	1
118		Nirsha	Manager _Office	1
119		Khoodia	Attendance Point	2
120			Manager Office	1
121		Badjna	Manager Office	1
122			29 Incline	1
123			31 Incline	2
124			Badjna OCP	1
125		Hariyajam	Manager Office	1
126			5/6 Incline	1
127			27 Incline	1
128		Chapapur	Chapapur OC	1
129			Manager Office	1
130			Kali Mati Incline	2
131		Shyampur B	Incline	2
132			Manager Office	1
133		Total for Mugma Area		40
134	**Pandaveswar Area**	Pandaveswar Area Office	Pandaveswar Area Office	2
135			Panthanagar Hospital (internet)	1
136			Pandaveswar Area Store	1
137		Pandaveswar Group of Mines	Pandaveswar colliery Agent Office	1

138			Pandaveswar colliery pit -A	1
139			Pandaveswar colliery pit -B	1
140			Kendra Colliery	1
141		Khottadih OCP	Khottadih OCP Agent Office	1
142			Khottadih OCP Attendnace room	1
143			Khottadih SS Siding	1
144		Khottadih colliery	Khottadih colliery Agent office	1
145			Khottadih colliery pit 1	1
146			Khottadih colliery pit 2	1
147		South samla Colliery	South samla Agent office	1
148			South samla colliery pit	1
149		Manderboni Colliery	Manderboni Agent Office	1
150			Manderboni Colliery pit	1
151		Madhaipur Colliery	Madhabpur Agent Office	1
152			Madhaipur 9 no pit	1
153			Laskarbandh Colliery	1
154		Dalurbandh Colliery	Dalurbandh Colliery	1
155		Kankartala Colliery	Kankartala Colliery	1
156		Total for Pandaveswar Area		23
157	**Rajmahal Area**	Rajmahal Area	Area Office	2
158			Pit Office	2
159			Workshop	1
160			Silo	1
161			CHP	1
162			Main Gate (OCP/ WS)	1
163			Town Admn	1
164			Hospital	1
165			Lalmatia Dispensary	1
166		Hurrah C	Hurra C	1
167		Simlong	Simlong Area Office (Required)	1
168		Chaparvitta Colliery	Chaparvitta Colliery	1
169		Total for Rajmahal Area		14

170	**Salanpur Area**	Salanpur Area Office	Area Office ground floor	1
171			Area office first floor	1
172			Area Store	1
173			Area dispensary	1
174		Mohanpur Colliery	Agent Office	1
175			Attendance room (at mine)	1
176			Pahargorah point	1
177		Itapara Colliery	Agent office	1
178			Mine office	1
179		Dabor colliery	Agent office	1
180			Attendance room	1
181			Workshop/ store	1
182		Bonjemari colliery	Agent Office	1
183			Attendance room	1
184			RHS dendua	1
185		Gourandi colliery	Agent Office	1
186			Pit office/ workshop	1
187		G Begunia colliery	Agent Office	1
188		Barmondia colliery	Agent/Manager office	1
189		Total for Salanpur Area		19
190	**Satgram-Sripur Area**	Bhanora Group of Mines	Bhanora West Block Incline	1
191			Bhanora West Block Agent Office	1
192			Bhanora OCP	1
193			Bhanora Old Office	1
194			Charanpur Ocp	1
195			Girmint Office	1
196		Nigha Group of Mines	Ningha Colliery	1
197			Ningha Agent Office	1
198			SSI (Ningha) OCP	1
199			SSI Colliery	1
200		Kalipahari Group of Mines	Kalipahari Colliery	1
201			Ghusick Colliery	1
202			Ghusick Workshop	1
203			Agent Office	1
204		Satgram-Ratibati Group of Mines	Satgram Project cap lamp room	1

205			Satgram Project Agent/ Manager Office	1
206			Satgram Incline Manager Office	1
207			Satgram Incline Cap lamp room	1
208			Chapuikhas colliery	1
209			Ratibati Agent Office	1
210			Kuardi colliery 14 no	1
211			Tirat colliery	1
212			Kuardi Manager Office	1
213			Mithapur/ Nageswar colliery	1
214		Nimcha ® colliery	Nimcha colliery	1
215			Agent Office	1
216			Nimcha Depot Office	1
217			Nimcha Highwall	1
218			Amkola OCP	1
219			Amkola Colliery	1
220		Kalidaspur project	Agent office	1
221			colliery	1
222			Jorsa CISF camp	1
223		J K Nagar ® Mines	J K Nagar colliery	1
224			J K Nagar project	1
225			Agent Office	1
226			J K Nagar siding/ depot	1
227			Pure searsole colliery	1
228			Modern satgram colliery	1
229		Area	Satgram-Sripur Area Office	1
230			Sripur Office	1
231		Total for Satgram-Sripur Area		41
232	**Sodepur Area**	Parbelia	Agent/ Manager Office	2
233			Cap lamp room	1
234		Dubeswari	Manager Office	1
235			Cap lamp room	1
236		Chinakuri I/II	Agent/ Manager Office	1
237			Cap lamp room	1
238		Chinakuri III	Cap lamp room	1
239		Methani	Agent/ Manager Office	1

240			Cap lamp room	1
241		Bejdih	Manager Office	1
242			Cap lamp room	1
243		Patmohana	Manager Office	1
244			Cap lamp room	1
245		Narsamunda	Manager Office	1
246			Cap lamp room	1
247		Dhemomain Incline	Manager Office	1
248		Dhemomain Pit	Agent/ Manager Office	1
249			Cap lamp room	1
250		Area Store	Area Store	1
251		Area Office	GM Building Entrance	1
252			GM Secretariate lobby	1
253		Sodepur ®	Sodepur ®	1
254		Total for Sodepur Area		23
255	**Sonepur Bazari Area**	Sonepur Bazari Area	Area Office	1
256		Sonepur Bazari Project	Project Office	1
257			Base Workshop	1
258			Glass House near shovel section	1
259			Glass House control room	1
260			New CHP	1
261		Total for Sonepur Bazari Area		6
262	**SP Mines Area**	Chitra	Girja	1
263			T D Quarry	1
264			Main Workshop	1
265			Auto Section	1
266			Stores	1
267			Jamtara Railway Siding	1
268		Area Office	Area Office	1
269		Total for SP Mines Area		7
270	**ECL, HQ**	ECL HQ	Technical Building	8
271			Administration Building	2
272			Sanctoria Hospital	1
273			Personnel/ TA Office	2
274			HRD office, Dishergarh	1
275			Filter plant office HQ Dishergarh Ghat	1

276			CVO and Environment Barachak House	2
277		Sodepur Central Store	Sodepur Central Store	2
278		Sodepur Central workshop Sunderchak	Sodepur Central workshop Sunderchak	1
279		Neamatpur Central Workshop	Neamatpur Central Workshop	1
280		Central Hospital Kalla	Central Hospital Kalla	3
281		Ukhra Regional Workshop	Ukhra Regional Workshop	1
282		Ratibati Workshop	Ratibati Workshop	1
283		Closed	BEFW	1
284		MRS	MRS all locations	5
285		Mugma Regional Workshop	Mugma Regional Workshop	1
286		Poniati Regional Workshop	Poniati Regional Workshop	1
287		VTC Salanpur Area	VTC Salanpur Area	1
288		VTC Sonepur Bazari Area	VTC Sonepur Bazari Area	1
289		VTC SP Mines	VTC SP Mines	1
290		VTC Rajmahal	VTC Rajmahal	1
291		VTC Kenda Area	VTC Kenda Area	1
292		VTC Kunustoria Area	VTC Kunustoria Area	1
293		VTC Kajora Area	VTC Kajora Area	1
294		VTC Bankola Area	VTC Bankola Area	1
295		VTC Sodepur Area	VTC Sodepur Area	1
296		Benali VTC and Workshop Satgram-Sripur Area	Benali VTC and Workshop Satgram-Sripur Area	2
297		VTC Pandaveswar Area	VTC Pandaveswar Area	1
298		VTC Jhanjhra Area	VTC Jhanjhra Area	1
299		VTC Mugma Area	VTC Mugma Area	1
300		Total for ECL, HQ		48
	Total			**330**

b. <u>Implementation of CCTV Surveillance System with network connectivity for overall monitoring at SP Mines Area, Sonepur Bazari Area, ECL, HQ with associated units:</u>

i. Out of the total 843 CCTV cameras, 472 cameras (comprising 331 Bullet, 91 Dome, and 50 PTZ cameras) will be installed for overall surveillance, covering areas such as coal heaps, road weighbridges, railway sidings, stores, magazines, workshops, Biometric Attendance Points, and other vulnerable locations.

ii. The CCTV cameras will be connected to the tendered Network Video Recorder (NVR) for local-level monitoring. For monitoring at the Control Room (at the SP Mines Area, Sonepur Bazari Area or ECL HQ as applicable), the bidder must establish both wired (using CAT 6 or fiber cables) and wireless (using radios) network connectivity, so that all the cameras under SP Mines Area, Sonepur Bazari Area, and ECL, Hq can be monitored centrally from the control room of SP Mines Area, Sonepur Bazari Area, and ECL, HQ respectively. This applies to SP Mines Area, Sonepur Bazari Area, and ECL HQ with associated units, excluding the Vigilance Office Barachak and Kalla Hospital, where ECL will provide the network for monitoring of their cameras at control room of ECL, HQ.

iii. A Video Management Software (VMS) will be installed at the Control Room of SP Mines Area, Sonepur Bazari Area, and ECL HQ for live viewing and record-keeping of the CCTV cameras in their respective locations.

iv. The CCTV cameras at these locations will be integrated with the tendered VMS to be supplied by the bidder, including the associated license costs, which fall under the bidder's scope. This integration will also apply to the cameras installed at Biometric Attendance Points, with the corresponding license costs included. These license costs have already been accounted for in the Bill of Quantities (BOQ).

v. The bidder will be required to implement completely new network infrastructure in these locations, as no existing network infrastructure will be provided.

vi. The bidder will be responsible for the operation and maintenance of the CCTV cameras, ensuring they remain fully operational according to the Service Level Agreement (SLA) for their intended purpose. The bidder will provide continuous support throughout the 6-year contract period.

vii. The bidder will also be responsible for installing a Network Management System (NMS) to monitor the operational status of CCTV surveillance components with network components at SP Mines Area and Sonepur Bazari Area.

viii. In this part the bidder will not be responsible for integration of the existing CCTV cameras.

ix. Following are the locations for installation of CCTV Cameras

SN	Location	Group	Location Name	Bullet	Dome	PTZ	Integration
1	SP Mines Area	Chitra East OCP	Girja	8	0	5	Integration of all 85 cameras to VMS at Area Control room of SP Mines with complete networking under the scope of the bidder
2			Khoon Patch including Weighbridge	5	1	5	
3			Tulsi Dabor Patch	4	0	4	
4			Damgarah Patch	8	0	7	
5		Jamtara Railway Siding	Jamtara Railway Siding	8	0	5	
6		Workshop and DDU	Workshop and DDU	8	0	0	
7		Area Stores	Area Stores	4	0	0	
8		Weighbriges	Weighbriges	10	3	0	
9		**Total for SP Mines Area**		**55**	**4**	**26**	
10	Sonepur Bazari Area		Project Office + Dispensary	3	0	0	Integration of all 165 cameras to VMS at Area Control room of Sonepur Bazari Area with complete networking under the scope of the bidder
11			Weighbridge 1	3	1	0	
12			Weighbridge 2	3	1	0	
13			Weighbridge 3	3	1	0	
14			Weighbridge 4	3	1	0	
15			Weighbridge 5	3	1	0	
16			Weighbridge 6	3	1	0	
17			Weighbridge 7	3	1	0	
18			Weighbridge 8	3	1	0	
19			Weighbridge 9	3	1	0	
20			Weighbridge 10 (Kalimandir)	5	1	0	
21			Weighbridge 11 (Dragline) + Dragline Yard + Office	8	1	0	
22			TRS	2	0	4	
23			New Siding	0	0	5	
24			New Coaldepo	3	0	3	
25			Andrew Yule Substation	5	0	0	
26			New CHP	15	0	0	
27			Mahalakshmi View Point	0	0	1	
28			DG Shed	4	0	0	

29			Base Work Shop (+ DDU, + Store)	45	0	5	
30			Magazine	4	0	0	
31			Glass House	3	0	0	
32			Bhatmura Sub Station	4	0	0	
33			Bhaluka Point	3	0	2	
34			Area Office	3	0	0	
35		**Total for Sonepur Bazari Area**		**134**	**11**	**20**	
36	ECL,HQ	ECL,HQ	ECL,HQ	46	0	0	Integration of all 152 cameras to VMS at Control room of ECL, HQ with complete networking under the scope of the bidder
37			CISF Seetalpur	7	0	0	
38			Seetalpur Guest House	7	1	0	
39			Sodepur Central Store	18	2	2	
40			Sodepur Central Workshop	22	2	1	
41			Neamatpur Workshop	10	1	1	
42			Sanctoria Hospital	0	32	0	
43		Vigilance Office Barachak		15	5	0	Integration of all 70 cameras to VMS at Control room of ECL, HQ with ECL's network
44		Kalla Hospital		17	33	0	
45		**Total for ECL, HQ**		**142**	**76**	**4**	
	Total			**331**	**91**	**50**	

c. **<u>Implementation of CCTV Surveillance System with network connectivity for overall monitoring and integration of existing IP based CCTV cameras at Rajmahal Area</u>**

i. Out of the total 843 CCTV cameras, 41 cameras (comprising 24 Bullet, 3 Dome, and 14 PTZ cameras) will be installed for overall surveillance, covering areas such as coal heaps, road weighbridges, railway sidings, stores, magazines, workshops, Biometric Attendance Points, and other vulnerable locations.

ii. The CCTV cameras will be connected to the tendered Network Video Recorder (NVR) for local-level monitoring. For monitoring at the Area Control Room, the bidder must establish both wired (using CAT 6 or fiber cables) and wireless (using radios) network connectivity.

iii. A Video Management Software (VMS) will be installed at the Control Room for live viewing and record-keeping of the CCTV cameras in their respective locations.

iv. The CCTV cameras at these locations will be integrated with the tendered VMS to be supplied by the bidder, including the associated license costs, which fall under the bidder's scope. This integration will also apply to the cameras installed at Biometric Attendance Points, with the corresponding license costs included. These license costs have already been accounted for in the Bill of Quantities (BOQ).

v. The bidder will be required to implement completely new network infrastructure in this location, as no existing infrastructure will be provided. The new network infrastructure shall integrate the new CCTV cameras to be supplied as per this tender as well as the existing CCTV cameras to the tendered VMS to be installed at Area Control Room of Rajmahal Area.

vi. The bidder will be responsible for the operation and maintenance of the CCTV cameras, ensuring they remain fully operational according to the Service Level Agreement (SLA) for their intended purpose. The bidder will provide continuous support throughout the 6-year contract period.

vii. The bidder will also be responsible for installing a Network Management System (NMS) to monitor the operational status of CCTV surveillance components (including existing cameras) and networking components at Rajmahal Area.

viii. This part also includes integration of existing IP-based cameras and implementation of network connectivity for its integration. The existing cameras shall be added to the VMS on ONVIF and the necessary costing for licensing has already been accounted for in the Bill of Quantities (BOQ).

ix. Following are the locations for installation of new CCTV Cameras and integration of existing CCTV cameras is mentioned below:

SN	Location	Bullet	Dome	PTZ	New/Old	Total
1	Coal Heap of Kendua	4	0	1	New	41
2	Coal Heap of IR Coal Depot	4	0	1		
3	Coal Heap Near IN pit WB	4	0	1		
4	Lalghutwa Side	1	0	2		
5	Lahandia side departmental view	0	0	2		
6	Taljhari side	0	0	3		
7	New Intermediate pump	2	0	0		
8	BLS Coal face	0	0	1		
9	Repeater Station Tower	0	0	1		
10	Hill Top Substation	0	0	2		
11	New Road W/B (03 Nos.)	9	3	0		
12	Silo Rail WB	1	0	0	Existing IP based	117
13	Inpit Road Weighbridge	2	0	0		
14	Nima Road Weighbridge	4	0	0		
15	CHP	0	0	4		
16	Silo Side	0	0	1		

17	Magazine	0	0	1		
18	Area Main Store	16	0	2		
19	NE Yard(Area Store)	4	0	2		
20	DDU	4	0	0		
21	CISF Workshop Main Gate	5	0	0		
22	Mine Entry Point	5	0	1		
23	Dumper Parking Area	0	0	4		
24	Lalmatia Chowk	0	0	1		
25	Workshop	0	0	1		
26	Dozer Section	9	0	0		
27	UrjaNagar Colony	26	0	0		
28	Rajmahal House, Urjanagar	10	0	0		
29	E&M and Fire Section	2	0	0		
30	Dhankunda Sub.St.	0	0	1		
31	Electrical Repair Shop	0	0	1		
32	CHP DH-3	2	0	1		
33	Coal heap Kendua Depot	3	0	0		
34	Coal heap RJ Siding	3	0	1		
	Total	120	3	35		158

5. <u>Scope of work for implementation</u>

a. To design, supply, install, integrate and conduct testing and commissioning of all tendered items, and to provide timely Training to ECL personnel. Also, to provide one year's onsite warranty and 05 year's CAMC after expiry of one year's onsite warranty period.

b. Command and Control System for CCTV Surveillance System at Rajmahal Area, Sonepur Bazari Area, SP Mines Area and ECL, HQ shall be designed and installed with Centralized server, Video Storage, NMS and Workstation PCs to view, command, control with provision of recording footages of all cameras installed under respective area. Also, minimum of 30 days video Recording at respective area control shall be ensured.

c. To establish 24x7 Live viewing, command, controlling and monitoring at Central Control Room at Rajmahal Area, Sonepur Bazari Area, SP Mines Area and ECL, HQ based on sequencing and site viewing.

d. To implement Groupings based on Branch, Divisions & important locations at Area as well as ECL HQ Control Room for streaming purposes.

e. To design and implement facilities for manually checking CCTV footages and its functionalities through control systems at different levels. NMS (Network Management System) shall be designed and implemented for all network elements for monitoring (viz. pinging of different network elements at regular interval through NMS software etc.) comprising of:

f. An integrated dashboard of E-maps showing all alerts/alarms/incidents at any point of time triggered or generated from Mine Level or area office level, status report of Connect and Disconnect Status of all active equipment installed under Network from mine level to Area Office Level and corresponding task to be executed through Sequence of Procedures.

g. Alert messages to be generated and escalated as per accountability matrix. The alert messages should also be escalated automatically to HQs in charge after a particular period if not addressed during that period.

h. To design and provide Mobile and PC applications to view live footages and alerts notifications for 50 users. The internet services for the same will be provided by ECL

i. To provide complete support for Total Contract period of 72 months (12 months + 60 months) i.e. OEM's warranty period of 12 months plus CAMC period of 60 months.

j. To ensure that the overall CCTV system shall conform to the ONVIF (Open Network Video Interface Forum) specification which guarantees interoperability between network video products regardless of the manufacturer.

k. The offered CCTV camera and NVR shall be of the same OEM.

l. CCTV OEM: should have ISO9001:2008 and ISO14001:2004 OR latest, to be submitted with ISO certificates.

m. To ensure and provide self-declaration that:

- **i.** while selecting CCTV OEMs of equipment, Hardware and Software (for this tender), the bidder has ensured that the Hardware, Software and Firmware (for this tender) including "Intellectual Property Rights" of Hardware and "Source Code" of Firmware & Software (including Cameras/NVR) selected are actual manufacturers and are not getting 3rd party manufacturing like branding and reselling in India through importing/trading and rebranding such equipment from Country that shares a Land Border with India.
  However, any bidder/OEM from a country which shares a land border with India will be eligible to bid only if the bidder is registered with the competent authority.
- **ii.** while selecting CCTV OEMs of equipment, Hardware and Software (proposed to be supplied for this tender) the bidder has ensured that OEMs are not suspended by ONVIF, even though the products of the OEMs may still appear on the ONVIF website.

n. To ensure that the OEM of CCTV cameras & NVR selected should have the MAC addresses (of all makes and models of IP CCTV quoted for this tender) registered in the name of the OEM. The MAC address should not be in the name of any 3rd party. The MAC address of all cameras should not be registered in the name of any 3rd Party/ company or any entity sharing land border with India until unless specifically allowed by the Government of India.

o. The proposal shall be submitted on turnkey basis execution.

p. Bidder shall have to provide access to ECL or third party authorized by ECL for establishment of connectivity of Area Control room with CCTV cameras as well as Control room of ECL HQ.

q. The central server of Area as well as ECL HQ Control room shall have provision of Remote viewing and controlling of video footages using client software installed on PCs or mobile phones through Intranet / internet.

r. It is the responsibility of the bidder to connect and integrate the CCTV Surveillance system to the nearest MPLS-VPN location through wired/wireless media using the existing resources wherever applicable.

C. Part B: Integrated solution for Access Control System and Weighbridge Automation System

1. Objective:

The primary objective of this part is to implement and maintain a comprehensive, integrated solution, with providing a unified interface/ dashboard, for monitoring the real-time data, notifications, and alerts for streamlining and optimization of coal transportation, accurate weighment and movement of only authorized vehicles within ECL. This includes implementation of Access Control System at entry exit check posts of ECL and Weighbridge Automation System with Truck Centering System at Road Weighbridges of ECL. It also involves real-time data transfer to ECL's SAP-ERP server and the Central Server at ECL, HQ, as well as integration with other systems such as GPS-based Vehicle Tracking System (VTS), Integrated Command and Control Centre (ICCC), and provisions for future integrations with state and central government portals, including JIMMS, FastTag, Vahan Portal, etc., at no additional cost to ECL.

2. Key Features:

The system must automate key features such as vehicle registration, verification, and identification, with record keeping etc., using RFID readers and ANPR cameras. It must ensure that only authorized vehicles can enter and exit the mines, siding, and coal stock areas, as well as be permitted for weighment at road weighbridges, based on predefined rules and centralized database. Also, the system will automate the weighment process at road weighbridges, ensuring vehicles are correctly positioned for accurate weighment.

Additionally, the system shall feature comprehensive log maintenance, tracks and keeps record of historical data, with providing customizable reports, generation of real-time alerts and notifications, and efficient data transfer to the central server at ECL, HQ, SAP-ERP server, ICCC with local storage, all with minimal human intervention, enhancing the efficiency and security of coal transportation in ECL.

3. Integrated Solution and its associated parts:

The Integrated solution is to be implemented with three individual parts as mentioned below with their major functionalities, quantities and locations to be implemented.

SN	Part	Major Functionality	Quantity and locations to be implemented
1	**Access Control System**	This system operates based on RFID Reader, ANPR cameras, and boom barrier with two-factor authentication (ANPR and RFID). Its main function is to automate vehicle access at entry exit check posts. This part is currently not implemented in ECL.	To be implemented at total 153 entry exit check posts in all areas of ECL
2	**Truck Centering System**	This system operates based on multiple sensors to detect the proper positioning of trucks on the road weighbridge platform. It ensures that vehicle is properly centred while the weighment is recorded. The system also includes visual and auditory indicators to guide drivers in achieving the correct positioning, with a manual override option available to the weighbridge operator. This part is currently not implemented in ECL.	To be implemented in 153, with 123 existing and 30 upcoming, Road Weighbridges in all areas of ECL
3	**Weighbridge Automation System**	This system operates based on RFID Reader, and boom barrier with single-factor authentication (RFID). Its main function is to automate weighment process at road weighbridges with no manual entry regarding weighment. This component is currently operational in ECL, and the bidder is required to integrate it with the overall system with further details mentioned in the subsequent points.	To integrate the 153, (123 existing and 30 upcoming) Road Weighbridges in all areas of ECL with overall system.

4. **<u>Implementation of Access Control System and Truck Centering System with new infrastructure and integration of Weighbridge Automation System utilizing existing infrastructure</u>**

Currently, the "Weighbridge Automation System" is operational in ECL, while the other two systems "Access Control System" and "Truck Centering System" are not yet implemented.

This project requires the bidder to implement the "Access Control System" and "Truck Centering System," including the supply, installation, commissioning, and maintenance of entirely new infrastructure. Additionally, the bidder must integrate the existing "Weighbridge Automation System" with the other two systems, utilizing the current hardware without the maintenance and operation of current hardware, but with supplying, installing, and maintaining new weighment client application software to ensure the systems operate in an integrated and synchronized manner. The bidder will assume full responsibility for the operation of all the systems.

For the comprehensive understanding of the existing infrastructure, the following are the details of the existing hardware infrastructure of road weighbridges related to RFID Reader, Signal Light and Boom Barrier

SN	Item Description	Make Model	Year of Purchase Quantity	Technical Specifications
1.	RFID Reader	Dadhwal DWIRF	2021 185	• Operating Frequency: 865~867 MHz Ingress protection: - IP 68 or better • Operating range: Read up to 08 Meters and write up to 05 Meters. • Communication interface: RJ45 • Air interface: EPC global UHF ISO-18000-6C/ ISO-18000-6B • **RFID Features:** ▪ Interference elimination. ▪ High data speed and anti-collision for multi tag reading ▪ Advanced Anti-Jamming ▪ built-in Antenna. ▪ 1,00,000 RF TAG IDs storage. ▪ Operating Temperature Range: upto 60 Degree Centigrade ▪ Humidity (non-condensing): should be able to work at 90% RH maximum. • **RF Antenna:** ▪ No polarization constraints. Capable of reading tags in any orientation. ▪ Read range of 0-8 meters
2.	Signal Light	Dadhwal DWILIT	2021 185	• Two high-intensity light bars: red and green (RG). • Standard Specifications
3.	Boom Barrier	Dadhwal 617	2021 248	• The length of the Boom Barrier is 5 meters • Automatically triggered/Open after getting information from Software with manual bypass • Safety arrangement for preventing boom from closing onto the vehicle using sensor • Degree of protection – IP 54 or better • Operating Temperature – up to + 60 °C • Humidity (non-condensing): 90% RH maximum. • Communication: - Relay interface to connect any Access Control or R.F identification reader device

				• Power input single phase 230 V AV + 10 % 50 hz or as per system requirement • Barrier operates with Photo-electric /Loop detector

Note: *The above the general specifications for more details the bidder are advised for site visit*

It is to be noted that if the bidder deems it necessary to modify the existing infrastructure of the Weighbridge Automation System—whether by replacing existing hardware or making other changes to ensure successful handover—such modifications shall be carried out at no additional cost to ECL, provided that the functionality & technical specification of such supplied items will be same or higher. Also, in such case, the comprehensive maintenance of such supplied items will be the responsibility of the bidder for the whole contract period of 06 Years (01 Warranty +05 year CAMC)

The new weighment client application software should incorporate all the functionalities and features specified in the scope of work and shall be customizable during the contract period of six years without additional cost to ECL and shall be independent to the make and model of the existing as well as upcoming infrastructure of the road weighbridges of ECL.

Further, as mentioned in above table this project involves the implementation of Truck Centering System and integration of Weighbridge Automation System at total of 153 road weighbridges, with 123 existing and 30 upcoming. The bidder is required to supply all the necessary equipment regarding Truck Centring System for all 153 road weighbridges at the time of delivery. However, the installation and commissioning of the Truck Centring System at the 30 upcoming road weighbridges, along with the supply and installation of weighment client application software along with its integration with other systems, will be carried out in accordance with the availability and installation schedule during the contract period of six years.

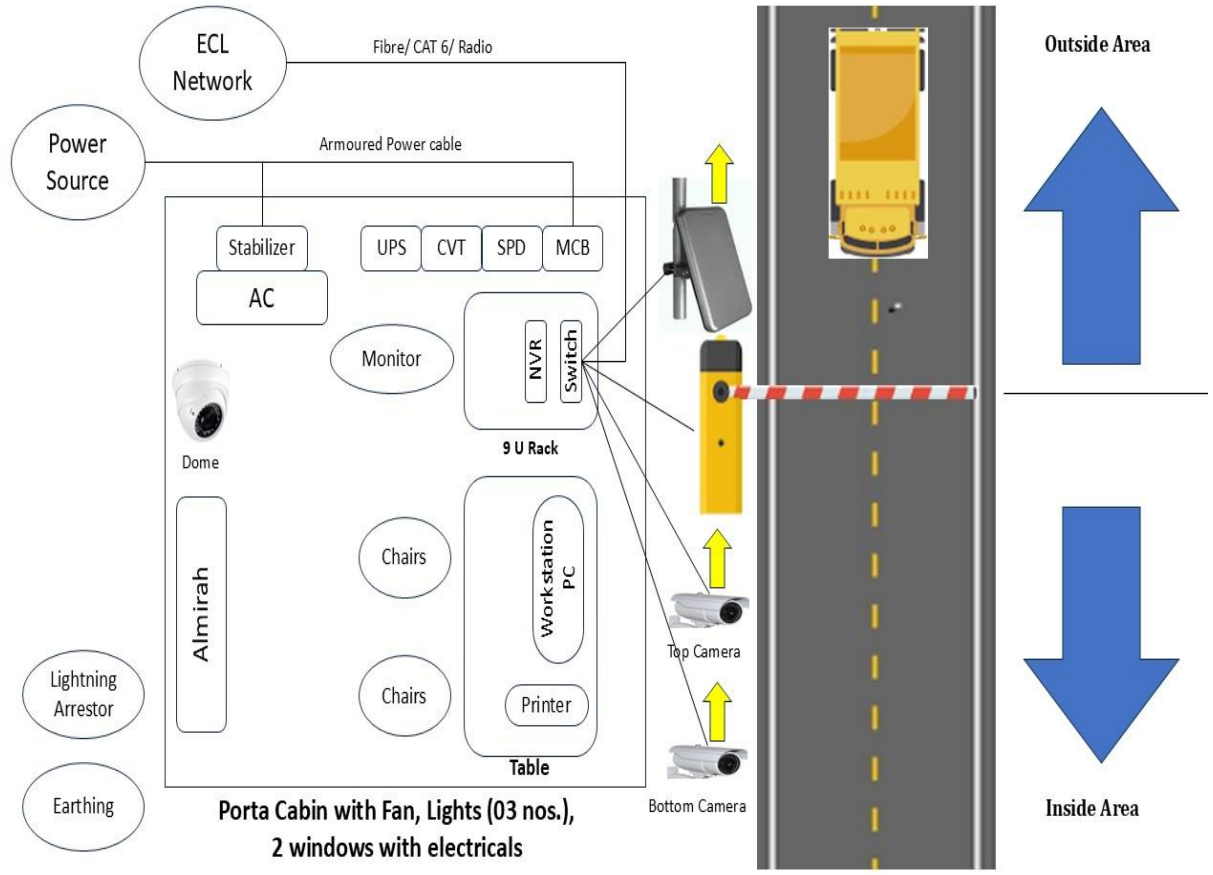
5. Implementation of total 04 types of Access Control System:

For implementation of Access Control System, total Four (04) different types of entry exit check posts are to be implemented in ECL, as mentioned below.

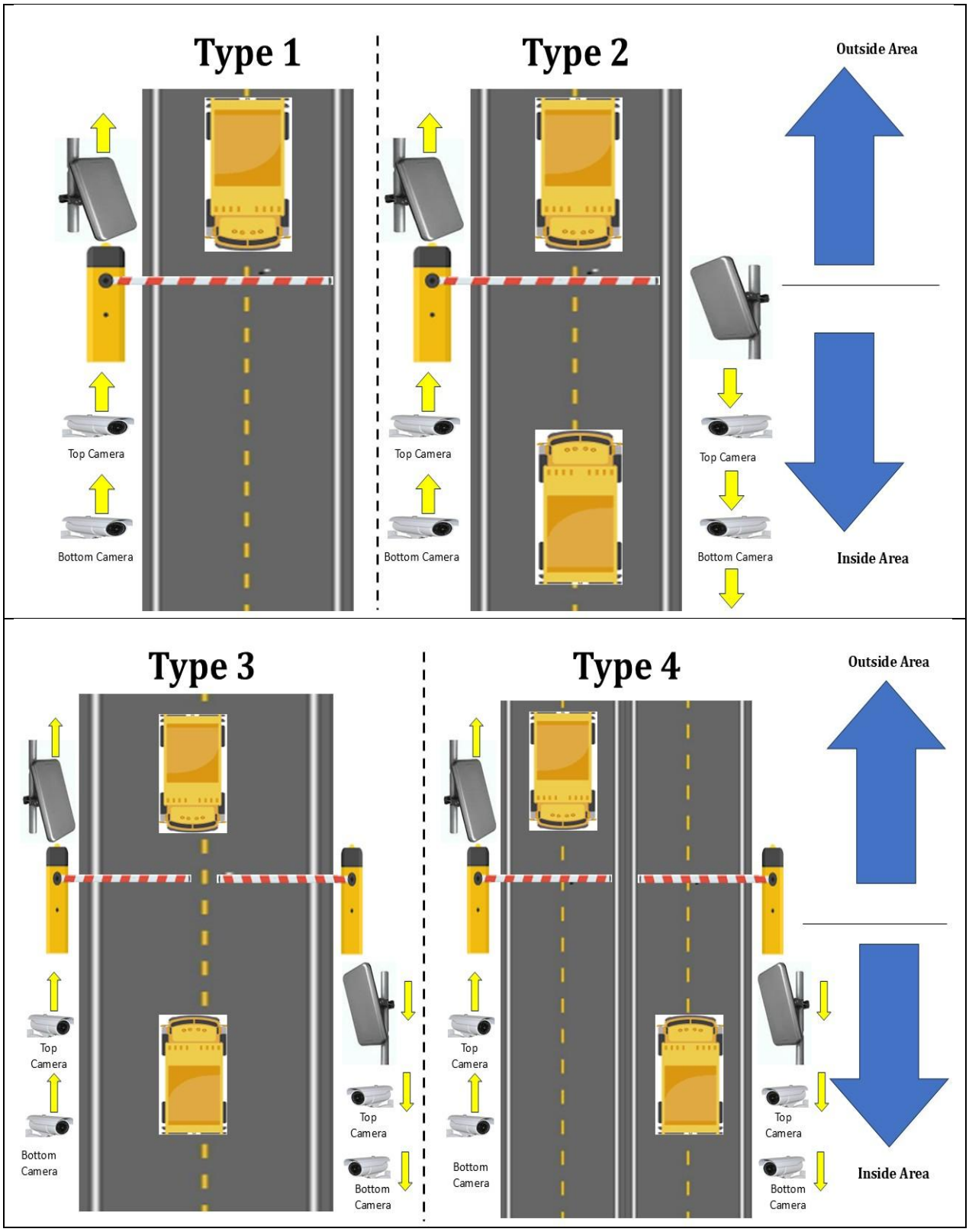
Type 1	Type 2	Type 3	Type 4
Only Entry/ Only Exit Check Post with single Boom Barrier, single RFID reader and single setup	Common Entry-Exit Check Post with single Boom Barrier but two RFID readers and single setup	Common Entry-Exit Check Post with two Boom Barrier, two RFID readers and single setup	Common Entry-Exit Check Post with two Boom Barrier, two RFID reader but two setups
For all-time Entry or Exit of vehicles from one direction.	For Entry as well as Exit of vehicles from two directions	For Entry as well as Exit of vehicles from two directions but with two boom	For all-time Entry as well as all- time Exit of vehicles from two directions simultaneously but

		barriers to cover the gate.	through two different lanes and two independent RFID and Boom Barrier setups. In such case one lane will permanently be used for all-time Entry and second lane for all-time Exit

<u>Following is the pictorial representation of the different types of the Access Control System</u>



Pictorial Representation of Porta Cabin (setup) and other items



<u>Note:</u> *The yellow-coloured arrow shows the direction in which the device is facing.*

Following is the major items associated to the types of Access Control System

SN	Item description	Unit	Type 1	Type 2	Type 3	Type 4
	Total Quantity of Type	**153**	**16**	**95**	**36**	**6**
1.	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos	2	4	4	4
2.	5 MP IP Dome Camera	Nos	1	1	1	2
3.	NVR (8 channel) loaded with 4 TB Storage	Nos	1	1	1	2
4.	Monitor 32 inches	Nos	1	1	1	2
5.	Workstation PC	Nos	1	1	1	2
6.	Printer	Nos	1	1	1	2
7.	Client Software	Nos	1	1	1	2
8.	16 port POE managed Switch	Nos	1	1	1	2
9.	CAT 6 Armoured STP cable	Mtrs	100	200	250	200
10.	CAT 6 UTP cable	Mtrs	50	50	50	100
11.	Armoured Power cable	Mtrs	50	50	50	100
12.	9U rack	Nos	1	1	1	2
13.	CVT/ Voltage Stabilizer	Nos	1	1	1	2
14.	MCB	Nos	1	1	1	2
15.	Power Surge Protector	Nos	1	1	1	2
16.	Lightning arrestor with all accessories	Nos	1	1	1	2
17.	Earthing	Nos	3	3	3	6
18.	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos	1	1	1	2
19.	Long Range RFID Reader/Writer with pole and mounting	Nos	1	2	2	2
20.	Signal Light for Traffic control	Nos	1	2	2	2
21.	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor (with IP 54 protection)	Nos	1	1	2	2
22.	LED Flood Light along with all accessories	Nos	1	1	2	2
23.	Portable Cabins	Nos	1	1	1	2
24.	10 m Tubular Swaged pole	Nos	1	1	2	2

6. **Integration with other systems:**

The integrated solution and individual parts must be integrated with the following systems

GPS-based vehicle tracking system	Through this integration the system must verify and manage vehicle access by checking the status of GPS devices such as installed/ not installed, active/ inactive, authorized for the concerned area or not, registered in concerned area or not, whether blacklisted or not, and provide appropriate restrictions, alerts, and warnings etc. Also, it shall provide details regarding loading/ unloading with location and timestamp to the GPS- based Vehicle Tracking System.
ICCC	Through this integration the system shall provide the real-time alerts and notifications to the ICCC for overall monitoring and surveillance with location details, timings, etc.
SAP-ERP System	Through this integration the system shall transmit real-time data directly and/or through a central server at ECL, HQ to the SAP-ERP server (as per the need of ECL) and function according to the responses received from the server.
Upcoming Systems	The system must support integration with both state and central government portals, including the JIMMS portal (for mining challans to Jharkhand Govt. portal), FastTag, Vahan portal (for reading the FastTag and getting the detail of the vehicles), and any other relevant platforms/ portals as required by ECL. This includes integrating APIs and SDKs from future systems and providing APIs and SDKs for future systems.
The bidder will be responsible for the development, customization, and implementation of these integrations during the contract period of six years without additional cost to ECL. The ECL will provide necessary support in obtaining the APIs from the relevant government platforms and will bear any transaction fees associated with using these government portals/ platforms.	

7. **Customization and upgradation:**

The system must be customizable, upgradeable and capable to incorporate new functionalities to meet ECL's specific requirements, all without incurring additional costs during the 06-year contract period.

8. **VAPT for software:**

ECL reserves the right to get the Software tested and certified for VAPT from any 3rd party authorized lab approved from Govt. of India. It will be responsibility of bidder to address & close all the vulnerabilities in defined timelines without any additional cost to ECL.

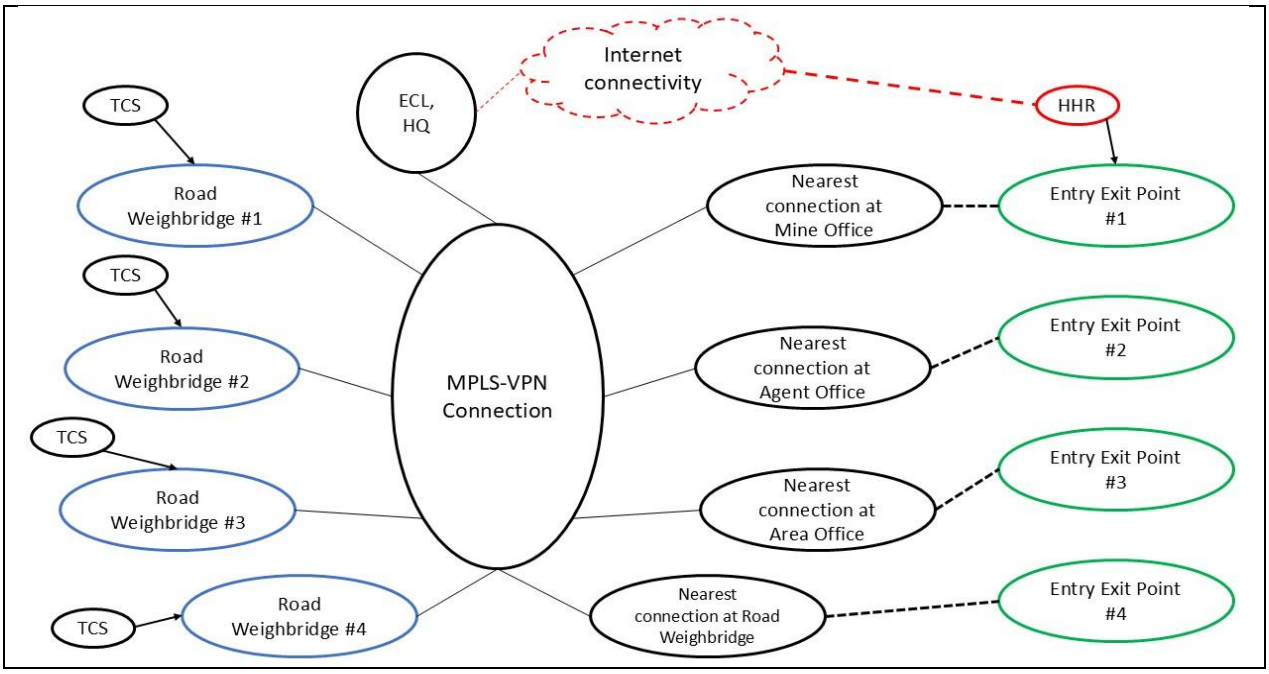
9. Connectivity for implementation and need for working in offline mode:

Connectivity:

All the road weighbridges for the implementation of Weighbridge Automation System with Truck Centering System are equipped with dedicated MPLS-VPN connections from ECL. However, the entry and exit point check posts, for the implementation of Access Control System, must be connected to the nearest MPLS-VPN link (at Area Office/ Agent Office/ Mine Office/ Road Weighbridge etc.) by the successful bidder, using either wired media (CAT 6 cable or fiber cable) or wireless media (radios). These MPLS-VPN connections are accessible from and to ECL HQ. The existing infrastructure such as towers, poles etc wherever available, is to be utilized for wired and wireless connection.

The bidder will install Central Servers at ECL HQ, with redundancy, for implementation of integrated solution. However, the Truck Centering System at the road weighbridges must be directly integrated with the Weighment client Application Software at Road Weighbridges, without relying on the Central Server. The software at central server shall work in integration to synchronize the data of Access control system as well as Weighbridge Automation System with each other and provide facility of working the client software in offline mode in case of connectivity failure.

Following is the pictorial representation of the connectivity



TCS means Truck Centering System
HHR means Hand Held RFID Reader
The solid black coloured lines indicate MPLS-VPN connection
The dashed black coloured lines indicate wired or wireless connection to be established by the bidder
The dashed red coloured lines indicate the public internet connection

Need for Working in Offline Mode:

The integrated solution should be designed to ensure continuous operation in case of connectivity failure to ECL HQ and shall work in offline mode, including vehicle identification and authentication. Once the connectivity is restored, all data and transactions captured during offline operation should be automatically transmitted and synced to the Central Server without requiring any manual intervention. The current data should be uploaded first and the remaining data should be synchronized in the background for immediately starting the current operation. Additionally, for reading FastTag and obtaining vehicle details from the government portal, as well as acquiring GPS device data, the system must maintain an up-to-date local database, updated daily or according to the schedule provided by ECL.

The successful bidder shall provide and integrate the hand-held RFID readers with SIM-based data connectivity for vehicle identification, ensuring functionality even during MPLS-VPN connectivity failures.

It is important to note that these MPLS-VPN connections will not have public internet connectivity, except for ECL, HQ. The bidder shall design and implement the system to support required integrations with other systems accordingly.

10. Scope of work for implementation of Integrated solution, Access Control System, Truck Centring System and Weighbridge Automation System:

a. To design, supply, install, integrate and conduct testing, commissioning and maintenance of all the tendered items for implementation of Integrated solution, Access Control System and Weighbridge Automation System with Truck Centering System.

b. To design and provide the following key features and functionalities for Overall Integrated Solution, Specific to Access Control System, Specific to Truck Centering System and Specific to Weighbridge Automation System.

i. Overall key features and functionalities for Integrated Solution

1. It shall automate and monitor the movement of authorized vehicles, ensure proper placement and accurate weighment on the road weighbridge. It shall capture and store data locally while enabling seamless real-time transfer to the Central Servers and the SAP-ERP server. The system shall eliminate manual entries, ensure data replication, and integrate with the GPS-based Vehicle Tracking System and ICCC to provide a comprehensive overview of each vehicle
2. It shall provide clearly reflected Gate challan details, including entry/exit type, arrival/departure time, weighment data, and loading status.
3. The hand-held RFID readers with SIM-based data connectivity for vehicle identification, shall be deployed at coal dumps, railway sidings, entry exit check posts, and road weighbridges as needed. These readers will be integrated with the

Access Control and Weighbridge Automation Systems. The readers must also have the capability to mark/ tag vehicle loaded by its operator when the loaded vehicle departs from any coal dump after loading of coal and shall also operate during network failures. The bidder is responsible for providing SIM cards and covering the rental charges for the RFID readers as part of the scope of work. ECL shall pay no extra charges for the same.

4. Central servers (main server and backup/ redundancy server) along with all hardware, software accessories, master application & redundancy will be installed at ECL, HQ. The functionality of the server shall include storage, analysis, interpretation of data, and generation of alerts and reports thereof, of all data and transactions collected from all check posts at Mines entry & exit points and road weighbridges connected to MPLS-VPN network of ECL as well as data generating through Handheld RFID reader (with Sim based connectivity). The main server and backup/ redundancy server shall be synchronized on real-time basis.
5. Client Application Software shall be installed at entry exit check posts as well as road weighbridges for Operation of Access Control System and Weighbridge Automation System respectively.
6. It shall operate offline with local registration and authentication along with report generation in case of network failure, and must sync with the central server when connectivity is restored. Request-initiator and listener services shall enable automatic data posting without manual intervention.
7. The client applications must be time-synced with central server. Syncing and replication of central database and timing with client application (at sites) should be done on regular intervals as per the schedule provided by ECL.
8. The client application at entry exit check posts for ANPR authentication shall function locally without dependency on central server and shall function in case of connectivity failure.
9. For the implementation of ANPR functionality at the entry and exit check post, a Workstation PC has been provided. The bidder is expected to implement the system based on this provided Workstation PC. However, if the bidder determines that an additional Local Processing Unit (LPU) is necessary for the proper functioning of the ANPR system, the cost for procuring and installing the LPU will be borne solely by the bidder. ECL will not be responsible for covering any additional charges related to the installation of the LPU.
10. The application software (master application at central server and client application at entry exit check posts and road weighbridges) shall allow the creation of multiple user logins with restricted access based on location. Each user will only be able to view details for their specific location, while a master login will have access to all location details. The software will also include location-based filters and the capability to generate location wise customized reports as per ECL's requirements.

11. The system should have ability to blacklist/ hotlist vehicles in order to block entry or exit of vehicle. The blacklist so created shall be consolidated at the central server level and replicated across all entry exit check posts and all the road weighbridges.
12. The system shall have online records of each step performed in the system with hierarchy-based approval mechanism for exceptions and over-rides.
13. The system shall have online MIS and Dashboards to track status of material dispatched, quality, quantity, grade and DO/SO wise dispatch, customer wise sales etc.
14. It shall have facility for Designing, setup and customization of online and offline reports pertaining to data entry, trip counting, detention monitoring, coal intake, DO lapses, performance matrix for transporters and operators etc.
15. It should have separate screens with user defined inputs for Gate Entry/Exit, Weigh Bridge, stockyards, registration centre, railway sidings etc.
16. Redundant entries must be automatically populated on the screen. No separate data entry is to be done.
17. System will have step-wise work flow. If a previous step is missing, entry should be restricted at the next step. Any vehicle can be blocked before proceeding to the next stage of entry, tare weighment, gross weighment and exit cycle.
18. Hierarchical provision to edit/delete the information.
19. The system shall have facility for access of critical MIS reports through mobile apps (Android, iOS and other major OS). Mobile apps to be provided without additional cost to ECL.
20. The system shall be capable of integrating with the SAP-ERP system of Coal India with importing of data from and posting of weighment/challan data (in a specified format) into SAP-ERP system.
21. The system shall have provisioning of a database of load carrying capacity (can be fetched from vahan portal via integration) of loading vehicles. Accordingly, the system shall have capability to integrate with any state/central govt. portals if required for capturing details/authorisation of vehicles i.e. mparivahan, fasttag for verification and getting information of registration, fitness, taxes etc. of the vehicle without any additional cost to ECL.
22. The system shall be integrated with the GPS based Vehicle Tracking System, for sharing of data regarding weighment timing, mine entry exit timing, status of loading, installation of GPS device, whether registered in the concerned area, trip details, Active and Inactive status etc. This is required for internal coal transportation of the vehicles.
23. The system shall maintain a local database to operate in offline mode, ensuring functionality in case of connectivity failure with the GPS-based Vehicle Tracking System and upcoming integrations with government portals for retrieving vehicle details through FastTag.
24. The system must register RFID tags uniquely and identify and prevent issuing of two RFID tags to the same vehicle or two vehicles the same RFID Tag. Ideally the

Tag must be issued to one area however, there should be provisioning of issuing the same RFID tags to multiple areas simultaneously as per the need of ECL.

25. The RFID tag will be of read - write type and shall be used for vehicles engaged in coal transportation to Railway sidings / washeries and for road sale of coal to various E- auction / FSA / MOU coal consumers or any other vehicle as per requirement including tippers, Vehicles carrying explosives or any other material and Administration vehicles. Application must distinguish between the them.
26. Passive RFID Tags and reflective acrylic stickers with standardized font shall be pasted onto the windshield of the vehicles to facilitate automated data entry and auto-identification of vehicles.
27. Yellow-coloured reflective acrylic sticker of size (4" X 16") with 4" large size black-coloured standardized font displaying the vehicle registration number needs to be affixed on top of the windshield so that the clear view of the vehicle registration number can be recorded through CCTV camera installed at weighbridges and entry / exit check posts. Reflective acrylic stickers to be pasted on vehicles for ANPR purposes shall be printed by the Successful Bidder during RFID tag registration. Cost for the same is to be borne by the Bidder only. The printing of vehicle number on the reflective acrylic sticker shall be under the responsibility of the bidder at no additional cost to ECL.
28. The tag will be of passive nature, tamper-proof & non-cloneable. If tag is destroyed or lost, fresh tag to be issued with record of reasons & details as per previous tag. Lost & mutilated tags will be deactivated prior to activation of new tags.
29. The tag will be unique and password protected and will be linked with the database to carry information (incorporated in the RFID database) like: Vehicle registration no., Type (Tipper/Truck), Registered laden weight (tons / kgs.), no. of wheels, name of the owner with address, Email, contact no. etc., name of the driver and address with contact no shift wise. Photo uploading facility of drivers (If required), Name and address of the transporter/ coal consumer with contact no, E- mail id etc, Details of the transporting work order/ sales delivery order indicating order no and date, party details, quantity of transportation/ sale, grade of coal, source and destination for lifting of coal, validity start and end date of the order, value of the order., etc., Validity of the tag (start date to end date), Trip cycle in hrs./ day to. Further In case of road sale, one weighbridge code with name to be provided, in case of coal transporting minimum two weighbridge code & weighbridge name to be provided.
30. Once the truck is assigned the aforementioned details, the same shall automatically be written by the RFID writer on to database. After the data is written and confirmation to this effect is given by the system, the vehicle will be permitted to enter the mines.
31. Boom Barriers with Traffic Singal Light shall allow movement of only authorized vehicles at entry exit check posts and at the road weighbridges and shall function automatically with manual override.

32. The CCTV cameras deployed for ANPR shall capture the vehicle's registration number and store the image in both the local and central databases. The image frame will be captured in real-time either from the local NVR channel or directly from the CCTV cameras (in the event of an NVR breakdown), with the connection to the Workstation PC via a switch. Additionally, the cameras will also be utilized for surveillance of entry and exit points.
33. Bidder must submit all the software with source code used in the system in soft copy /CD ROM and all manuals for maintenance, administration (2 hard copy & 2 soft copy) at the time of commissioning.
34. The products/equipment proposed/offered should be rugged and proven for 24x7 working in dusty environment of coal mines.
35. It must have provision for user-based response on various exception/alerts with time stamp.
36. All the Central servers and all applications/software including storages at ECL HQ/ Area Hqs shall be cloud ready from day one i.e. entire system or any part of it can be migrated as a DR to cloud in future at any point of time during the contract period, if required by ECL. The cloud will be provisioned by ECL.
37. Following Salient features of different MIS reports shall be implemented:
    a. Area-wise Current Operation Status of all vehicles indicating time of tare & gross weighment at road Weighbridge at sending end and receiving end, weighment, reconciliation of weight at sending and receiving end, time taken to complete trip, time spent in loading and unloading Area etc. as per requirement
    b. DO wise lifting and balance report
    c. Net quantity of material dispatched from mines for road sale and internal vehicles. Filters should be for grade, DO No, transporter, party, type etc.
    d. Net quantity received at sidings and relevant shorts
    e. Value and quantity of coal dispatched, taxes liability, etc.
    f. Abnormal Delay in reaching destination with probable suspicious location identification using data analytics.
    g. Complete trip detail with images of vehicles.
    h. Customised reports of trips and despatches
    i. Total Distance Travelled Reports along with Trip Report (daily/Weekly basis)
    j. Number of round trips per vehicle per day /week/month/custom interval reports.
    k. Onboard weight per vehicle per trip per day/week/month/custom interval report.
    l. Time stamping of vehicle for its presence at weighbridge & Boom Barrier at mine and siding end.
    m. Capable to generate any other customized report as per the requirement of ECL

ii. <u>Key features and functionalities specific to Access Control System:</u>

1. It shall work in two factor vehicle authentication mode as mentioned below with capability to run on single factor authentication in case failure of one of the factors.
   a. **Factor 1:** Reading Vehicle's Number Plates with Automatic Number Plate Recognition (ANPR) cameras
   b. **Factor 2:** Reading RFID tags with Long-range passive RFID Reader/writer.
2. Further it should have provision for enabling/disabling of ANPR authentication at entry exit check posts. If disabled, RFID tags shall be used for vehicle authentication. Further option to restrict road sale vehicles to RFID-only authentication if required by ECL shall be there.
3. Two nos. of "5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost" will be used for ANPR. One at the bottom side and another on top side of the pole. The bidder shall quote the product inclusive of ANPR licenses. The licenses for ANPR asked in the BOQ are additional ANPR licenses which will be deployed at other locations than the Entry Exit Check posts.
4. The Boom barriers shall work automatically for Vehicle entry and exit with manual override.
5. The system should automatically detect the vehicle approaching the entry exit check post (inward or outward moving vehicle) and capture the vehicle's registration number on vehicle's number plate via CCTV cameras using ANPR based on OCR (Optical Character Recognition) and RFID tags through RFID reader. Further the live video feed of CCTV cameras shall be displayed in the entry exit check post through NVR and Monitor.
6. The system shall support real-time i.e., instant detection of vehicles with processing Instant recognition of license number plates, recording of vehicles, reading vehicle's number plate, reading vehicle's RFID tag, looking up the server database and therefore triggering of alarms/ alerts based on the vehicle status and category as specified by the database. The system usage shall be privilege driven using password authentication.
7. The System shall store and enter the images of vehicle and its license plate, along with date timestamp and its location details, into the user specified database at local Workstation PC as well as central server. These images shall be able to be printed on the challan and shall be stored along with the challan details.
8. The system shall be robust to variation in number plates in terms of font, size, contrast and colour and should work with good accuracy.
9. The system shall compare the RFID data, ANPR data and master database and allow entry if data matches; otherwise, block entry and log alerts locally and centrally.

10. The system should have ability to mark the vehicle loaded or unloaded at the entry exit check-post, if required by the operator posted at entry exit check post but its log should be generated in the system for safety and to prevent any misuse.
11. The system should have ability of generating report of all vehicles that have entered/ exited during a specified time slot; List of all vehicles that are within premises (i.e., Entered but not exited yet); List of all vehicles that are within the premises for more than a user specified time limit; Incidence list of any particular vehicle showing how many times and when that vehicle has entered/exited the premises; Provision to generate an Alert if a vehicle that is marked a 'Not permitted' appears at the Entry/exit gate & at the road weighbridge; Provision of additional report which will be finalize in consultation with concerned users during implementation and during contract period.
12. Each buyer is currently assigned a DO/SO against which it collects the material from the mines. They are only authorized to pick material of a particular grade. Based on this information, entry should be restricted in un-authorized mines/yards.

iii. <u>Key features and functionalities specific to truck centering system:</u>

1. In ECL Road Weighbridges the platform size varies from 9 meter to 16 meter. Accordingly, the successful bidder must implement a truck centering system to ensure proper alignment of trucks in the centre of the platform.
2. Separate signal lights with Audio System at road weighbridges shall be used to direct the drivers for proper placement of trucks automatically with manual override option to operator.
3. Weighment at each weigh bridge, must be recorded only if the vehicle is properly centred on the road weighbridge. The weighment challan shall be generated when the truck centering system sends affirmatory command in context of vehicle position.
4. The status of vehicle position on platform, ready for weighment, overload, and weighment details shall be displayed on the display for weighbridge operator as per ECL requirement.

iv. <u>Key features and functionalities specific to weighbridge automation system:</u>

1. The system must automatically detect the vehicle approaching the road weighbridge through RFID tags read by RFID readers with instant detection of vehicles, looking up the server database and therefore triggers alarms/ alerts based on the vehicle status and category as specified by the database. The system usage shall be privilege driven using password authentication.

2. Single Desktop client Application Software shall cater to both Road Sale and Internal dispatch mode of transactions. The transaction mode shall automatically switches depending on the DO/SO number.
3. GST challans shall be handed over to every road-sale vehicle as per applicable Govt. norms. Weight as recorded via weigh-bridge digitizer shall automatically captured by the Desktop weighment client application software. There will be an audit trail of changes/rejections, if any.
4. The GST challan format shall be finalized after consulting with ECL.
5. The system shall fetch the weighment data automatically from the digitizer. No manual entry of weight, whatsoever, under any circumstances are allowed. Further, the system shall display the weighment exactly as it is read from the weigh-bridge on to the computer screen.
6. The system shall have a database of load carrying capacity (can be mentioned during vehicle registration or can be fetched from vahhan portal etc. via future integration) of loading vehicles so that over loading can be checked during weighment and alert can be shown on display at road weighbridge for effective operation of weighbridges. Accordingly, the system shall have capability to integrate with any state/central govt. portals if required for capturing details/authorisation of vehicles i.e. mparivahan etc. without any additional cost to ECL.
7. ECL downloads Delivery Order DO / Sales Order (SO) from SAP-ERP (in .txt format/ excel format). The DO/ SO has all the details of the coal to be shipped like party name, GST No., Shipping Address, Rates, Various Taxes, Grade of coal, Type of DO/ SO, Validity Date, Quantity etc. This information must be directly imported from the .txt or Excel file into the weighment application via an API link from the SAP-ERP system (or another method as specified by ECL), with a manual override option available. The weighment and challan data must then be posted into the SAP-ERP database in a specified format, as determined by ECL. Weighment shall be done by the client weighment application software. Against each weighment an API call is to be made to the SAP-ERP system and on receiving success message from SAP weighment is to be saved.

   Note: - The format of DO/SO & all the necessary details shall be shared with the bidder during execution of the project (if required). The successful bidder shall comply and provide a weighment software as per the prevailing requirements of ECL.
8. Each weighbridge is assigned specific DOs/SOs. Coal can only be lifted from authorized weighbridges corresponding to a particular DO/SO. The weighment client application software must include this functionality.
9. During 1st weighment, driver comes down from the vehicle and operator shall lock the tare weight in the weighment client application software. CCTV cameras shall

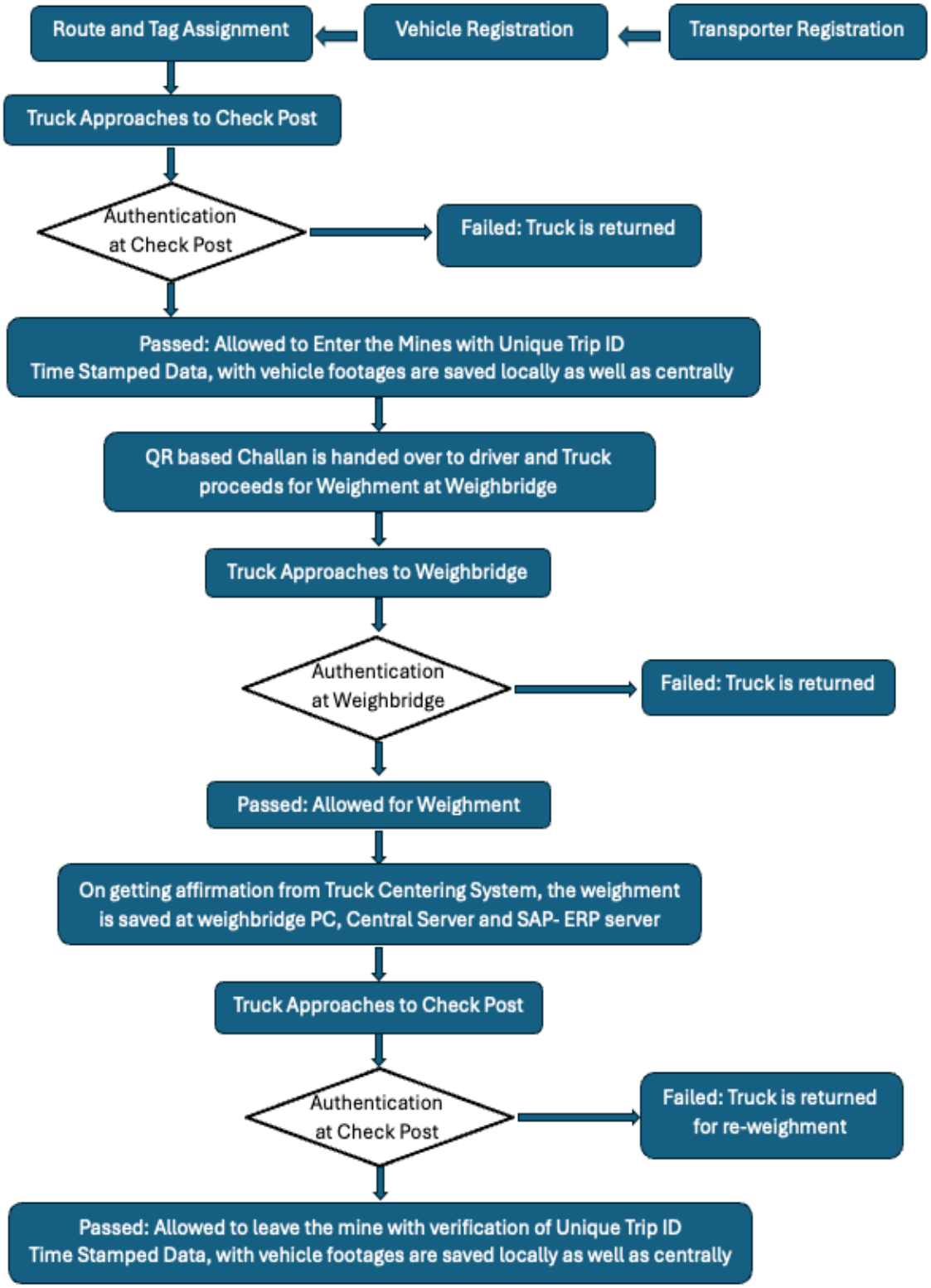
capture the images of trucks from front of the vehicle, rear of the vehicle and top of the vehicle and save with the tare weight details.

10. Against each second weighment an API call is to be made to the SAP-ERP system and only on receiving success message, weighment information along with CCTV images, of trucks from front of the vehicle, rear of the vehicle and top of the vehicle, are to be printed & generated on the acknowledgment slip and saved to local as well as central database
11. The application shall restrict manual entry of vehicle registration number and its weighment. Only RFID Tags read vehicle registration number are to be auto-populated on the computer screen.
12. Two consecutive vehicles shall not allow to be weighed till the digitizer goes to zero.
13. There shall be Flexibility setup time difference between weighment of same vehicles after a certain time-gap only. The average time taken by the vehicle to reach and come back from a particular destination along a particular route are to be considered in calculating the time-difference.
14. It shall have provision for disallowing weighment of vehicles if – vehicle is blacklisted, unauthorized to enter in mines, vehicle appears for weighment before scheduled time difference or delayed beyond permissible time, the CCTV, Vehicle Tracking System, etc. is non-functional/non-responsive
15. Vehicles with expired DO/ SO validity status are not to be allowed to enter or get weighed.
16. Vehicles with lapsed DO/ SO (i.e. where the allotted quantity has been lifted) are not to be allowed to get weighed.
17. Auto-fill entries shall be made against a particular DO/ SO (like party, grade, transporter, party) upon reading a vehicle RFID Tag at weighbridge
18. Posting of data from sites to central database shall occur on real time. The application shall be able to check for missing data by comparing data entries of local and central databases.
19. Each buyer is currently assigned a DO/SO against which it collects the material from the mines. They are only authorized to pick material of a particular grade. Based on this information, entry should be restricted in un-authorized mines/yards.
20. In case of raw coal transfer from Mines to Washery, at washery end a Purchase Order (generated in SAP) is to be downloaded directly into the weighbridge client application software or should be fetched from SAP through an API link. Likewise, the sales process, Raw Coal is to be received against the Purchase Order with the API call to be done during the second weighment in Washery and on receipt of success message weighment slip is to be generated.

21. Average of tare weight: The system should keep record of last 5 tare weights recorded of a vehicle. If the latest tare weight is beyond the threshold its weighment must not be allowed. Proper error message is to be displayed. Weight exceeding RLW weight should not be allowed for weighment.
22. Successful bidder is required to create a web based online portal which will show the consolidated weighment information of ECL like daily road dispatch area wise, weighbridge wise, consumer wise, coal type wise, error reports etc, internal coal transportation reports along with dashboard. The dashboard should be interactive.
23. Weighment client application software should not allow more number of vehicles against a Sales Order than those allotted for the day. Weekly allotment details should be fetched directly from the SAP system and no manual intervention in allotment is to be allowed.
24. Changing the date and time of the client application software, should not be allowed and should be password-protected as it might affect the time of record of weighment.
25. Application must allow capturing of tare and gross weight on different weighbridges as permitted by mines manager.
26. For sidings, weighment details captured at mine's end should be displayed on the receiving end's weighbridge. If the difference in Net weight exceeds a threshold, alerts shall be generated locally as well as at central server and there should be provision to block the weighment, sending notification and alerts to concerned authorities, as well as ICCC. Further logs of all such generated alerts be kept in the local machine as well as central server.
27. For Weighment at both ends (Stock yard and sidings), if the difference in Net weight exceeds a threshold, alerts should be generated locally and sent to central server as well as ICCC. Further logs of all such generated alerts be kept in the local machine as well as central server.
28. The application should have a system for weighment of explosive vehicles too. In case of explosive vehicles, the loaded vehicle is weighed first and then later the empty/second weight is done. The difference should be the final output as the net quantity of explosive for that particular vehicle. There should be no recall of tare for explosive vehicles and a transaction is to be considered as complete when both the loaded and empty weight are done. The system should register temporary tags or FastTag in case of Explosive vehicles.
29. A soft-copy of the invoice printed with CCTV images should be kept as a log.

11. Following is the Proposed Work-Flow and System Methodology of the Integrated system:

a. FLOW DIAGRAM



b. <u>SYSTEM METHODOLOGY:</u>

i. ECL operates two modes of coal transportation: One is Internal Coal Transportation and Second is Road Sale Dispatch. In Internal Coal Transportation, coal is first transported from the mines to the siding and then finally transported to the customers via Rail mode. This transportation is carried out using trucks provided by an agency contracted for coal transportation by ECL. In contrast, Road Sale Dispatch involves directly transporting coal from the mines to the customers by road, with the vehicle being provided by the customers. The frequency of internal coal transportation is significantly higher than that of road sale dispatch. Additionally, the likelihood of the trucks used in road sale dispatch reappearing is much lower.

ii. The Permanent RFID Tags shall be provided to the Internal coal carrying vehicles and Temporary RFID tags shall be provided to the Road sale vehicles or Explosive vehicles before allowing into the mine area, at any suitable location including entry exit check posts and road weigh bridges. However, the system shall able to function with Permanent/ Temporary RFID tags to any vehicle. The necessary information such as truck registration number, consumer/ transporter name, permit details, driver details, address and other required details as per ECL requirement to be recorded into the system and RFID Tag. The RFID Tag will be affixed suitably on each truck wind shield so that it should be retrievable later on by RFID Reader.

iii. In case of Road sale dispatch/Explosive vehicles, the temporary RFID tag can be reused after completion of each trip. For each trip of road sale, the validity of RFID Tag will be mentioned in RFID Tag & Database. The software solution should be designed accordingly. The road sales truck driver shall report to mine/sub area office for getting temporary RFID tag. The officer at the mine/sub area office shall record the information of the truck like truck registration no. Consumer no, permit details etc. Based on the information stored on the central server and a temporary RFID card having a unique no mapped with the information collected is issued to the truck driver. Now the truck moves to the mines through entry check post. The validity of the temporary RFID tag will expire as soon as the road sale vehicle exits the exit check post. The driver will return it to the exit check post.

iv. The Boom barrier should normally be in closed condition with traffic light in red condition. Whenever any vehicle approaches the Mine entry gate, RFID reader shall read the RFID tag on the vehicle for vehicle details. The ANPR camera installed at entry exit Check post shall capture the vehicle registration number. The live video feed shall be displayed in the Workstation PC. The system shall authenticate the vehicle on the basis of RFID Tag information and vehicle details from video image of ANPR camera with data base. On the basis of authentication, the signal light should become green and boom barrier should be opened to allow the authorised vehicle to enter into the mine area. A QR based entry receipt along with a trip no. shall be generated for vehicle. The vehicle number from RFID tag and ANPR camera should be compared and matched for

authentication for opening of boom barrier. If there is no Tag or not a valid tag on the Truck then the boom barrier shall not be opened and vehicle will be sent back.

v. In case of network failure, local authentication should be available on the entry exit check post Workstation PC and through a SIM-based handheld RFID reader connected to the central server. The data may then be uploaded/updated to the central server and the local Workstation PC once the network is restored.

vi. On the basis of authentication, QR based challan shall be prepared at entry exit check post and print out of the same shall be handed over to the truck driver. After completion of authentication vehicle is entered into the mine area and shall proceed to road weigh bridge for first/tare weighment.

vii. In the event of an RFID/ANPR authentication failure at the entry exit check post, the vehicle will be sent back. Additionally, any blacklisted vehicles registered in both the local and central server databases will not be allowed entry and should be displayed at the entry exit check post.

viii. FASTag and Vahan portal-based solution may also be used in addition to the proposed RFID based system for optimal results and reduced congestions through future integration with the Govt. Portals. It may be noted that FASTag solution will only work for vehicles/equipment with a FASTag. For this, any kind of integration with NHAI/NPCI etc. shall be taken care by the bidder which shall be covered in the scope of successful bidder.

ix. Similarly, whenever the vehicle approaches the Mine exit gate after loading and weighment completion, the RFID and ANPR details will be authenticated again for opening of boom barrier to proceed to siding.

x. The necessary reports should be generated at entry exit check post at local Workstation PC as well as at central server. The reports may be in respect of list of in and out vehicles at the entry exit check post vehicle wise, transporter wise and customised as per ECL requirement including the list of vehicles plying in the mine area at any given point of time. It is to note that with generation of necessary reports with customisation as per ECL requirement, the paper work/ manual intervention if any shall be avoided at entry exit check post. Further, the reports with local database shall be generated in case of network failure.

xi. A new weighment client application software shall be installed at Weighbridge PC over the existing infrastructure at road weighbridge through which vehicle details shall automatically populated on the client application software without any manual intervention for entering vehicle number and vehicle details.

xii. The Weighbridge automation system identifies the vehicles on the road weighbridge using RFID tag information through RFID reader. The authentication is done after due verification from local vehicle data base/centralised data base. Integrated weighment and RFID software makes coal weighment & vehicle identification completely man less by automatically recognizing through RFID Reader and other details through RFID Tag on the windshield of truck. It shall use sensors, traffic lights and speakers to guide the

driver about the correct position of vehicles and weighment status with the truck centering system.

xiii. After completing the identification process, image capturing and weighment operation, a virtual challan would pop in PC at WB with the details of vehicle, transporter, order and tare weight etc. and video image of the truck/vehicle with number details shall be embedded on the challan along with QR code.

xiv. Now the vehicle shall proceed to loading area of mines. After getting loaded the truck/vehicle comes back again to the road weighbridge, where the identification and weighment process shall be done as mentioned above. The final challan is popped up in the road weighbridge PC, the gross weight is now recorded in the challan with vehicle image and QR code. The print out of the challan is handed over to the truck/vehicle driver. The challan design should be customisable based on specific requirements throughout the contract at no extra cost to ECL. The vehicle and gross weighment details may also be triggered/flagged at destination weighbridge for reconciliation.

xv. Vehicle now proceeds towards exit check post for siding where it is again authenticated, after authentication, the vehicle is allowed to move towards siding. In case if any vehicle approaches without taking any final weighment from the last road weighbridge, the boom barrier will not be opened at exit gate and system will immediately generate an alert. The system shall also send same alert to the predefined designed officer for necessary corrective action with proper log in to the system.

xvi. At the siding check post also, the vehicle is authenticated by RFID Reader, boom barrier opens and vehicle moves on the road weighbridge platform and gross weight of vehicle is recorded. The gross weight of the vehicle at the mine and sidings weighbridge is cross checked and difference in weight if any is recorded. The coal now dumps at coal yard. The truck now comes back to the weighbridge platform after getting authenticated, the tare weight of the truck is recorded on the same virtual challan. Net weight of coal should also be recorded in each virtual challan. If any discrepancies found, exception alerts and report will be generated.

xvii. In case of any abnormal delay in reaching loaded vehicle from mine road weighbridge to Siding Road weighbridge, an alert and report may be generated from system for corrective action.

xviii. Wherever vehicles are having RFID tags of ECL as well as FASTTag in such cases system should identify only ECL RFID Tag for authentication.

xix. In the event of a network failure, local authentication should be available at both the entry/exit check post and the road weighbridge, as well as through a SIM-based handheld RFID tag reader. The data can be uploaded or updated to the central server once the network is restored

xx. Similar functionality shall be implemented for receiving end weighbridges and entry exit check posts like siding, coal stocks etc., ensuring timely arrival of vehicle, reconciliation of net weight and generate alerts for deviations or delays, both locally and centrally on real-time basis.

xxi. Weighment record will also be shown in gross, tare & net value of coal in Kgs, as measurable unit for weighbridges is in KG.

xxii. A challan/ weigh slip of each loaded vehicle shall be generated with a provision for hard copy printout through weighbridge printer. The challan along with the entire vehicle details and snapshots are to be stored with proper directory management for the entire duration of the project. A copy of the weighment challan data in the required format shall be posted to the SAP-ERP server through a private API over MPLS-VPN network in real time. Similarly, a copy of the weighment challan along with snapshots taken, shall be posted to the RFID application server at ECL, HQ for data replication and report generation.

xxiii. An important point to be noted is that the safety sensors of Boom-Barriers should be high enough to prevent accidental dropping of barrier during passage of cattle or humans. Here, the alarm module of Vehicle Tracking System shall be activated till the vehicle unloads its material at the siding/unloading point.

xxiv. In ECL either side can be used as entry side or exit side for weighment, i.e., weighment from any side of the weighbridge.

xxv. The entire process flow will be customised by the successful bidder as per specific requirements of ECL throughout the contract period at no additional cost to ECL

12. <u>Following is the Mine wise and Area wise distribution of quantities types of Access Control System with its type:</u>

SN	Area Name	Colliery Name	Mine Name	Type 1	Type 2	Type 3	Type 4
1	Bankola Area	Khandra Colliery	VK Unit	0	2	0	0
2			NKJ Unit	0	1	0	0
3		Shankarpur Colliery	Millenium Unit	0	1	0	0
4			3/4 pit	0	1	0	0
5			Railway Siding (West) side of siding 1	0	1	0	0
6		Nakrakonda Kumardihi-B Colliery	Kumardihi-B CM	0	1	0	0
7			Nakrakonda OCP	0	1	1	0
8			POCP-2 Siding	0	1	0	0
9		Shyamsunderpur Colliery	SSPur CM Project	0	1	0	0
10			SSPur Siding (Siding 2)	0	1	1	1
11		Kumardihi-A/ Tilaboni Colliery	Kumardihi - A	0	1	0	0
12			Ols SSPur/ Tilaboni UG	0	2	0	0
13			Tilaboni Siding side of siding 1	0	0	1	0
14		Bankola Colliery	Pit no.: 8	0	1	0	0
15			No.: 2 Incline	2	0	0	0
16			No. 1 Incline	0	1	0	0

17			Sankalp Khani Incline	0	1	0	0
18		**Total for Bankola Area**		**2**	**17**	**3**	**1**
19	**Kajora Area**	Jambad OCP	Jambad OCP	0	0	2	0
20		Jambad UG Colliery	Jambad UG Colliery	0	1	0	0
21		Porascole West Colly	Porascole West Colly	0	1	0	0
22		Porascole East Colly	Porascole East Colly	0	1	0	0
23		Khas Kajora 5/6 no pit	Khas Kajora 5/6 no pit	0	1	0	0
24		Khas Kajora 10/11no pit	Khas Kajora 10/11no pit	0	1	0	0
25		Central Kajora Colly.	Central Kajora Colly.	0	1	0	0
26		Naba Kajora Siding	Naba Kajora Siding	0	1	0	0
27		Porascole Siding	Porascole Siding	0	1	0	0
28		Madhusudanpur Siding	Madhusudanpur Siding	0	1	0	0
29		**Total for Kajora Area**		**0**	**9**	**2**	**0**
30	**Kenda Area**	Bahula, NJ Unit	Bahula UG	0	1	0	0
31		Bahula, RJ Unit	Bahula UG	0	1	0	0
32		Bahula Railway Siding	Bahula UG	0	3	0	1
33		Siduli UG Colliery	Siduli	0	1	0	0
34		Siduli OC (H)	Siduli	0	2	0	0
35		CL Jambad OC (D)	CL Jambad	0	1	1	0
36		CL Jambad OC (H)	CL Jambad	0	2	0	0
37		CBI(OC)/SOCP	Shankarpur OCP	0	1	0	0
38		CBI(UG)	Chora Block Incline	0	3	0	0
39		New Kenda OCP	New Kenda OCP (H)	0	2	0	0
40		Chora 10 Pit	Chora 10 Pit UG	0	1	0	0
41		Chora 7&9 pit	Chora 7&9 pit UG	0	2	0	0
42		**Total for Kenda Area**		**0**	**20**	**1**	**1**
43	**Kunustoria Area**	Parasia Colly./Siding	Parasia Colliery	0	2	0	0
44		Belbaid Colly./Coal Stock	Belbaid Colliery	0	1	0	0
45		Belbaid Siding	Amritnagar Group of Mines	0	2	2	0
46		Kunustoria Colly./Coal Stock	Kunustoria Colliery	0	1	0	0
47		Kunustoria Colly. Pit no. 3		0	1	0	0
48		Bansra Colly., C-Pit	Bansra Colliery	0	1	0	0
49		Bansra Siding	Bansra Group of Mines	0	1	1	0
50		Northsearsole OCP	Northseasole OCP	0	1	1	0
51		Amritnagar Colly.	Amritnagar Colliery	0	1	0	0
52		Narayankuri HW Project	Narayankuri HW Project	0	1	2	0
53		**Total for Kunustoria Area**		**0**	**12**	**6**	**0**
54		**Sonepur Bazari Project**	Old CHP Gate	0	1	0	0

55	**Sonepur Bazari Area**		Dragline yard Gate	0	0	1	0
56			Project office Gate	0	1	0	0
57			Bazari Kali Temple Gate	0	1	0	0
58			Railway Siding Eastern Gate	0	0	0	1
59			Old 4 MVA Gate	0	1	0	0
60		**Total for Sonepur Bazari Area**		**0**	**4**	**1**	**1**
61	**Rajmahal Area**	Rajmahal OCP	Rajmahal OCP	0	0	0	2
62		Hura - C	Hura - C	0	0	0	1
63		**Total for Rajmahal Area**		**0**	**0**	**0**	**3**
64	**Satgram Sripur Area**	Nimcha 4 no Pit	Nimcha 4 no Pit	0	1	0	0
65		Nimcha 7/8 no Pit	Nimcha 7/8 no Pit	0	1	0	0
66		Amkola OCP	Amkola OCP	0	1	0	0
67		Nimcha High Wall	Nimcha High Wall	0	1	0	0
68		Satgram Incline	Satgram Incline	0	1	0	0
69		Satgram Project	Satgram Project	0	1	0	0
70		Chapuikhash	Chapuikhash	0	1	0	0
71		Kalidaspur Project	Kalidaspur Project	0	1	0	0
72		J.K Nagar Colliery	J.K Nagar Colliery	0	1	0	0
73		J.K Nagar Project	J.K Nagar Project	0	1	0	0
74		Pure Searsole	Pure Searsole	0	1	0	0
75		Bhanora UG	Bhanora UG	0	1	0	0
76		Charanpur OCP	Charanpur OCP	0	1	0	0
77		Nimcha Siding	Nimcha Siding	0	1	0	0
78		JMCN Siding	JMCN Siding	2	0	0	0
79		**Total for Satgram-Sripur Area**		**2**	**14**	**0**	**0**
80	**Pandaveswar Area**	Dalurbandh Colliery/OCP	Dalurbandh Colliery/OCP	0	0	1	0
81		Khottadih OCP	Khottadih OCP	0	0	1	0
82		Khottadih UG Colliery	Khottadih UG Colliery	0	0	3	0
83		Pandaveswar Colliery	Pandaveswar Colliery	0	0	1	0
84		Madhaipur Colliery (09 no Pit)	Madhaipur Colliery (09 no Pit)	0	0	1	0
85		Manderboni Colliery	Manderboni Colliery	1	0	1	0
86		**Total for Pandaveswar Area**		**1**	**0**	**8**	**0**
87	**S P Mines Area**	Girja Check Post	Girja Check Post	2	0	0	0
88		Girja Marshleen Yard	Girja Marshleen Yard	0	0	1	0
89		Near Patch E	Near Patch E	0	0	1	0
90		Check Point near Khoon WB	Check Point near Khoon WB	2	0	0	0

91		Damgarah patch south end	Damgarah patch south end	0	0	1	0
92		Damgarah patch north end	Damgarah patch north end	0	0	1	0
93		Bhawanipur Gate	Bhawanipur Gate	0	0	1	0
94		Jamtara Railway Siding	Jamtara Railway Siding	2	0	0	0
95		**Total for SP Mines Area**		**6**	**0**	**5**	**0**
96	**Jhanjra Area**	Jhanjra Project	MIC Coal Transport Gate	2	0	0	0
97		Jhanjra Project	3&4 Transport & Main Gate	0	1	0	0
98		Jhanjra Project	POCP-1 Siding Gate	2	0	0	0
99		**Total for Jhanjra Area**		**4**	**1**	**0**	**0**
100	**Sodepur Area**	Narsamuda Colliery	Narsamuda Colliery	0	1	0	0
101		Chinakuri Mine III Railway Siding	Chinakuri Mine III Railway Siding	0	1	0	0
102		**Total for Sodepur Area**		**0**	**2**	**0**	**0**
103			Chapapur II (New Hired Patch)	0	0	1	0
104		Badjna	31 no Incline	0	0	2	0
105			29 no Incline	0	0	1	0
106		Badjna OCP	Badjna OCP	0	0	1	0
107		Hariajam Colliery	27/28 Incline	0	1	0	0
108			5/6 Incline	0	1	0	0
109		Nirsha OCP	Nirsha OCP	0	0	1	0
110		Central Pool/ Railway Siding	Central Pool/ Railway Siding	1	0	1	0
111		Khoodia	Khoodia	0	0	1	0
112		Lakhimata Colliery	Lakhimata Colliery	0	0	1	0
113		Shyampur B	Shyampur B	0	1	0	0
114		Rajpura OCP	Rajpur OCP	0	1	0	0
115		Barmuri OCP	Barmuri OCP	0	0	1	0
116		**Total for Mugma Area**		**1**	**4**	**10**	**0**
117	**Salanpur Area**	Mohanpur	Mohanpur	0	4	0	0
118		Itapara	Itapara	0	1	0	0
119		Gourandi (A)	Gourandi (A)	0	1	0	0
120		G. Begunia	G. Begunia	0	2	0	0
121		Bonjemihary	Bonjemihary	0	2	0	0
122		Dabor	Dabor	0	2	0	0
123		**Total for Salanpur Area**		**0**	**12**	**0**	**0**
	Total			**16**	**95**	**36**	**6**

13. **Following is the Area wise Distribution of quantities of Truck Centering System:**

SN	Area Name	Total Existing WB	Total Upcoming WB	Total WB
1	Bankola Area	13	2	15
2	Kajora Area	7	3	10
3	Kenda Area	9	2	11
4	Kunustoria Area	9	2	11
5	Sonepur Bazari Area	11	0	11
6	Rajmahal Area	4	3	7
7	Satgram Sripur Area	13	7	20
8	Pandaveswar Area	10	0	10
9	S P Mines Area	6	2	8
10	Jhanjra Area	6	1	7
11	Sodepur Area	8	0	8
12	Mugma Area	14	2	16
13	Salanpur Area	13	6	19
	Total	**123**	**30**	**153**

D. Part C: Command & Control Centre in Areas and ECL HQ

1) The project encompasses the establishment of an Integrated Command & Control Centre (ICCC) at ECL Headquarters and Area Command & Control Centre (also referred to as Area Control Rooms/ Area e-Surveillance & Control Rooms) at 13 Areas of ECL to manage and monitor operations across the entire ECL.
2) The Command & Control Centres will monitor video feeds from various CCTV cameras installed in vulnerable locations within the mines and other establishments of ECL. Equipped with state of art AI based AI based Video Analytics, the ICCC will detect, and alert specified events as detailed in the Technical Specification Document.
3) The ICCC at ECL HQ shall be continuously monitored 24x7 by trained Service Personnel of the supplier. The successful bidder shall deploy 3 operators for round-the-clock monitoring in the ICCC at ECL HQ during the Warranty Period.
4) The successful bidder must regularly perform continuous testing and configurations to ensure the system functions effectively. Additionally, the system should include a feature that logs all software modifications.
5) The ICCC system should offer a dedicated portal featuring an area-wise dashboard. Area authorities should be able to access this portal using area-specific login credentials to view information, generate reports, and file Action Taken Reports (ATRs) for specific events or exceptions within their respective areas. The quantity of such credentials shall be at least 30.
6) The ICCC System Dashboard should have provision for providing escalation matrix for automated sending of exceptions to designated authorities as per SOP defined by ECL from time to time. The notifications shall be sent on email, SMS and App notification. The ICCC software should have a mobile app for field responders.
7) Command & Control Centres shall be established at all Areas, while an Integrated Command & Control Centre (ICCC) shall be set up at ECL HQ.

(i) **Area Command & Control Centres**

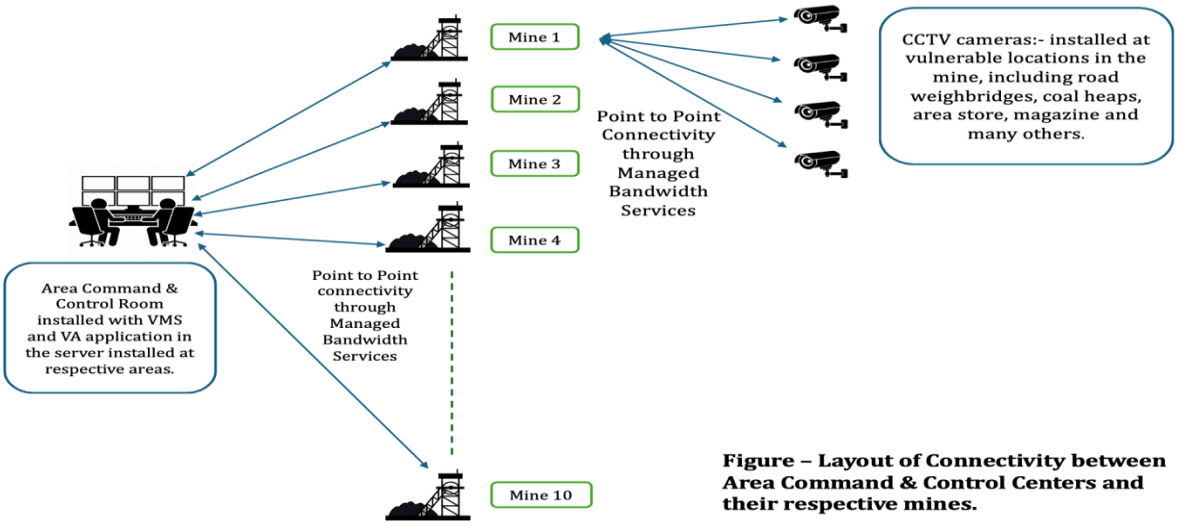


Figure – Layout of Connectivity between Area Command & Control Centers and their respective mines.

- As shown in the diagram, each Area Command & Control Centre is connected to its respective mines through Point-to-Point Connectivity on Managed Bandwidth Services. Within the mines, CCTV cameras are strategically installed at critical locations such as road weighbridges, coal heaps, stores, magazines and many others to enhance surveillance and security.
- The AI based Video Analytics Application will be hosted on servers installed in each Area Command & Control Room, with one server per area and ECL HQ.
- AI-based Video Analytics will be deployed on CCTV cameras installed across various locations within each Area, covering both existing and newly installed cameras as per ECL's requirements. The AI based Video Analytics Application software will be implemented throughout all ECL Areas, and the bidder is required to provide a centralized dashboard that will serve as an integrated platform for managing all AI based Video Analytics functions.
- Video Management Software (VMS) has already been installed in 10 out of 13 areas, with all video feeds transmitted via Managed Bandwidth Services. Video Management Software (VMS) for rest of the 3 locations i.e., Sonepur Bazari Area, S P Mines, and Rajmahal has been covered under this project. The bidder has to integrate the existing VMS to the VA as well as ICCC. The bidder is responsible for integrating the existing VMS (make: IDIS, model: ISS - I NEX) with the VA and ICCC. If integration with the

existing VMS is not feasible, the bidder must install their own VMS software on the existing hardware provided by ECL, if necessary.

- The area-wise allocation for AI based Video Analytics deployment is detailed below:

TABLE-A: PART-I

SN	Description of Analytics	Bankola Area	Kajora Area	Kenda Area	Kunustoria Area	Sonepur Bazari Area	Rajmahal Area	Satgram-Sripur Area
		1	2	3	4	5	6	7
1	Camera Tampering Detection	30	13	19	18	9	26	30
2	Crowd detection	25	8	8	30	31	10	31
3	Automatic Number Plate Recognition	0	0	0	0	1	1	0
4	Object counting or people/ vehicle counting	0	0	0	0	1	1	0
5	Classification of objects	0	0	0	0	1	1	0
6	Loaded/Unloaded condition of the vehicles	0	0	0	0	1	1	0
7	Perimeter Violation	20	10	19	6	0	0	11
8	Loitering Detection	10	0	6	7	0	2	6
9	Fire & Smoke Detection	18	7	11	6	0	12	7
10	Traffic Congestion detection	0	0	0	0	1	1	0
11	PPE KIT detection	10	5	4	8	0	6	5
12	Safety hazards detection	0	0	0	0	1	1	0
13	Insufficient illumination	0	0	0	0	1	1	0
14	Accidents Alerts	0	0	0	0	1	1	0
Total		**113**	**43**	**67**	**75**	**48**	**64**	**90**

TABLE-A: PART II

SN	Description of Analytics	Pandaveswar Area	SP Mines Area	Jhanjra Area	Sodepur Area	Mugma Area	Salanpur Area	ECL HQ	Total QTY.
		8	9	10	11	12	13	14	15
1	Camera Tampering Detection	10	15	30	35	38	34	3	310
2	Crowd detection	3	12	10	37	34	14	7	260
3	Automatic Number Plate Recognition	0	1	1	0	0	1	1	6
4	Object counting or people/ vehicle counting	0	1	1	0	0	1	0	5
5	Classification of objects	0	1	1	0	0	1	0	5
6	Loaded/Unloaded condition of the vehicles	0	1	1	0	0	1	0	5
7	Perimeter Violation	2	4	7	3	0	7	6	95
8	Loitering Detection	4	2	3	4	8	4	4	60
9	Fire & Smoke Detection	3	5	9	2	18	2	5	105
10	Traffic Congestion detection	0	1	1	0	0	1	0	5
11	PPE KIT detection	0	1	5	6	7	0	0	57
12	Safety hazards detection	0	1	1	0	0	1	1	6
13	Insufficient illumination	0	1	1	0	0	1	1	6
14	Accidents Alerts	0	1	1	0	0	1	0	5
Total		**22**	**47**	**72**	**87**	**105**	**69**	**28**	**930**

Note: - The allocation of AI based Video Analytics deployment is subject to change throughout the contract period and must be carried out as per ECL's requirements.

- AI based AI based Video Analytics Software should be capable of performing the following analytics:

  (a) Vehicle Detection and classification
  (b) Loaded/Unloaded Status of vehicles
  (c) Number plate detection
  (d) Automatic Number Plate Recognition
  (e) Rule-based vehicle counting
  (f) Wrong way driving detection
  (g) Over speeding detection
  (h) Accident Detection
  (i) Traffic Congestion detection
  (j) Detection of number of people present inside the truck during tare and gross weighment
  (k) Detect and highlight incidents of vehicles bypassing weighment at weighbridge
  (l) Detect and highlight incidents of vehicles bypassing Boom-barriers
  (m) Crowd detection and miscellaneous event detection
  (n) Unusual / Anti-social activity detection: Detection of fire, weapons, objects-theft and incident of vandalism
  (o) Detect cases tempering of CCTVs (change in orientation, obfuscation of live view, change in focus etc.) or camera disconnection
  (p) Detection of safety hazards: Identify whether a person is wearing a safety kit as per prevalent safety norms. Also, other incidents pertaining to unsafe behavior (e.g. Human presence near speeding vehicles, water logging, saggy electrical wires etc.)
  (q) Weighbridge and Boom Barrier tampering
  (r) Insufficient illumination detection

- During the period of contract, the successful bidder shall be responsible for customization of the solution as per requirement of the company, including additional AI based Video Analytics on mutually agreed basis.

(ii) **<u>Integrated Command & Control Centre at ECL HQ</u>**

The Integrated Command & Control Centre (ICCC) shall be established at ECL HQ. ECL will provide a suitable room with adequate space at its headquarters for the establishment of monitoring centre.

The following essential infrastructure will be installed at the Integrated Command & Control Centre (ICCC) at ECL HQ:

SN	Item Description	Unit	Quantity
1	ICCC Software Platform	Set	1
2	Servers for ICCC	Nos.	3
3	Videowall	No	1
4	30 KVA UPS	No	1

The Integrated Command & Control Centre (ICCC) at ECL HQ will be equipped with all the essential infrastructure to ensure seamless operations. The ICCC software platform will be hosted on servers at the ECL HQ, enabling centralized monitoring and management. It will be accessible across all areas via Managed Bandwidth Services (MBS). A videowall will be set up to display live surveillance feeds and analytics with clarity, while a 30 KVA UPS will provide backup power, ensuring uninterrupted system performance for stable and efficient operations.

8) <u>Integration with various surveillance system in ECL</u>

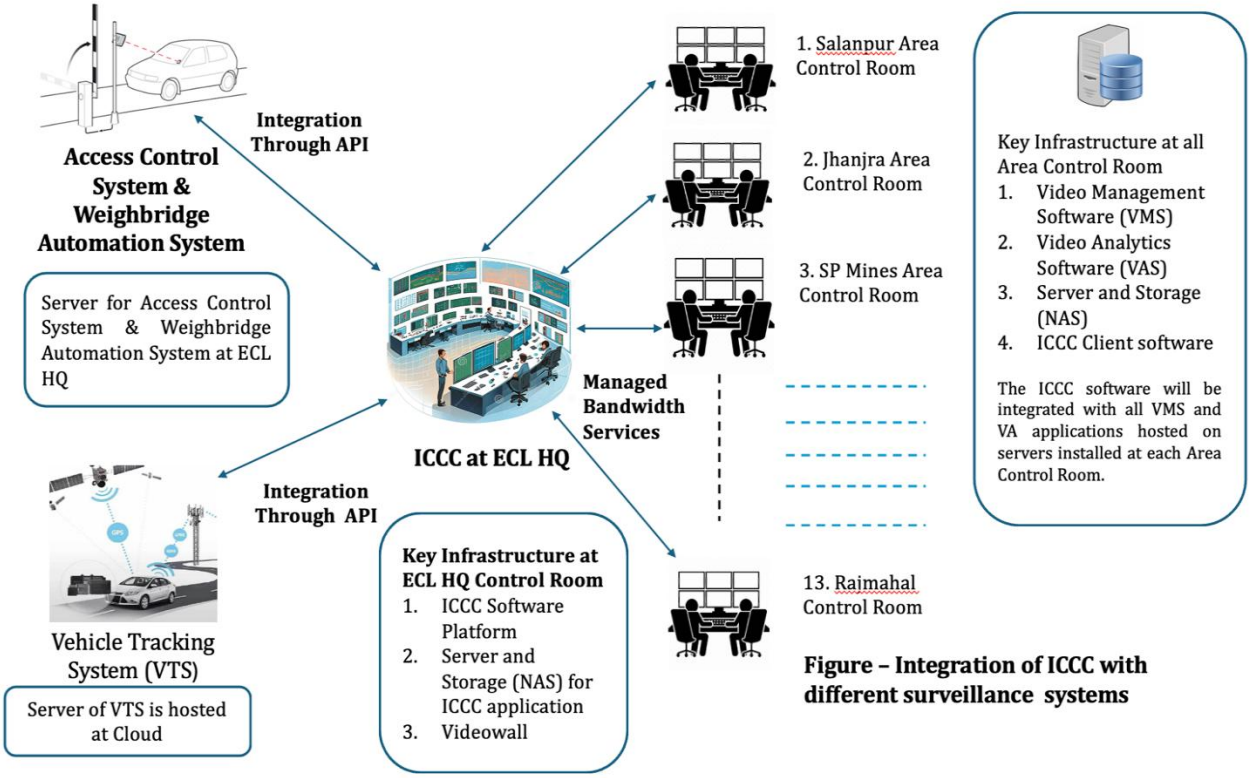


Figure – Integration of ICCC with different surveillance systems

An Integrated Command & Control Centre (ICCC) will be established at ECL HQ as the central hub, seamlessly connected to all Area Command & Control Centres (also referred to as Area Control Rooms/ Area e-Surveillance & Control Rooms) through Managed Bandwidth Services. Each Area Command & Control Centre equipped with a Video Management System (VMS) and a AI based Video Analytics Application Software (VAS), hosted on local servers shall be fully integrated with the ICCC Software Platform. Additionally, the Weighbridge Automation System (WAS) server is already installed at ECL HQ, while the Entry-Exit Access Control System (ACS) server will be installed at ECL HQ. Both systems will be integrated into the ICCC Software Platform. The successful bidder will be responsible for developing the necessary APIs/software to enable seamless integration, ensuring a unified dashboard that facilitates real-time data monitoring, reporting, and data correlation across multiple systems.

Integration Scope

(i) **CCTV Surveillance System:** The bidder shall be responsible for developing the necessary APIs/software to integrate both the existing and new VMS (Video Management System) and VAS (AI based Video Analytics System) with the ICCC Software Platform at its own cost. The integration will be carried out via Managed Bandwidth Services, connecting the Area Command & Control Centres to the ICCC at ECL HQ. CCTV cameras will not be directly accessible at ECL HQ; therefore, integration with the ICCC Software Platform must be done exclusively through the VMS. If the bidder is unable to integrate the existing VMS, they must deploy their own VMS including {CVMS license for existing cameras (1144 Nos.) as well as for CCTV Cameras for Biometric Machines (330 Nos.)} while utilizing the existing infrastructure at the Areas and ECL HQ (including Server Storage, NAS, etc.) to ensure seamless integration.

(ii) **Vehicle Tracking System (VTS):** **ECL will provide the API** for VTS integration. ECL has implemented a GPS-based Vehicle Tracking System (VTS) installed on coal-carrying vehicles used for transportation from mines to sidings. This system is hosted on a cloud-based server.

(iii) **Access Control System (ACS) & Weighbridge Automation System (WAS):** Since the bidder will be deploying its own. The bidder will be responsible for API development to ensure successful integration with the ICCC Software Platform.

(iv) **Futureproofing and IoT Integration:** The project is designed to be scalable and adaptable, ensuring seamless integration with future IoT systems. This future-ready solution will enhance security, operational efficiency, and centralized management, making ECL's operations more robust and technologically advanced. The bidder shall quote the price for future integration of the IOT system as per the requirement of ECL.

Note: -

(i) ECL will provide the API only for the Vehicle Tracking System (VTS) and any future IoT systems. For the systems, including the CCTV Surveillance System (both existing and new), Access Control System, and Weighbridge Automation System (WAS), the bidder shall be solely responsible for developing the necessary APIs/software at its own cost. If required, the bidder must also install its own software, including the Video Management System (VMS), to ensure successful integration.

(ii) If existing systems such as VTS, WAS, or SAP-ERP are renewed during the six-year contract period, the bidder shall be responsible for integrating them at no additional cost to ECL.

9) Bill of Quantities (BOQ)

The items listed below in the Bill of Quantities (BOQ) are relevant to this specific part of the project.

SN	Item Description	Quantity	Unit
1	Integrated Command and Control Centre (ICCC) software platform	1	Set
2	Video Wall along with all hardware & software accessories	1	No.
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	310	Nos.
	(ii) Crowd detection	260	Nos.
	(iii) Automatic Number Plate Recognition	6	Nos.
	(iv) Object counting or people/ vehicle counting	5	Nos.
	(v) Classification of objects	5	Nos.
	(vi) Loaded/Unloaded condition of the vehicles	5	Nos.
	(vii) Perimeter Violation	95	Nos.
	(viii) Loitering Detection	60	Nos.
	(ix) Fire & Smoke Detection	105	Nos.
	(x) Traffic Congestion detection	5	Nos.
	(xi) PPE KIT detection	57	Nos.
	(xii) Safety hazards detection	6	Nos.
	(xiii) Insufficient illumination	6	Nos.
	(xiv) Accidents Alerts	5	Nos.
4	Servers for ICCC and Video Analytics	16	Nos.
5	Online UPS of 30 KVA with 2-hour backup at Full Load	1	No.
6	42 U Rack for Server	1	No.
7	Furniture Set	1	Set
8	Air Conditioner, 2 Ton, 5 Star	3	Nos.
9	Unit Rate for future integration of different verticals with ICCC	1	No.
10	Technical Manpower for 01 Year	3	Set
11	Video Storage	1	Nos.
12	Workstation PC	3	Nos.
13	24 Port Layer 2, Managed switch	2	Nos.
14	16 port POE managed Switch	13	Nos.
15	Floor mounted Rack 24U enclosure	13	Nos.
16	1 KVA online UPS with 120 min backup with isolation transformer	13	Nos.

E. INSTRUCTIONS TO THE BIDDER

1. The successful bidder shall provide complete support with timely training to the ECL personals for total Contract period of 72 months (12 months + 60 months) i.e. Warranty period of 12 months plus CAMC period of 60 months from the date of successful commissioning of overall project.
2. Software designed & developed for RFID, ANPR, Boom Barrier based Access control system, RFID and Boom barrier based weighbridge automation system with truck centering system, Integration of presently running GPS based VTS system, CCTV Surveillance System, Web based On-line Portal related to vehicle management and weighment, Mobile app, Report generation & their customization, alert generation, hardware & peripheral accessories required for above mentioned software & services should be capable of catering present ECL requirement.
3. All the software designed & developed should be independent of hardware make & model and capable of integration with present deployed systems like RFID-Boom barrier-based weighbridge automation system, CCTV surveillance system, GPS based Vehicle tracking System as well as must have capability to integrate with other upcoming systems during the tenure of contract period without additional cost to ECL.
4. The ownership of all data generated, captured and reports generated by the system during the entire contract period shall remain with ECL and the same should be made available to ECL in softcopy in desirable format after completion of contract period for future use.
5. ECL utilizes two types of networking connectivity: MPLS-VPN and Point-to-Point managed bandwidth services. The MPLS-VPN network, with lower bandwidth, connects ECL HQ, Area Offices, and mines, enabling seamless communication between all locations. It will be used for the Access Control System at Entry-Exit Check posts and the Weighbridge Automation System at road weighbridges. On the other hand, the Point-to-Point managed bandwidth services, which offer higher capacity, provide separate connections between the Mines and Area Offices, and between the Area Offices and ECL HQ, specifically designed to transmit video footage from CCTV cameras. There is no direct connectivity between Mines and ECL HQ through this service. The ICCC will also utilize the Point-to-Point managed bandwidth for its operations. Bidders are required to design the system in accordance with these connectivity specifications as outlined in the scope of work.
6. **OWNERSHIP RIGHTS:**
   - **a)** All the application software developed, and hardware used for the mentioned application & services in the bid document will become property of ECL after completion of contract.
   - **b)** The selected Bidder shall handover all passwords (login, database etc.), authentication etc., all data, material, information, flow diagram and documents (including design documents, database design / diagrams, the source code with Third Party controls/ software, etc.) collected/ generated during development / implementation of the project to ECL.
   - **c)** The application software and the source code with the version control system shall be submitted separately to ECL.

7. The proposed solution including hardware and software for the RFID based boom barrier system, CCTV system and Network connectivity should have standards of RFID/Boom barrier system implemented at NHAI corridors.
8. The solution offered should have the capacity to add-on the components if any new requirement(s) come during the contract period without any additional cost to ECL.
9. All the Central servers and all applications/ software including storage at ECL HQ/ Area HQs shall be cloud ready from day one i.e. entire system or any part of it can be migrated as a DR to cloud in future at any point of time during the contract period, if required by ECL. In this regard, the necessary infrastructure shall be provided by ECL.
10. It shall be the responsibility of the successful bidder to integrate the proposed system seamlessly through any upcoming/ future networking infrastructure of ECL without any additional cost to ECL.
11. The successful bidder shall ensure that the functioning & operation of the existing IT-Initiative system in ECL shall not be disturbed till the proposed solution is commissioned and handed over.
12. Bidder shall have to provide access to ECL or third party authorized by ECL for establishment of connectivity of Area Control room with CCTV cameras as well as Control room of ECL HQ.
13. ECL has considered all equipment for fulfilment of objectives of the system. However, if any items/ equipment/ system is required in addition to the equipment mentioned in the bid document for fulfilment of the objectives, the same should be supplied, installed, and maintained by the Successful Bidder at its own cost. Hence all such items/services which are not included in the bill of materials, but may be required during installation & commissioning, shall be provided by Successful Bidder without any additional cost to ECL.
14. All the software must be provided with latest version and from the original source. Source of software products should be specified in case of third-party software. The supplier shall attach a copy of software license/distributorship license for supply of such products. The supplier shall indemnify the owner for possible damage due to copyright violation, if any.
15. The required OS and DBMS licenses for achieving the full functionality will be under the scope of the bidder without additional cost to ECL.
16. **PRICE**: The quoted amount will remain fixed for the entire contract's duration. The rates for each item offered in this tender will apply to any future procurements, should ECL require additional materials during the warranty and CAMC period. However, any reduction in material rates must be passed on to ECL.
17. During the period of the contract, any facility/feature/ software upgradation of any supplied item which are offered by the respective OEM/ Developer will have to be made available to ECL without any additional cost to ECL, even if the same is not mentioned in the bid document.
18. **Wiring/cabling standard:** All wiring/cabling/ terminations/ patches must follow EIA/TIA standards wherever applicable. All Cabling should have proper non-perishable identification at the terminals for easy tracing. Proper documentation with diagram should be made for all wiring/cabling/terminations. All indoor wiring/Cablings should be properly secured, duly terminated, easily maintainable and aesthetically sound.
19. **Placement/Fittings etc.:** All Equipment installed should be properly placed which should be secure, aesthetic, convenient for maintenance, healthy from operational and fault liability point of view and follows standards (Industry and/or recognized authority) wherever

applicable. It will include placement of all switches in suitable Racks, other hardware including Power Supplies in suitable racks/enclosures/stands etc. Suitable enclosures, if any is needed beyond the list of equipment mentioned for the Area, as per site requirement, will have to be provided by the bidder without any additional cost to ECL.

20. POWER SUPPLY:

a) Unregulated Power Supply (nominal 220V AC) will be provided by ECL for installation of entire system. However, the Successful Bidder will arrange for any additional power wiring/ distribution, and for regulation/stabilization of voltage required. All such works must be technically and aesthetically sound and standard based, wherever applicable.

b) The bidder will ensure to provide necessary protection system/module for the entire system from lightning, even if not mentioned in the list of equipment. In addition to the material as provided in the list of equipment, the vendor is free to add additional equipment or protection system to strengthen the protection system at his own cost.

c) The Successful Bidder will ensure that the protections provided in the equipment work properly in case of any abnormal electrical condition. ECL will make best efforts to ensure a healthy power supply, however in case of any damage to the equipment due to abnormal power supply or any other electrical disturbance, the damage must be rectified/replaced by the Successful Bidder without any additional cost to the company. The equipment so replaced against the damage should be of the same make and model or higher specifications, without affecting the performance of the system.

d) Power supply to the system & damage due to lightning, over-voltage, etc.: The service provider will ensure to provide necessary protection system like Lightning Arresters, Earthing (The contractor shall strengthen the earthing and to bring it to the desired level i.e. <= 2 ohm) and Ethernet Surge Protectors for the entire entire system from Lightning, even if not mentioned in the bid document. In addition to the material as provided in the bid document, the service provider is free to add additional equipment or protection system to strengthen the protection system at their own cost

Hence, any damage to the equipment, system, or cables caused by lightning or abnormal power supply or power surge must be rectified, repaired, or corrected by the service provider at no additional cost to ECL throughout the entire contract period of 6 years.

21. Earthing: The installation, testing, and commissioning should include lightning arresters, chemical earthing, and earth pit digging as required. If the successful bidder deems additional protections, earthing, or arresters necessary, they must provide these at no extra cost to ECL. **Bidders are advised to bid accordingly.**

22. The Successful Bidder will arrange for all tools/tackles, testing instruments, software etc. required for installation, testing and maintenance of the system throughout the contract period (06 years). All the hardware and software along with accessories and fitting hardware such as brackets, poles together with associated trenching duct laying, cable pulling, masonry

& civil work, site restoration, Power, CAT-6 cabling, integration of the system are included in the scope of work.

23. The Successful Bidder will be fully responsible for the safety of its personnel engaged in installation work. They will ensure that they and their personnel abide by the provisions of all relevant acts applicable, relating to the work. They will also ensure that he abides by all the rules and regulations of central and state governments and any local authority while executing the work. ECL will have the right to call for any documentation in this regard, instruct the Successful Bidder and take any action as deemed fit, in case of deficiency. The Successful Bidder will be solely responsible for all liabilities in this regard. The Successful Bidder will indemnify ECL from any consequences caused due to their failure in this regard and will be solely responsible for bearing them.

24. The Successful Bidder shall make the necessary arrangement at the ICCC for logging the complaints received. The Firm shall ensure a single point of contact with escalation matrix for all the complaints related to respective Project/Unit/HQ. It shall be the responsibility of the Firm to mobilize their technical team and back-end support for restoration of the break down. The Firm is required to provide 24 x7 help desk mentioning telephone number, E-Mail address.

25. The application software, middleware, and drivers for various gadgets in use shall be property of ECL and have to be handed over to ECL for their further use, if any. All system & network security guidelines & approvals as per rules within the prevalent law/act must be followed.

26. All hardware and software offered by the bidder must be IPV6 compliant with backward compatibility i.e. IPV4 compliant and configuration needs to be done either in IPv4 or IPv6 or combination of both as per the requirement of ECL as per the guidelines of Ministry of Telecommunication during the contract period. Each Component / Peripheral of the System must be compatible with each other. There should not be any compromise on the specification.

27. Bidder shall provide all patches and updates of firmware, during the entire contract period as and when released by OEMs without any additional cost to ECL. They have to provide a total technical solution with all necessary hardware and software including installation, commissioning, operation and maintenance for the entire contract period (including warranty period).

28. The Bidder should not bid / supply any equipment / item, that is likely to be declared end-of-sale/obsolete within the rated service life of the equipment/item from the date of supply of that equipment/item. In case of such happening, such devices should be replaced with a similar or higher model of supplied make, by the Bidder, without sacrificing the performance of the system, at free of cost.

29. The Bidder has to ensure the compatibility of all the material supplied with the existing network in addition to ensuring compatibility among them. Any additional material (cables, connectors, interfaces and other parts of the equipment, etc.) not exclusively mentioned in the Buyer specification Document but required for successful installation and operation of

the system or integrating to the existing network, are in the scope of supply of the Successful bidder at its own cost. **Tenderers are advised to bid accordingly.**

30. The fibre laying is over the ground and are intended to be used within ECL's territory. However, in case of road crossing the permission from local authorities shall be taken by the successful bidder.
31. **SITE SURVEY:** Bidders are encouraged to visit and examine the work site and its surroundings at their own responsibility, cost, and risk. They should obtain all necessary information for preparing their bid and conduct a detailed survey before submission. The network design and system implementation should prioritize flexibility for future expansion and ease of maintenance. The cost of visiting the site is at the bidder's expense. It will be assumed that bidders have visited the site/area, familiarized themselves with the prevailing conditions and potential fluctuations, and have factored these into their bid, regardless of whether they actually visit the site or not. Bidders cannot make claims against the buyer for non-compliance due to a lack of prerequisite information, as it is solely their responsibility to obtain all necessary information about the site, surroundings, working conditions, weather, etc., before submitting their bid.
32. **PATENT RIGHTS:** The Bidder shall indemnify all Projects / Units/HQ of ECL against all 3rd party claims of software piracy & infringement of intellectual propriety rights. In the event of any claim asserted by a third party of infringement of copyright, patent, trademark or industrial design rights arising from the use of the Goods and services or any part thereof in India, the Bidder shall act expeditiously to extinguish such claim. If the Bidder fails to comply and ECL is compelled to pay compensation to a third party resulting from such infringement, Bidder shall be responsible for the compensation including all expenses, court costs and lawyer fees to ECL and as per the company norms further action may be taken against the bidder.
33. **PACKING: -** All equipment and materials supplied against this contract shall be suitably protected, covered in waterproof packing and the packing should be strong enough to prevent damage or deterioration during transit handling and storage at the site till the time of final commissioning. The bidder shall be responsible for any loss or damage during transportation, handling and storage due to improper packing. Care should be taken to make separate packages for each installation location.
34. Receipt inspection at yard shall be carried out by Buyer QC team and/or any external agency nominated by Buyer jointly with Seller representative. All requisite documentation including packing list, certificates, approved drawings etc. are to be submitted to Buyer in advance for enabling inspection. Detailed Packing list (considered as actual supply quantity) shall clearly mention individual items in package and shall be the basis for physical inspection on receipt.
35. **Splitting of Contract: -** Since this is a case-consolidated contract covering all aspects i.e., Supply, Installation, Testing and Commissioning as well as Comprehensive Annual Maintenance Contract after expiry of warranty period, it will be **considered as a single unit** and there will be no splitting of the contract.
36. **A HINDRANCE REGISTER** shall be maintained by both department and the contractor at site to record the various hindrances, encountered during execution of Supply, Installation,

Testing and Commissioning. Hindrance register will be signed by both the parties. The contractor may also record his observations in the Hindrance Register. In case the contractor has a different opinion for hindrance and a dispute arises then the matter would be referred to the HOD (E&T), ECL or the next higher authority whose decision would be final & binding on the contractor.

37. The successful bidder must provide 24/7 technical support at ECL HQ during the one-year warranty period with three qualified and experienced technical personnel possessing a Diploma or Bachelor's degree in Engineering(Electronics, IT, Computer Science, or related field) and 1-5 years of experience in system maintenance, technical support, or field service operations. These personnel must be skilled in hardware troubleshooting, networking, CCTV, and command center systems, capable of performing assigned duties efficiently, and proficient in training ECL personnel in ICCC operations, including report generation and system debugging. Additionally, the bidder must provide comprehensive training to ECL's technical and monitoring staff in relevant ICCC domains while ensuring compliance with all applicable labour law guidelines concerning ECL. Bidders should consider these requirements when preparing their bids.

Role of Manpower to be deployed at ECL HQ

(i) The deployed manpower will monitor all types of alerts on dashboard, and they will identify the authenticity of the alerts and report to concerned person for further necessary action as per Standard Operating Procedure (SOP) of the ECL. The deployed manpower will also follow the instructions/orders related to ICCC given by HOD(E&T) time to time during the contract.

(ii) The deployed manpower may work in shifts and manage the ICCC system and shall comply the statutory rules and regulations. The posting & shift duty of manpower may be changed during entire contract period as per the requirement of ECL. The deployed manpower shall report to the HOD(E&T), ECL or his authorized representative.

(iii) The Successful Bidder will maintain sufficient spares, tools and accessories required for repair/ replacement/relocation/maintenance and configuration of active and passive components of the System to achieve guaranteed availability during the entire contract period.

(iv) In case of the deployed manpower is on leave, it is the responsibility of bidder to provide the substitute. Bidder will provide name & contact no. of manpower and documents regarding the same need to be submitted at the time of Installation & commissioning.

(v) Arranging accommodation for manpower is the responsibility of the bidder at their own cost.

(vi) A logbook containing the details of the incidents of fault/problem reported, attended, rectified with time should be maintained by deployed manpower and made available to ECL. All the entries must be verified, and counter signed by the Nodal officer of ICCC/GM (E&T)/HOD or his authorized representative at the site.

The responsibilities of the deployed technical personnel do not include tasks associated with the Comprehensive Annual Maintenance Contract (CAMC), and these personnel shall not be involved in CAMC-related work.

In addition to the above, the Successful bidder will have to keep provision for an on-call network engineer (for resolving network issues), an on-call server engineer (for resolving hardware issues) and post a full time Service-Delivery Manager (SDM), a representative of Successful bidder, immediately after the signing of the Contract. The details of nominated SDM (the name and contact details, including address, telephone number, mobile number, email address) should be conveyed in writing to ECL within 10 days of receipt of order. The Service-Delivery Manager will be required for the entire duration of the contract to co-ordinate the project at ECL HQ. The SDM shall be in direct employment and under administrative control of the Successful bidder and the ECL Management shall have no responsibility /liability whatsoever in this regard.

The responsibilities of the Service-Delivery Manager as a part of support are as follows (indicative but not exhaustive):

a. Act as a Single Point of contact (SPoC) for the entire project.
b. Responsible for the entire execution & management of the project after receipt of work order.
c. Overall monitoring of project
d. Co-ordination for Delivery/installation/Commissioning/Testing of ICCC System.
e. Call flow management, Quality Service Delivery.
f. Service Level Agreement management and reporting.
g. Submission of periodical Reviews and reports required by ECL.
h. Crisis management and Emergency response procedures

38. HANDOVER OF DOCUMENTATION: Upon commissioning of the system, the Successful Bidder will hand-over the following documents, in hard copies (02 Sets) and soft copy (in .pdf or any other acceptable format), to the HoD (E&T), ECL or his authorized representative:

(i) Detailed schematic Network diagram of the complete installation.
(ii) Escalation matrix for service support
(iii) Detailed diagram of all sub-systems of the Network
(iv) Detailed Cable/Wire Network as installed
(v) All configuration details including different related lists
(vi) Any license/permission or related documents
(vii) Test results
(viii) Any operation and/or maintenance procedures etc. (along with soft copy in pdf format)
(ix) List of Equipment supplied/installed and other items used (e.g. Wires/Cables), Software, Portable Equipment and Spares. The list must include Make, Model, Brand, Version, Sl. No. etc., as applicable.
(x) Relevant Manuals, CDs etc.

(xi) Any other relevant document/Information as required as per the Bid document, or, as may be required for operation/maintenance of the system or, otherwise, relating to individual Item/the System.

All the Documents should bear the reference Bid number and Supply Order/ Contract Number of ECL have a Heading and signed with seal by the authorized representative of the Successful Bidder. However, any original document like License may be submitted with forwarding letter only from the Successful Bidder. All the documents to be submitted to ECL within 30 days of Commissioning of the System.

ECL will have the right to call for any further documents/clarification from the Successful Bidder on Products supplied at any stage. The Successful Bidder must submit such documents within the required time of receipt of such communication.

39. TRAINING

The successful bidder will provide onsite training to Operational/Technical Personnel of ECL and other users during installation and maintenance of the system. In addition, the Party will also arrange training for up to 6 Technical Personnel (in batches) of ECL on the Server, Switches & Software at Developer/OEM/OEM authorized centre/premises, free of cost, if desired by ECL. The duration of such training will be for a minimum of six days, and the course content should cover operation/monitoring of equipment and software.

40. PRE-COMMISSIONING INSPECTION AND TESTING: Upon installation, the System and its Components will be jointly (in presence of both the Successful Bidder and ECL representatives) tested as follows:

i. All equipment and installations will be individually inspected physically.

ii. Individual active equipment will be tested with Power. Passive items also will be tested wherever feasible.

iii. Different Sub-Systems will be tested wherever feasible.

iv. Then, the system will be tested in an integrated manner.

v. The parameters of testing will be jointly decided keeping in view the technical specifications, scope of work and Special Terms & Conditions of the bid document.

vi. For testing the video analytics features of the system, ECL shall ensure that there is proper connectivity exists between all fundamental components. If connectivity has been established for the recording/ storage device(s) of a particular Area, then the testing shall be undertaken with the camera feeds of that particular Area.

vii. All the tendered video analytics features tested during inspection should give expected results with at least 90% accuracy.

viii. On acceptance of the test results by HOD(E&T), ECL or his respective authorized representatives, as satisfactory, trial run will start.

41. TRIAL RUN, SYSTEM ACCEPTANCE & COMMISSIONING: On acceptance of testing of the system, the system will be put under trial run.

i. The duration of trial operation of the complete equipment shall be seven (07) days.
ii. The trial operation shall be considered successful, provided that each item of the equipment can operate continuously at the specified operating characteristics, for the period of trial operation.
iii. The bidder should be able to continuously demonstrate the system functionality at the desired accuracy.
iv. Upon successful completion of the trial run and testing at ECL HQ & its associated units as well as all the Areas, the entire system will be considered commissioned. The HOD(E&T) will then issue an Installation and Commissioning Certificate for the complete system.
v. If the overall commissioning of the project is delayed for one month or more and an extension for delivery is granted without imposing a delay penalty, then a period up-to 01 month (included in the warranty period of one year) will be provided for the restoration of the breakdown, during which no penalty will be imposed.
vi. Warranty of the whole system will be considered from the date of Commissioning of complete system at ECL.
vii. During the period between the actual date of installation of equipment and commissioning of the system, the system will be maintained in all respects, including those detailed in the CAMC, by the Successful Bidder at their own cost.
viii. Acceptance of the System by ECL will not absolve the Successful Bidder from all of its liabilities. In addition to whatever has been stated in Bid document, the Successful Bidder will be liable for any deficiency in the System traceable to be due to deficiency on the part of the Successful Bidder in Supply/Execution or any other aspect, during the warranty and CAMC Period.

42. Consignee: -

(a) For ECL HQ: Depot Officer, Sodepur Central Stores, PO – 713359, Dist.: Paschim Bardhaman, West Bengal
(b) For 13 Areas Offices as per following table, Consignee: Depot Officer of respective Area Stores

SN	Name of the Area	Address of Depot Officer of respective Area
1	Bankola Area	Bankola Area Store, ECL PO-Ukhra, Dist.- Paschim Bardhaman, West Bengal - 713363
2	Kajora Area	Kajora Area Store, PO-Kajoragram, Dist.- Paschim Bardhaman, West Bengal - 713338
3	Kenda Area	Kenda Area Store, PO-Bahula, Dist. Paschim Bardhaman, West Bengal - 713322

4	Kunustoria Area	Kunustoria Area Store, PO-Topsi, Dist. Paschim Bardhaman, West Bengal- 713362
5	Sonepur Bazari Area	Sonepur Bazari Area Store, ECL Project Office, Dist- Paschim Bardhaman, West Bengal - 713378
6	Rajmahal Area	Rajmahal Area Store, PO-Lalmatia, Dist. Godda Jharkhand - 814165
7	Satgram-Sripur Area	Satgram-Sripur Area Store, PO-Devchandnagar, Dist.- Paschim Bardhaman, West Bengal-713332
8	Pandaveswar Area	Pandaveswar Area Store, PO-Pandaveshwar, Dist.- Paschim Bardhaman, West Bengal-713346
9	S. P. Mines Area	S. P. Mines Area Store, PO-Chitra, Dist.-Deoghar, Jharkhand - 815351
10	Jhanjra Area	Jhanjra Area Store, PO-Laudoha, Dist.- Paschim Bardhaman, West Bengal - 713385
11	Sodepur Area	Sodepur Area Store, PO-Sunderchak, Dist.- Paschim Bardhaman, West Bengal-713360
12	Mugma Area	Mugma Area Store, PO-Mugma, Dist.- Dhanbad, Jharkhand, PIN- 828204
13	Salanpur Area	Salanpur Area Store, PO-Lalganj, Dist.- Paschim Bardhaman, West Bengal - 713359

43. **TRANSPORTATION & TRANSIT-INSURANCE: -** The bidder shall be solely responsible for transporting the materials to the designated consignee location and, after acceptance, further transporting them to the installation site at its own expense. All transportation and transit insurance necessary for the supply and commissioning of the system shall be arranged by the bidder at no additional cost. Any damage to the materials or system during transit and until the final handover shall be the sole responsibility of the vendor. Any insurance required to mitigate such risks shall be procured and borne by the vendor.

44. **Delivery Schedule:** The supply, installation, testing, and commissioning of the complete system must be completed within **240 days** from the date of issuance of the Purchase Order. This timeline does not include the trial run period of 07 days.

45. **Paying Authority:** GM (F) I/C, Finance dept. ECL HQ. However, the bill raised for supply installation, commissioning shall be certified by the respective area on satisfactory performance to HOD (E&T) ECL HQ

46. **Payment Terms: -**
    i. 80% value of the equipment {except (i) **Integrated Command and Control Centre, (ICCC) software platform (ii) Video Analytics Application Software (iii) Vide Management Software Solution (VMS) (iv) Enterprise CVMS with Video Analytic License (v) Network Management Systems (vi) Central Server application (vii) Client Software**} and accessories and 100% taxes and duties and other charges shall be made within 21 days after receipt and acceptance of items with all accessories by

the consignee and also receipt and acceptance of Security Deposit / Performance Bank guarantee (as specified in the tender document) by ECL.

ii. Balance 20% payment of the equipment and 100% of installation & commissioning charges shall be released within 21 days from submission of bills after successful completion of installation, commissioning and acceptance of commissioned equipment at sites by respective Areas of ECL and subsequent issuance of certificate for final Installation, integration, commissioning and acceptance of the system for entire ECL by HOD (E&T) ECL to their entire satisfaction based on acceptance certificate issued by Areas. For successful commissioning of equipment the supplier has to depute competent person at site and provide technical assistance required for commissioning and initial testing of the equipment.

iii. 100% value of items **(i) Integrated Command and Control Centre, (ICCC) software platform (ii) Video Analytics Application Software (iii) Vide Management Software Solution (VMS) (iv) Enterprise CVMS with Video Analytic License (v) Network Management Systems (vi) Central Server application (vii) Client Software** and other software shall be made after successful completion of installation, commissioning and acceptance of commissioned system at site by respective Areas of ECL and subsequent issuance of certificate for final Installation, integration, commissioning and acceptance of the system for entire ECL by HOD (E&T) ECL to their entire satisfaction upon presentation of acceptance certificates issued by the Areas.

iv. Payment related to Technical Manpower Services, CAMC Services, and Future Integration Services are excluded from the above terms and will be processed upon completion of the respective activities as outlined in the contract.

v. Payment for technical manpower services will be processed on a quarterly basis. This payment will be based on certified records of manpower availability and attendance, as verified by the Nodal Officer.

vi. Payment for future integration services related to various verticals with the Integrated Command and Control Centre (ICCC)— such as Biometric Attendance System will be released upon successful completion of the integration, ensuring seamless interoperability with the ICCC.

vii. The payment of CAMC charges will be made on a quarterly basis after successful completion of the work as per terms and conditions of CAMC with applicable SLA. The availability of the systems handed over by ECL to the Successful bidder will be calculated on a monthly basis and invoices for the same will be submitted in triplicate to ECL HQ after ratification of HQ representatives.

viii. Last quarterly bills shall be settled after end of the contract period after adjusting all outstanding dues.

ix. No interest is payable on any amount whatsoever to the successful Bidder/ contractor. The company reserves the right to recover/ enforce recovery of any overpayments detected after payment as a result of post payment audit or technical examination or by any other means, notwithstanding the fact that the amount of disputed claims, if any, of the contractor exceeds the amount of such over payment and irrespective of the facts whether such disputed claim of the contractor are the subject matter of arbitration or not. The amount of such overpayments may be recovered from the subsequent bills under the contract, failing that from contractor's claim under any other contract with the company or from the contractor's security deposit or the contractor shall pay the

amount of overpayment on demand.

x. Amount payable/ recoverable for any subsequent change in GST will be made to/from the successful bidder/contractor after departmental verification of such changes of tax law issued by statutory authority.

xi. No Mobilisation advance shall be paid to the successful bidder/contractor.

47. OTHER TERMS AND CONDITIONS

i. The Successful bidder should liaise with statutory authorities to obtain all the necessary clearances (if any). Tenderer should also indicate the License fees etc. in the offer, which ECL has to pay to the concerned Govt. Organizations. Arrangement for its payment will be the responsibility of the Supplier.

ii. The technical specifications listed in the bid document are the minimum required specifications. Items of higher specifications will also be acceptable. The bidder has to provide entire working solution as defined in the scope of work. If the solution of the bidder requires items of higher specifications, than those mentioned in the bid, then the bidder has to quote for such items of higher specifications without additional cost to ECL. Bidders have the option to propose a product with higher specifications for evaluation. However, it's crucial that any such higher specification product fully meets the criteria set out in this tender document and is both compatible with and meets the requirements of ECL. Any suggested deviations from the specified product must be thoroughly explained, along with reasons for its suitability and alignment with the objectives of the tendered solution. Non-compliance with these requirements could result in the rejection of the bid. Furthermore, if the documents submitted by the bidder create any ambiguity that is not adequately clarified by the bidder, the bid shall be promptly rejected. Also, the decision to accept any offered higher specification product falls under the authority of ECL. Failure to declare Make and Model of materials offered/quoted and/or non- submission of required details/ technical document/OEM Datasheet may lead to rejection of the bid. The successful bidder/Service Provider shall arrange all the subscription/Licenses required for all software (which are covered or not covered in BOQ) for smooth working of the offered ICCC solution as applicable.

iii. The Successful bidder should be able to provide online availability calculation and performance reports through a web dashboard, automatically. There should be an interactive platform for recording suggestions and remarks.

iv. There shall be a provision to integrate / share information to the ERP server, i.e., SAP or any other data management service as and when desired by the management at no extra cost. All system & network security guidelines & approvals as per rules within the prevalent law/act must be followed.

v. The application software of the offered system shall be capable of communicating and exchanging data with SAP/ any other relevant system as desired by the ECL management.

vi. There should be a provision for backing up the entire database onto a cloud/server/any other service arranged by ECL for use in the future.

vii. Industry standards have to be followed for integration with other industry standard solutions provided by other vendors.

viii. The ownership of all the data captured and reports generated by the system during the entire contract period shall remain with ECL and the same be made available to ECL in soft copy, as and when required, for future uses.

ix. It shall be the responsibility of the successful bidder to make necessary adjustments and/or repairs/maintenance to keep-up the project in ECL as per specification / requirement in proper and satisfactory working condition.

x. Any software modification/upgradation as well as data integration, within the existing hardware, deemed necessary by ECL, shall be done with no additional cost to ECL.

xi. Any other software package required by ECL, **related to ICCC System,** to be installed outside of existing hardware (not mentioned in the current BOQ and scope of work), during the tenure of the contract, may be provided by the successful bidder, if desired by ECL, after proper justification of the item rate. The scope of work shall be modified accordingly to include various aspects of the new system.

xii. **EMPLOYMENT OF WORKMEN/COMPETENT PERSONS**: It is the responsibility of Successful Bidder to employ engineers/technicians or competent-technical persons as may be required and to follow all the norms and practices set under statutory requirements and provisions made in the labour law and other applicable laws and acts, as defined from time to time at subject locations of ECL.

xiii. **ACCIDENT OR INJURY TO WORKMAN: -** In no manner ECL shall be liable to the Bidder or any member of the Bidder's staff or any person or to Govt. or other bodies for injuries or death caused as a result of accidents either within or outside the installation site in the course of work. The Bidder shall be responsible for such contingencies and will make good all claims for compensations, claimed by the labour or staff or as decided by appropriate authority / tribunal or other competent agencies and discharge, all liabilities under the workman's compensation Act and other relevant laws of the land. The Bidder shall also indemnify the company and disburse all the sum that may be awarded in respect of claims for compensations out of or consequent to any staff or labour working under you or any other involved persons pursuant to the provisions of workman's compensation Act (VIII of 1923 and IX of 1933) or any subsequent modification or amendment to the act thereof or any other enactment or law in force at relevant time.) All costs incurred in connection with any such claims should be made good by the Bidder and the company reserves the right to pay in the first instance such amount of compensation is payable under the said act and to recover the amount so paid from the Bidder.

xiv. Safety Provisions: Workers should be trained for the job before being engaged by the contractor, if not trained, and a code of safe practice to be framed and circulated to all concerned workers for general safety awareness for the proposed job.

xv. Bidder has to adopt proper safety aspect as per ECL & other industrial norms and ensure safety of their personnel and others during erection, installation, commissioning and maintenance of the system. The Successful Bidder shall be responsible for safety & security of the men & materials engaged in this work & all types of necessary safety precautions should be followed during execution of the work. The Successful Bidder will also have to arrange all materials, safety tools like helmet, shoes, safety belt etc. In no matter ECL shall be liable to the Successful Bidder or any member of the contractor's staff or any other person deputed by the Successful Bidder in case of injuries or death caused as a result of accidents either within or outside the working

areas of ECL.

xvi. The Successful Bidder shall provide for full medical treatment/compensation to the contractor's staff and labour in case of accident on duty and ECL shall have no responsibilities whatsoever towards the staff/labour deployed by the Successful Bidder. In the event of any injury and or accident that may occur to the workmen engaged by the Successful Bidder compensation shall be paid as per statutory norms by the Successful Bidder.

xvii. ECL will not be liable in respect of any damage or compensation payable to any workmen or any other person employed by the Successful Bidder. The bidder shall also ensure compliance of all guidelines, restrictions and directives by the manpower employed by the bidder regarding COVID-19 or any other in force at the time of commencement of work or issued at any time during the entire contract period.

xviii. It will be the sole responsibility of the Successful Bidder (to whom the contract will be awarded), to secure necessary permissions, clearances from whatever bodies/organizations necessary to complete the contractual task and this is to be done by the Successful Bidder at their own cost and responsibility. ECL will be responsible for any delay on this part.

xix. All workers working near moving machines should wear fluorescent jackets.

xx. All the rules, regulations and laws under mines act to be complied.

xxi. Traffic rule to be enforced strictly. In the open cast mine pedestrian should utilize road meant for them.

xxii. **TERMINATION, SUSPENSION, CANCELLATION & FORECLOSURE OF THE CONTRACT**

In case of failure by the bidder to comply with any of the terms and conditions of this document or of the awarded work order and/or in case of unsatisfactory performance by the contractor and/or owing to loss of viability in successfully maintaining/ upgrading or implementing the project, ECL reserves all the rights to terminate the contract wholly, without any advance notice and also to impose suitable penalty including stoppage of payment of previous bills.

ECL also reserves the right for foreclosure of the CAMC, in case the CAMC is no longer required. Further, if the progress of the work or of any portion of the work is unsatisfactory, the Engineer-in-charge, after giving the bidder 15 days' notice in writing, without cancelling or terminating the contract, shall be entitled to employ another agency for executing the job or to carry out the work departmentally or contractually through tendering/limited tendering process, either wholly or partly, debiting the bidder with cost involved in engaging another agency or with cost of labour and the prices of materials, as the case may be. The certificate to be issued by the engineer-in-charge for the cost of the work so done shall be final and conclusive and the extra cost, if any, shall be borne by the bidder. However, when this clause is invoked, the penalty will not be applicable.

F. SCOPE OF WORK DURING WARRANTY PERIOD AND CAMC PERIOD

1. The warranty for the supplied materials shall remain valid for 01 (One) year from the date of overall commissioning of the system. The CAMC of 05 (Five) years shall come into force after the expiry of 01 (One) year of warranty period from the date of overall commissioning of the system.
2. The bidder is responsible for providing proper maintenance of the entire system, repair/replacement of each item damaged/ malfunctioned due to technical reasons. **However, equipment damaged due to theft, burglary, hit-and-run, strike/riots, accident, fire, earthquake or other Acts of God, will be exempted from this clause.** If a breakdown of the system occurs due to any of the above-mentioned reasons, then a joint inspection will be carried out by ECL official(s) & vendor representative(s). Additionally, any damage to the equipment or system caused by lightning or abnormal power supply must be rectified, repaired, or corrected by the bidder at their own expense.
3. The bidder is responsible for repair, replacement, servicing, and maintenance of the entire system, including preventive maintenance. This includes system testing, repairs, and upkeep of all equipment to ensure proper functionality. It also encompasses the maintenance and repair of all tendered items, including batteries of all types **(with batteries to be replaced in case it gets faulty during the six-year contract period)** and other associated active and passive components. Additionally, any necessary digging, cable laying for maintenance, preventive maintenance expenses, routine inspections, and attending to breakdown calls as required are included.
4. The Bidder shall be responsible for providing all materials, equipment, and services, specified or otherwise, which are required to fulfil the intent of ensuring operability, maintainability, and reliability of the complete equipment covered under this scope of work. This work shall follow all applicable standards, statutory regulations and safety requirements in force from the date of award of this contract.
5. The bidder is responsible for covering the cost of on-site manpower, all necessary spare parts (on a landed-price basis), labor charges, and any other associated expenses during both the Warranty and CAMC periods. No additional payments shall be made for maintenance beyond the CAMC charges, which will be paid on a quarterly basis, as these costs are included in the contract.
6. **The bidder will have to establish a helpdesk with the 24x7 availability along with the contact number and escalation matrix.** The bidder shall provide a dedicated portal/web interface/site/email/WhatsApp to ECL for logging calls (trouble tickets/ docket number), viewing status of calls (trouble tickets/ docket number), escalation of trouble tickets (docket number) and monitoring performance parameters.
7. All supplied software which includes firmware, operating system pack, other utilities, backup software, drivers, etc. if any, supplied by the successful bidder to be activated, registered and updated at all installed locations by the supplier. All service packs, patches, updates bug fixes, related with the operating system and other software covered under the bill of materials for the respective job, are to be supplied by the supplier without any additional cost, if required,

during the full contract period (01 year Warranty and 05 years CAMC period). Antivirus license will be renewed by the bidder at Bidders own cost.

8. All Travel and other expenses for the persons/service engineer/technician deployed by the bidder are included in his/her bid. The bidder shall also undertake to train the persons deployed by the employer for day-to-day operation.
9. **Data Handling and Retrieval Responsibility:** The Successful Bidder shall be responsible for the proper handling and retention of all recorded data in the system procured under this tender, limited to the data stored in the Surveillance HDD (or NAS/SAN/Storage device) installed in the Server / NVR / Workstation.
   i. In the event that the recorded data becomes corrupted, inaccessible, or lost due to any means, the bidder shall bear full responsibility for retrieving and recovering the data. The bidder shall take immediate action to rectify any issues that may lead to data loss or corruption to minimize the impact on the system's functionality. The retrieval process shall be carried out at no additional cost to the tendering entity, solely for the data stored in the device.
   ii. Furthermore, the bidder shall implement suitable backup and recovery mechanisms to safeguard the recorded data from potential data loss scenarios. The backup strategy should be robust and should not affect the continuous monitoring and recording capabilities of the offered system.
   iii. The procuring authority shall not be held liable for any data loss, corruption, or damages incurred by the bidder due to mishandling, negligence, or any other reasons related to the recorded data.
   iv. By submitting a bid in response to this tender, the bidder acknowledges and agrees to take full responsibility for the proper handling, security, confidentiality and retrieval of recorded data as outlined in this clause.
   v. This clause forms an integral part of the tender document and shall be binding upon the Successful Bidder upon acceptance of their bid.
10. Comprehensive Maintenance of the supplied system is the essence of work. Bidder has to provide consistent availability of the entire system after installation & commissioning of the same.
11. It also includes configuration of hardware/software/firmware of entire system, network devices (if any), with repairing/replacement of defective parts.
12. Bidder is also required to provide renewal of Licenses for all the supplied systems, if required, including patches, updates, upgrades and its related inspection & supervision services free of cost.
13. Maintenance/replacement of all types of interconnecting cables, networking cables, connectors and power cables supplied along with the equipment shall be covered under this scope of work.
14. Service report needs to be made after attending breakdown/maintenance calls countersigned by Area/HQ nodal officer or his authorized representative at the site and copies of all the service reports including periodical maintenance visits must be submitted to Area/HQ nodal officer on monthly basis.
15. A logbook containing the details of the incidents of fault/problem reported, attended, rectified with time should be maintained and made available to ECL. All the entries must be

verified, and counter signed by the Area/HQ nodal officer or his authorized representative at the site.

16. **MANPOWER:** - During the Warranty and CAMC periods, the vendor shall deploy a minimum of 14 Service Engineers to ensure the smooth operation of the system with on-site maintenance support (01 service engineer in ECL HQ and 13 Service Engineers at 13 Areas of ECL). The successful bidder will provide name and contact number of service engineers deployed including identity document. In the case of the deployed engineers on leave, it is the responsibility of the bidder to provide the substitute as required. Arranging accommodation for the service engineers is the responsibility of the bidder at their own cost.
17. If any equipment is to be taken outside the premises of ECL, the successful bidder should replace the existing equipment with a spare working equipment of the same make and model or higher specification model of the same make, and configure to work in the network seamlessly, so that there should not be less number of equipment or equipment of lesser specifications installed than the equipment ordered, installed and commissioned at any point.
18. The prospective bidders must maintain minimum 10% spares for hardware and other related equipment along with the required qualified maintenance personnel at strategic locations to provide satisfactory maintenance support for the system offered to ensure SLA.
19. **AFTER SALES SERVICE AND MAINTENANCE**: The Successful Bidder shall also deploy additional service engineer/technician (wherever required) for early rectification of the issues /breakdowns, for cleaning of equipment or providing assistance and technical support, in order to maintain the guaranteed availability of the system at point of time of the contract and post them at convenient place in consultation with ECL to ensure smooth maintenance of commissioned ICCC system during warranty period (01 year) and CAMC period (05 years).
20. At the end of the CAMC period the bidder shall ensure that the system is in full working condition. In case the system is not in full working condition then the damages can be recovered from pending dues of the bidder.

G. Service Level Agreement (SLA) and Penalty

The successful bidder is required to ensure availability of the system on a monthly basis covered under the contract and shall guarantee that all equipment supplied pursuant to this contract and installed and maintained by the successful bidder, shall be available for use of ECL and shall meet the performance criteria in accordance with the terms & conditions of the guaranteed availability.

The SLA is applicable in Warranty period as well as CAMC period.

All the Areas and its locations of ECL are situated in very harsh & lightning prone environment. The Bidder is required to ensure adequate protection of the equipment (going to be installed at various locations) from lightning, surge, spike, over-voltage, short circuit, input reverse polarity, induced transient etc. It shall be the responsibility of Bidder to install all necessary protection equipment required in lightning prone areas, harsh & Industrial Load conditions for maintaining the SLA. If any equipment damaged/fails on above account, the bidder is required to replace/ repair the equipment / devices free of cost during entire contract period.

1. **CAMC charges of the system of the corresponding area**

   The total CAMC charges for the entire system will be distributed area-wise based on the proportional capital cost of each area within the procured system. The E&T department at ECL HQ will calculate and communicate these allocations.

2. **Availability of the system**
   i. **CCTV Surveillance System**

   For SP Mines Area, Sonepur Bazari Area, Rajmahal Area and ECL, HQ, if the footage of CCTV camera is not recorded/viewed in the devices present in the Control Room (Area/ HQ, as applicable) due to breakdown of any hardware, software and connectivity, then penalty would be imposed on Bidder and the same would be deducted from the respective quarterly CAMC bills on pro-rata basis.

   For all other locations, if the footage of CCTV camera is not recorded/viewed in the NVR due to breakdown of any item, then penalty would be imposed on Bidder and the same would be deducted from the respective quarterly CAMC bills on pro-rata basis.

   The breakdown shall be attended within six hours of receipt of information via telephone call /fax / message etc. and must be rectified within 12 hours of receipt of information.

$$\textbf{Monthly Availability (A1)} = \frac{[\textbf{n x No. of days in month x 24} - (\textbf{H1} + \textbf{H2} + \ldots + \textbf{Hn})]}{\textbf{n x No. of days in month X 24}} \textbf{ X 100}$$

Where,

n – number of CCTV cameras
H – Total hours of footage of CCTV cameras not recoded/viewed in recording device of Area Control Room/ NVR (as applicable) in the month.

Note:

1. If the existing cameras connected to the VMS installed by the bidder are unable to record or display footage due to its related existing hardware issue, they will not be considered as a breakdown for penalty calculation.
2. If any equipment, such as a UPS, display, or other device, remains non-functional for more than 96 hours, the CCTV cameras associated to that devices will be deemed to be in a full breakdown state, even if it is partially operational

ii. **<u>Entry-Exit Access control system at Areas</u>**

If the Access Control System installed at check-post at mine entry & exit or any other location as decided by ECL is under breakdown due to fault of any hardware or software, then the penalty would be imposed on bidder and the same would be deducted from the respective CAMC bills.

The breakdown shall be attended within six hours of receipt of information via telephone call /fax / message etc. and must be rectified within 12 hours of receipt of information.

Computation of monthly average availability and penalty will be done separately for each RFID boom barrier Access control system on monthly basis as follows:

$$\textbf{Monthly Availability (A2)} = \frac{[\textbf{n x No. of days in month x 24} - (\textbf{TBD1} + \textbf{TBD2} + \ldots + \textbf{TBDn})]}{\textbf{n x No. of days in month X 24}} \textbf{X 100}$$

Where,
n – no. of Entry Exit check posts installed with Access Control System in the Area
TBD – Total hours of Access control system at check posts are under break down in the month.

Note:

(i) If any equipment, such as a UPS, camera, display, or other device, remains non-functional for more than 96 hours, the Access Control System will be deemed to be in a full breakdown state, even if it is partially operational.

iii. **<u>Weighbridge Automation System (WAS) and Truck Centering System (TCS)</u>**

If the Weighbridge Automation System (WAS) installed at road weighbridges experiences a breakdown due to software issues, or if the Truck Centering System (TCS) is non-functional due to hardware or software faults, resulting in disruption of weighment operations, a penalty will be imposed on the bidder. This penalty will be calculated on a pro-rata basis and deducted from the respective CAMC bills.

The breakdown shall be attended within six hours of receipt of information via telephone call /fax / message etc. and must be rectified within 12 hours of receipt of information.

Computation of monthly average availability and penalty will be done separately for Weighbridge Automation System and Truck Centering System at Road weighbridges system on monthly basis as follows:

$$\textbf{Monthly Availability}(\mathbf{A3}) = \frac{[\mathbf{n \text{ x No. of days in month x } 24 - (TBD1 + TBD2 + \ldots + TBDn)}]}{\mathbf{n \text{ x No. of days in month X } 24} \mathbf{X\ 100}$$

n – no. of Road Weighbridges installed with Weighbridge Automation System (WAS) & Truck Centering System in the Area
TBD – Total Breakdown hours in the month.

Note: -

(i) If any equipment, such as a UPS, Audio System, Sensor or any other device, remains non-functional for more than 96 hours, the Weighbridge Automation System and Truck Centering system will be deemed to be in a full breakdown state, even if it is partially operational.

(ii) Any disruption in the system's functionality caused by existing hardware issues of ECL in the Weighbridge Automation System or Fiber connectivity provided by ECL will not be considered as breakdown hours.

iv. **<u>Command & Control Center at ECL HQ and Areas</u>**

If any equipment installed at command Control Center is under break down due to fault of any hardware or software, connectivity then penalty would be imposed on Bidder and the same would be deducted from the respective quarterly CAMC bills on pro-rata basis. Computation of monthly average availability and penalty will be done for each Area Command Control Center on monthly basis as follows:

Availability of the system at each Area and ECL HQ during the month shall be calculated as follows:

$$\textbf{Monthly Availability}(\textbf{A4}) = \frac{[\textbf{no. of days in month x 24} - \textbf{H}\,]}{\textbf{no. of days in month x 24}}\ \textbf{x 100}$$

Where,

H = Total hours of which any equipment at each area command control center is under break down in the month.

3. **<u>Overall Monthly Availability of the systems in all the Areas shall be calculated as follows:-</u>**

Overall Availability for the month of an Area (A) = $\frac{\mathbf{A1 + A2 + A3 + A4}}{\mathbf{4}}$

4. **<u>Overall Monthly availability of the systems at ECL HQ shall be calculated as follows:-</u>**

Overall Availability of the system in ECL HQ (B) = $\frac{\mathbf{A1 + A4}}{\mathbf{2}}$

5. **<u>Planned Downtime</u>**

Planned Downtime means any time when the equipment is unavailable because of maintenance, configuration/reconfiguration or other services with the prior approval of Engineer In charge or his authorized representative. Such services may include but are not limited to restarting applications, rebooting servers, applying patches or fixes, reconfiguring storage allocation, reloading data etc. Planned downtime will not be added to total downtime hours for calculation of availability. Dust cleaning of all the equipment including cameras wherever required is to be done once in a month during the contract period.

6. **<u>Penalty</u>**

The bidder shall give an uptime guarantee of 95% of monthly average availability for the overall system in the area/ HQ (as applicable)=. In case availability falls below the above level, penalty shall be imposed.

$$\textbf{Penalty} = \frac{(\textbf{95} - \%\ \textbf{Monthly Availability})\ \textbf{x}\ (\textbf{Monthly CAMC Charges})}{\textbf{100}}$$

7. **<u>Penalty for any month for each Area/ECL HQ to be imposed will be calculated on the basis of the following:</u>**

   i. 95% and above: No penalty.
   ii. 75 % to 95%: Penalty amount for this range (i.e., 75 % to 95 % range) will be as calculated above.
   iii. For availability falling below 75% the maximum penalty amount for that month will be imposed i.e., total CAMC value per month.
   iv. If the availability falls below 75% in 3 (three) or more months out of 6 consecutive months during the contract period, ECL shall reserve the right to terminate the contract, and withhold the remaining CAMC value.
   v. If the breakdown occurs due to the unavailability of network connectivity, the system will not be considered as a breakdown for penalty calculation.

8. **<u>Payment of Penalty charges during Warranty and CAMC Period</u>**

   i. During the payment of quarterly bill of CAMC, the penalty amount, if any, shall be deducted from the CAMC bill of corresponding billing period.
   ii. During the warranty period, the penalty calculation will be based on the charges of the first year of the CAMC.
   iii. During the warranty period, the penalty amount as per SLA, the successful bidder shall deposit the amount to ECL. Failing to do so by successful bidder, the performance bank guarantee deposit shall be revoked.

H. ELIGIBILITY CRITERIA & PROVENNESS CRITERIA

1. PRE-QUALIFICATION CRITERIA

a) EXPERIENCE CRITERIA

The experience of only those orders/ contracts would be considered for assessing experience criteria which have been commissioned for 05 years period prior to the date of opening of tender as per the following criteria:

1. Single order of at least 35% of estimated bid value., or
2. Two orders of at least 20% each of estimated bid value or
3. Three orders of at least 15% each of estimated bid value

The definition of Similar work shall be as follows:

Successful Supply, installation, commissioning and maintenance of CCTV surveillance system or RFID based boom barrier system or RFID based weighbridge automation system or RF based LAN/WAN networking or Smart City Projects with ICCC or combination of any of these

The bidder must submit order copies along with commissioning and work completion certificates issued by the end user. Here, end user (buyer organisation) refers to the entity that directly utilizes the supplied product, software, or service for its intended purpose, excluding intermediaries, resellers, or contractors.

In respect to the above experience criteria, the bidder is required to furnish the following information certified by the end user:

1. Description of qualifying experience.
2. Order Number / Contract Agreement Number of each experience.
3. Name & address of Employer/ Order Issuing authority of each experience.
4. Percentage (%) share of each experience (in case the experience has been earned by the bidder as a partner in a partnership firm then the proportionate value of experience in proportion to actual share of bidder in that partnership firm will be considered against eligibility else it shall be taken as 100%).
5. Start date & end date of each qualifying experience

Note: The experience of only those orders would be considered for assessing experience criteria which have been commissioned during 05 years period prior to the date of opening of tender even though the supply order, delivery and acceptance of material is beyond 05 years prior to the date of opening of tender with the following scenarios:

1. If the order is for supply, installation and commissioning then the total cost of the order will be considered for evaluation of experience criteria.
2. If the order is for supply, installation and commissioning with maintenance of one or more years and its maintenance services is completed prior to the date of opening of tender, then the total cost of the order including supply, installation, commissioning along with the maintenance service charges will be considered for evaluation of experience criteria.
3. If the order is for supply, installation and commissioning with maintenance of one or more years and its maintenance services are still ongoing and not completed prior to the date of opening of tender, then the total cost of the order including supply, installation, commissioning along with the portion of maintenance charges executed prior to the date of opening of tender will be considered for evaluation of experience criteria.
4. If the order is on rental basis, then the rental charges of the services executed and corresponding value for 05 years period prior to the opening of the tender will be considered for evaluation of experience criteria

b) <u>FINANCIAL TURNOVER CRITERIA</u>

The average annual financial turnover during the last 3 (three) years ending 31st March of the previous financial year should be at least 30% of the estimated cost of the tender. The "Previous Financial Year" shall be computed with respect to the publication date of the tender.

In this regards the bidder needs to submit the following documents for necessary compliance with Turnover certificate having UDIN issued by a Practicing Chartered Accountant having a membership number with Institute of Chartered Accountants

1. Annual turnover of each of the last 3 years ending 31st March of the previous, financial year.
2. Name of the Chartered Accountant issuing the Profit and Loss A/c or the Turnover certificate.
3. Membership Number of the CA

2. PROVENNESS CRITERIA

The bidder has to quote only the proven products with the provenness criteria as mentioned below:

The product offered by the tenderer shall be considered proven provided that 01 no. of quoted model or similar product, for the following tendered items, must have been supplied in India to mining industry and /or to the other Industries (Private or Government/ Public Sector Undertaking, Indigenous/ Global) and all of them performed satisfactorily for a minimum period of one year from the date of commissioning. The performance of only those items would be considered for assessing provenness which have been commissioned 1 year prior to the date of opening of tender but not 05 (five) years prior to the date of opening of tender.

The provenness criteria is to be submitted for the following items:

SN	Item Description
1	Integrated Command & Control Centre (ICCC) Software Platform
2	Video Analytic Application Software
3	Servers for ICCC and Video Analytics
4	Video Storage
5	Workstation
6	5MP IP Bullet Camera (Motorized vari-focal lens)
7	NVR (32 channel) loaded with 16TB Storage
8	16 Port POE managed switch
9	Long Range RFID Reader/Writer
10	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor
11	3 KVA (or more) online UPS

"Satisfactory Performance" means that the supplied items must have achieved the guaranteed annual availability, if any, as per the performance guarantee clause of the supply orders/ contracts for a minimum period of one year from the date of commissioning. In case supply orders/ contracts do not have provision for guaranteed annual availability, the bidder will submit satisfactory performance report issued by end users as per benchmark regarding performance of item, if any, incorporated in the supply orders/ contracts against which these items were supplied.

In support of the provenness criteria, the bidder is requested to submit supply orders/ contracts along with its satisfactorily performance report for a minimum period of one year with due certification from the concerned OEM that the supplied products as mentioned in the supply orders/ contracts are the OEM's own products. In case the satisfactory performance report of any item by the end user is not available then a self-certification from

the concerned OEM regarding one-year satisfactory performance shall be submitted by the bidder

In case where provenness of the offered product is being ascertained on the basis of supply made in other subsidiaries of CIL or other PSU's/Govt. departments/Private Organization, the bidder shall submit a self-certificate as per format mentioned in Annexure-SP to fulfil provenness criteria.

In case, any specific purchase order(s) has/have not been fully executed and any complaint/ claim is pending, then details of such cases to be categorically mentioned with the reasons thereof so that decision making is in clear perspective without any hidden facts in the subject matter.

The authenticity of the self-certificate as well as other documents submitted by the bidder will solely be their responsibility and appropriate action will be taken by CIL/ Subsidiary Company if it is subsequently found to be misleading/false/forged.

However, the Purchaser reserves the right to obtain the performance directly from the end user of the item/product.

Note:

1. In case, Authorized Agent /Dealer is submitting the offer on behalf of the manufacturer, supplies made by the manufacturer (directly by themselves or through their resellers) will be considered for which bidder has to submit supply order copy(s) and satisfactory performance report from the end user(s).
2. The bidder is advised to submit complete set of Supply order(s) along with other relevant documents like technical particulars, amendments, if any.
3. No new supply orders shall be accepted as shortfall documents. Therefore, bidder must be careful while uploading supply order copies online.
4. "Similar Item" shall be such item, which fulfils the following:
    a. Performs almost identical operations as the quoted model.
    b. Should be equal to or higher than the tendered capacity.
    c. Uses sub-components, sub-systems and major assemblies of substantially similar design & construction to the model quoted - only ratings/ specifications may differ (Lower or Higher).

I. RELAXING PRIOR EXPERIENCE AND TURNOVER CRITERIA FOR START-UPS AND MSEs IN PUBLIC PROCUREMENT:

Definition and Eligibility of Startup shall be in line with OM vide letter no. F-20/2/2014 PPD (pt.) Dtd. 25.07.2016 of Under Secretary to GOI, Ministry of Finance, Department of Expenditure, Procurement Policy Division, New Delhi, with subsequent amendments, if any.

Definition of MSEs shall be as per Public Procurement 2012, with subsequent amendments, if any

Prior experience and prior turn over criteria are not applicable for Startup & MSEs in respect of either of the following situations and no further documents regarding provenness will be required to be submitted by this category of Bidders.

a. If Bidders have submitted documents to prove the Startup / MSE status for the tendered item without certificate towards quality, assurance and capability from some authority like MSME, NSIC, etc., the Tender Inviting Authority, if required, may assess the techno-commercial capability of the venders to manufacture and deliver goods as per the prescribed quality and technical specification before awarding contract to them. For this purpose, a 'PROFORMA for Equipment and Quality Control' (as per Annexure-A) has been enclosed in the tender documents and such MSEs/ STARTUPS should submit the details of plant & machinery, quality control arrangements, etc., in the above PROFORMA along with their bids for verification of their technical capability.
If required, a techno-commercial team of the organization may visit the manufacturing unit of the vendor without any undue delay for quick finalization of the tenders.

b. If favourable technical capability reports obtained earlier on such firms for supply of the item in question as per the required specification is available, these may be considered, provided date of such reports is not more than one year from the date of opening of bids.

In case there is deficiency in technical capacity of the firm, the same shall be communicated to them by the Tender Inviting Authority for improvement in the quality of their product for future tenders, clearly indicating that their offer cannot be considered for relaxation against the tender in question and to avoid any future complications. The issues relating to technical capability may be decided by the Head of the Technical Department.

c. If Bidders have submitted documents to prove the startup / MSE status for the tendered item and whose products are ISI marked / DGMS approved / covered under DGS&D Rate Contracts on them / current holding Rate Contracts with CIL or its Subsidiary for supply of the tendered item / supplied and proven in CIL or its Subsidiary Companies / Proven product of the Ancillary unit of a Subsidiary Company of CIL, they will be required to submit the applicable related documents, digitally signed copy (for documents submitted on e-procurement portal) /self-certified copy (for documents submitted in modes other than e-procurement portal), for relaxation from the criteria of prior experience and prior turnover.

- Valid BIS Marking License for the quoted items on them.

OR

- Rate contract as issued by CIL / any other Subsidiary for the quoted items on them.

OR

- Valid DGMS Approval certificate for the quoted items on them.

OR

- Proven Ancillary certificate issued by Subsidiary Companies for the quoted items on them.

The Document (s) / Certificate (s) furnished by the Bidders for ISI markings or DGMS approval for any relaxation should be valid as on date of tender opening and a self-certified copy of such document(s) / certificate(s) valid as on date of supply, must accompany their bill(s).

If the documents are related to OEM or principal, the self-certification of OEM or principal shall also be required.

Note: - ECL may prefer the vendors to have prior experience rather than giving orders to new entities for procurement of items related to public safety, health, critical security operations and equipment, etc. in line with clarification given by Department of Expenditure, MoF vide OM No. F.20/2/2014/PPD-(Pt) dated 20.09.2016.

J. DOCUMENTS TO BE SUBMITTED BY THE BIDDER

1. The bidder is required to submit the documents pertaining to the experience criteria as per the bid.
2. The bidder is required to submit the documents pertaining to the financial turnover criteria as per the bid
3. The bidder is required to submit the compliance report of the technical specifications of all the items in the Bid.
4. The bidder is required to submit the compliance of payment terms along with the compliance of service level agreement terms & CAMC of the bid.
5. The bidder is required to submit the complete drawing details of the tower Erection & Foundation Diagram of Tower
6. The bidder is required to submit the documents pertaining to the provenness criteria as per bid and with the following:
   - **a)** The bidder is requested to submit supply orders/ contracts along with its satisfactorily performance report for a minimum period of one year with due certification from the concerned OEM that the supplied products as mentioned in the supply orders/ contracts are the OEM's own products.
   - **b)** In case the satisfactory performance report of any item by the end user is not available then a self-certification from the concerned OEM regarding one-year satisfactory performance shall be submitted by the bidder.
   - **c)** The provenness criteria is to be submitted for the following items:

SN	Item Description
1	Integrated Command & Control Centre (ICCC) Software Platform
2	Video Analytic Application Software
3	Servers for ICCC and Video Analytics
4	Video Storage
5	Workstation
6	5MP IP Bullet Camera (Motorized vari-focal lens)
7	NVR (32 channel) loaded with 16TB Storage
8	16 Port POE managed switch
9	Long Range RFID Reader/Writer
10	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor
11	3 KVA (or more) online UPS

7. The bidder is required to submit the Statement of Make and Model as per the format mentioned below, and technical literature/ brochure/ datasheet/ writeup/ Technical Compliance from the concerned OEMs of respective items, of all parameters as mentioned in

the Technical Specification Document of the following tendered items. In case of any non-clarity of technical parameters necessary confirmations from the OEM will be sought.

SN	Item Description	Make	Model	Details of Make & Model along-with technical document/OEM Datasheet of the following offered items should be submitted by the bidder	If higher specification product is offered, then whether this product will compatible & fulfil the requirement of ECL
				Actual or Higher	**Yes or No**
1	Integrated Command and Control Centre (ICCC) software platform				
2	Video Wall along with all hardware & software accessories				
3	AI based Video Analytics Application Software				
4	Servers for ICCC and Video Analytics				
5	Online UPS of 30 KVA with 2-hour backup at Full Load				
6	42 U Rack for Server				
7	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost				
8	5 MP IP Bullet Camera (Motorized vari-focal lens)				
9	5 MP IP Dome Camera				
10	2 MP IP PTZ Camera				
11	NVR (4 channel) loaded with 1 TB Storage				
12	NVR (8 channel) loaded with 4 TB Storage				
13	NVR (16 channel) loaded with 8 TB Storage				
14	NVR (32 channel) loaded with 16TB Storage				

15	Monitor 32 inches				
16	55" LED Display				
17	Server				
18	Video Storage				
19	Workstation PC				
20	Printer				
21	Network management Systems				
22	Video Management Software solution				
23	24 Port Layer 2, Managed switch				
24	8 port unmanaged gigabyte switch				
25	Industrial 8 Port POE Switch				
26	8 port managed Switch				
27	16 port POE managed Switch				
28	Media Converter				
29	CAT 6 Armoured STP cable				
30	CAT 6 UTP cable				
31	Armoured Optical Fibre cable				
32	Floor mounted Rack 24U enclosure				
33	9U rack				
34	CVT/ Voltage Stabilizer				
35	1 KVA UPS with 60 min backup				
36	1 KVA online UPS with 120 min backup with isolation transformer				
37	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer				
38	CPE radio				
39	Point to Multipoint Radios				
40	PTP radio				
41	Long Range RFID Reader/Writer with pole and mounting				

42	Signal Light for Traffic control				
43	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor				
44	LED Flood Light along with all accessories				
45	Handheld RFID reader along with all hardware & software accessories				
46	Truck Centering System with controller and all accessories				
47	Audio System along with all hardware & software accessories				

Note: -

a) Failure to declare MAKE and MODEL of materials quoted and/or non-submission of technical document/OEM Datasheet may lead to rejection of the bid. Bidder should not offer Multiple make and Multiple models for the same item as the same may lead to rejection of the bid.

b) In order to participate in the bidding process, it is imperative for the bidder to provide a precise identification of the MAKE and MODEL for the items listed in above table. For instance, when bidding for a server, the bidder must specify a brand such as HP or Lenovo or DELL and so on. Similarly, when bidding for UPS, the brands should be clearly mentioned, like Vertiv or Luminous or APC etc.

c) The use of vague terms like "Reputed" or "Branded," or any other non-specific descriptions, will not be accepted as a valid MAKE specification.

d) Furthermore, it is essential that bidders do not propose multiple brands and models for the same item, as doing so may result/lead in the rejection of their bid.

8. The bidder is required to submit the Authorization in original from the manufacturer mentioning the specific tender no. for participating for the following items

SN	Item Description
1	Integrated Command & Control Centre (ICCC) Software Platform
2	Video Analytic Application Software
3	Servers for ICCC and Video Analytics
4	Video Storage
5	Workstation
6	5 MP IP Bullet Camera (Motorized vari-focal lens)
7	NVR (32 channel) loaded with 16TB Storage

8	16 Port POE managed switch
9	24 port Layer 2 Managed switch/ Industrial 8 Port POE switch/ 8 Port managed switch
10	CPE radio/ Point to Multipoint Radios/ PTP radio
11	Long Range RFID Reader/Writer with pole and mounting
12	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor
13	3 KVA (or more) online UPS
14	Online UPS of 30 KVA with 2 hours backup at Full Load

9. The bidder is required to submit the documentary evidence in respect of

a) Regulatory Approvals/Certifications for BIS for the items mentioned below from any BIS/NABL recognized/ accredited laboratory:

SN	Item Description
1	Videowall along-with all hardware & software accessories
2	Servers for ICCC and Video Analytics
3	Workstation PC
4	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost
5	5 MP IP Bullet Camera (Motorized vari-focal lens)
6	5 MP IP Dome Camera
7	2 MP IP, PTZ Camera
8	NVR (4 channel) loaded with 1 TB Storage
9	NVR (8 channel) loaded with 4 TB Storage
10	NVR (16 channel) loaded with 8 TB Storage
11	NVR (32 channel) loaded with 16 TB Storage
12	Video Storage
13	55" LED Display
14	Monitor 32 inches
15	Server
16	Printers
17	LED Flood Light along with all accessories
18	1 KVA UPS with 60 mins backup
19	1 KVA online UPS with 120 min backup with isolation transformer (inbuilt / external)
20	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer (inbuilt / external)
21	Long Range RFID Reader/Writer with pole and mounting
22	Handheld RFID reader along with all hardware & software accessories

b) Regulatory Approvals/Certifications for BIS/ CE for Boom Barrier.

c) MTCTE/TEC ER certificate for

i. Switches: (i) 24 Port Layer 2 Managed Switch, (ii) 8 port unmanaged gigabyte switch, (iii) Industrial 8 Port POE Switch, (iv) 16 port POE managed Switch, and (v) 08 port

managed Switch. The certificate should be submitted for the offered product or any product from the offered product family/ product series against requirement of switches

ii. Armoured Optical Fibre cable

iii. Radios: (i) CPE Radio, (ii) PTP radio, and (iii) Point to Multipoint Radios

d) Confirmation to the latest version of ETSI / FCC / IEEE & ETA / TAC (Type Approval Certificate) approval of WPC for Radios: (i) CPE Radio, (ii) PTP radio, and (iii) Point to Multipoint Radios

e) The offered radios: (i) CPE Radio, (ii) PTP radio, and (iii) Point to Multipoint Radios, should be from trusted telecom approved OEM / certified from CERT-in or their empaneled organizations and the same may be submitted at the time of supply. The necessary documentary evidence in this regard against the proof of status of application may be submitted from the OEM along with the bid

f) Regulatory approval from WPC, if applicable, for the operating frequency for (i) Long Range RFID Reader/Writer and (ii) Handheld RFID reader

g) Certificate from NABL accredited laboratory for IP 68 or better for Long Range RFID Reader/Writer

h) Equipment Type Approval from WPC for (i) Long Range RFID Reader/Writer and (ii) Handheld RFID reader

10. Declaration / Undertaking to be submitted by the bidder: The bidder is required to submit following self-declaration/ undertaking

a) For CCTV Cameras and NVR:

i. While selecting OEMs of equipment, Hardware and Software (for this tender), it is ensured that the Hardware, Software and Firmware (for this tender) including "Intellectual Property Rights" of Hardware and "Source Code" of Firmware & Software selected are actual manufacturers and are not getting 3rd party manufacturing like branding and reselling in India through importing/trading and rebranding such equipment from Country that shares a Land Border with India.

However, any bidder/OEM from a country which shares a land border with India will be eligible to bid only if the bidder is registered with the competent authority.

ii. While selecting CCTV OEMs of equipment, Hardware and Software (proposed to be supplied for this tender) it is has ensured that OEMs are not suspended by ONVIF, even though the products of the OEMs may still appear on the ONVIF website.

iii. The MAC addresses are registered in the name of the OEM. The MAC address should not be in the name of any 3rd party/ company or any entity sharing land border with India until unless specifically allowed by the Government of India.

iv. The CCTV cameras and NVRs conform to the ONVIF (Open Network Video Interface Forum) specification which guarantees interoperability between network video products regardless of the manufacturer.
v. The CCTV cameras and NVR complies with the Meantime Between Failure (MTBF) rating of 1,00,000 hours for all CCTV cameras and all NVRs.
vi. All the CCTV Cameras are of IP V 6.0 compliant.
vii. As per the Cyber-Security, the IP cameras offered must comply with essential requirement (ERs) for security prescribed by Meity as amended from time to time and should have security testing certificate for the offered IP cameras issued by STQC Laboratory or any other agency notified by MeitY from time to time, at the time of supply. The security testing certificate for the offered IP cameras issued by STQC Laboratory or any other agency notified by MeitY from time to time, shall be submitted at the time of supply.

b) For Integrated Command and Control Centre (ICCC) software platform:

i. The proposed Command & Control Application is published on at-least one of the leading global CSP marketplace i.e., AWS/Azure/GCP or any other MeitY empanelment Cloud Service Provider.

c) For LAN switches:

i. ECL reserves the right to get the switches tested and certified for VAPT from any 3rd party authorized lab approved from Govt. of India. It will be responsibility of bidder to address & close all the vulnerabilities in defined timelines without any additional cost to ECL

d) It is confirmed that bidder has fully understood the scope of work and all other requirements for Supply, installation, testing and commissioning of tendered Integrated Command and Control Centre (ICCC) in ECL as per the given details in the Tender/ bid document. "M/s…… is hereby agree/accept all the technical specifications / clauses mentioned in this tender document/GeM Bid with reference No. dtd. "We undertake that the documents enclosed herewith are genuine and no material/facts have been concealed or suppressed. We are not debarred / blacklisted by any Government organization in the field subject tender. We also understand that the contract is liable to be cancelled if found to be obtained through fraudulent means or by concealment of information/facts.

e) The bidder shall submit the undertaking to provide the Comprehensive Services during the warranty of one (01) year and CAMC period of five (5) years even after the end of support / end of life as declared by the OEM of the items supplied by the bidders.

f) End-of sale declaration for the whole contract period for all the quoted products and in case any product is discontinued during the contract period, declaration for replacement

of the product with product of the equivalent or higher specification without affecting the performance of the system without any extra cost to ECL.

g) All Software should be an open platform Application & should be complied with any camera OEM.

h) The successful bidder shall arrange all the subscription/Licenses required for all software (which are covered or not covered in BOQ) for smooth working of the offered ICCC solution. Bidder is liable to be rejected if there is/ are any deviation(s) from the bid requirement.

i) The bidder will maintain stock of spare parts of the items / system for the entire contract period from the date of commissioning of work. The Bidder will undertake that supplies of necessary maintenance equipment and spare parts will be made available for all the equipment and the complete System for the entire contract period on continuing basis.

j) "The bidder is responsible for providing proper maintenance of the system, repair/replacement of each and every item damaged/ malfunctioned due to technical glitch, power supply, lightning/thundering during the entire contract period from the date of successful commissioning without any extra cost to ECL".

k) Bidder shall provide all patches and updates of firmware, during the entire contract period as and when released by OEMs without any additional cost to ECL.

l) The offered equipment like Servers, Software, Workstations, IP Radios, IP Devices & other networking components must be fully IPv6 compliant. The equipment should be dual-stack compatible, capable of supporting both IPv4 and IPv6 protocols concurrently.

m) All the CCTV Cameras, IP Radios, Central Server, Workstation PC, Switches, IP Devices & other networking components are of IP V 6.0 compliant.

n) **Malicious Code Certificate:** The bidder should upload following certificate in the bid that we M/s. [Name of the Bidder/Firm]: -

i. This is to certify that the Hardware and the Software being offered, as part of the contract, does not contain Embedded Malicious code that would activate procedures to:

    1. Inhibit the desires and designed function of the equipment.
    2. Cause physical damage to the user or equipment during the exploitation.
    3. Tap information resident or transient in the equipment/network.
    4.

ii. The firm will be considered to be in breach of the procurement contract, in case physical damage, loss of information or infringements related to copyright and Intellectual Property Right (IPRs) are caused due to activation of any such malicious code in embedded software.

11. To provide declaration from the concerned OEMs that

a) From OEM of CCTV Cameras and NVR:

i. While selecting OEMs of equipment, Hardware and Software (for this tender), it is ensured that the Hardware, Software and Firmware (for this tender) including "Intellectual Property Rights" of Hardware and "Source Code" of Firmware & Software selected are actual manufacturers and are not getting 3rd party manufacturing like branding and reselling in India through importing/trading and rebranding such equipment from Country that shares a Land Border with India.

ii. However, any bidder/OEM from a country which shares a land border with India will be eligible to bid only if the bidder is registered with the competent authority.

iii. While selecting CCTV OEMs of equipment, Hardware and Software (proposed to be supplied for this tender) it is has ensured that OEMs are not suspended by ONVIF, even though the products of the OEMs may still appear on the ONVIF website.

iv. The MAC addresses are registered in the name of the OEM. The MAC address should not be in the name of any 3rd party/ company or any entity sharing land border with India until unless specifically allowed by the Government of India.

v. The CCTV cameras and NVRs conform to the ONVIF (Open Network Video Interface Forum) specification which guarantees interoperability between network video products regardless of the manufacturer.

vi. To provide undertaking from the OEM of CCTV cameras and NVR regarding Meantime Between Failure (MTBF) rating of 1,00,000 hours for all CCTV cameras and all NVRs.

vii. All the CCTV Cameras are of IP V 6.0 compliant.

viii. As per the Cyber-Security, the IP cameras offered must comply with essential requirement (ERs) for security prescribed by Meity as amended from time to time and should have security testing certificate for the offered IP cameras issued by STQC Laboratory or any other agency notified by MeitY from time to time, at the time of supply. The security testing certificate for the offered IP cameras issued by STQC Laboratory or any other agency notified by MeitY from time to time, shall be submitted at the time of supply.

b) From OEM of Integrated Command and Control Centre (ICCC) software platform:

i. The Integrated Command and Control Centre (ICCC) software Application is published on at-least one of the leading global CSP marketplace i.e., AWS/Azure/GCP or any other MeitY empanelment Cloud Service Provider.

c) From OEM of LAN Switches

i. Declaration/ documentary evidence that switches support CE/BIS, EMC: TEC/SD/DD/EMC-221/05/OCT-2016, EN / IEC 62368-1:2018 or latest, EN61000-

4-2, EN61000-4-3, EN61000-4-4, EN61000-4-5, EN61000-4-6, EN61000-4-8, RoHS Directive

d) From OEM of Long-Range RFID Reader/Writer

i. The MAC address of RFID Reader (outdoor) is registered in the name of OEM supplying RFID Reader (outdoor).

12. To provide documents from the concerned OEMs that

a) OEM of CCTV Cameras and NVR

i. The OEM must submit the test report regarding Meantime Between Failure (MTBF) rating of 1,00,000 hours from Govt. / NABL Accredited test labs / 3rd party test house of international repute such as UL, TUV etc. in respect of the following
   a. 5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost
   b. 5 MP IP Bullet Camera (Motorized vari-focal lens)
   c. NVR (8 channel) loaded with 4 TB Storage
   d. NVR (32 channel) loaded with 16 TB Storage

ii. ISO9001:2008 and ISO14001:2004 OR latest certificates

b) OEM of Integrated Command and Control Centre (ICCC) software platform

i. Web Security Audit certificate from a Cert-In authorized Security Auditor for Integrated Command and Control Centre (ICCC) software platform.

c) OEM of LAN switches

i. ISO 9001, ISO 14001/18001 Certification OR latest certificates

d) From OEM of Long-Range RFID Reader/Writer

i. RF Test Report of the Reader and the same must confirm the name of OEM

Annexure-SP

FORMAT FOR SELF- CERTIFICATION OF PERFORMANCE

Tender No: - ----------

"We certify that *the items covered in the purchase order(s)/ Rate Contract (s) copies enclosed with our offer have been fully executed and have performed satisfactorily as per the provisions of respective purchase order(s)/ Rate Contract (s) and all the complaints/claim(s) lodged by the purchaser, if any, have been attended to and no complaints/ claim(s) are pending*."

SN	Item description	Purchase order/ Rate Contract No	Purchase order/ Rate Contract Date (in DD/MM/YYYY format)	Date of Supply (in DD/MM/YYYY format)	Proof of payment against supply orders.

Name & Signature of bidder with date
Designation
Seal of the firm

ANNEXURE-A

PROFORMA FOR EQUIPMENT AND QUALITY CONTROL

(To be submitted by the Startups / MSE firms who claim the relaxation of the prior turnover and prior experience criteria, as per bid terms)

Reference: Eastern Coalfields Ltd Hqrs_____Tender No._____ Date_____
for supply of_____

1. Name and Address of the Firm
2. (a) Telephone No. office/factory/works (b) Fax No. / E-mail ID
3. Location of manufacturing works/factories owned by the firm (documentary evidence of ownership must be produced).
4. Brief description of the factory (i.e. area covered accommodation, Department into which it is divided, laboratory etc.)
5. Details of plant and machinery erected and functioning in each department (monographs and description pamphlets) be supplied if available.
6. Whether the process of manufacture in the factory is carried out with the aid of power or without it.
7. Process Flow Chart for the whole manufacturing process of the tendered item
8. Details and stocks of raw materials held.
9. Production capacity of items quoted for with the existing plants and machinery (a) Normal (b) Maximum
10. Details of Quality Assurance Plan and Quality Control Infrastructure such as laboratories etc.
11. (a) Details of technical supervisory staff in-charge of production and quality control
    (b) Skilled labour employed
    (c) Unskilled labour employed
    (d) Maximum number of workers (skilled and unskilled) employed on any day during 18 months preceding the date of application.
12. Whether stores were tested to any standard specification, if so, copies of original test certificate should be submitted in duplicate.

(Signature of Tenderer)

NB: Details against sl nos. 5 to 12 inclusive need be restricted to the extent they pertain to the items under reference.

ANNEXURE-B

Area Wise Bill of Quantity related to all parts

1. Bill of Quantity for Bankola Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	30	0	0	0	30
	(ii) Crowd detection	Nos.	25	0	0	0	25
	(iii) Automatic Number Plate Recognition	Nos.	0	0	0	0	0
	(iv) Object counting or people/ vehicle counting	Nos.	0	0	0	0	0
	(v) Classification of objects	Nos.	0	0	0	0	0
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	0	0	0	0	0
	(vii) Perimeter Violation	Nos.	20	0	0	0	20
	(viii) Loitering Detection	Nos.	10	0	0	0	10
	(ix) Fire & Smoke Detection	Nos.	18	0	0	0	18
	(x) Traffic Congestion detection	Nos.	0	0	0	0	0
	(xi) PPE KIT detection	Nos.	10	0	0	0	10
	(xii) Safety hazards detection	Nos.	0	0	0	0	0
	(xiii) Insufficient illumination	Nos.	0	0	0	0	0
	(xiv) Accidents Alerts	Nos.	0	0	0	0	0
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	88	0	88
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	29	24	0	53
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0

15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	18	0	0	18
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	24	0	24
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0
19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	3	24	0	27
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	24	0	24
25	Printer	Nos.	0	0	24	0	24
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	24	15	39
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	15	0	0	15
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	30	8	0	38
34	16 port POE managed Switch	Nos.	1	0	24	0	25
35	Media Converter	Pairs	0	0	16	0	16
36	CAT 6 Armoured STP cable	Mtrs	0	0	4550	750	5300
37	CAT 6 UTP cable	Mtrs	0	870	1200	0	2070
38	Armoured Optical Fibre cable	Mtrs	0	0	9400	0	9400
39	Armoured Power cable	Mtrs	0	0	1200	300	1500
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	24	15	39
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	24	15	39
44	MCB	Nos.	0	0	24	15	39
45	Power Surge Protector	Nos.	0	0	24	15	39
46	Lightning arrestor with all accessories	Nos.	0	0	24	15	39
47	Earthing	Nos.	0	0	72	15	87
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	24	15	39
51	CPE radio	Nos.	0	0	6	0	6
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	27	0	27

55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	4	0	4
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	44	0	44
57	Signal Light for Traffic control	Nos.	0	0	44	30	74
58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	27	0	27
59	LED Flood Light along with all accessories	Nos.	0	0	27	0	27
60	Portable Cabins	Nos.	0	0	24	0	24
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	4	0	4
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	850	0	850
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	1700	0	1700
64	Reflective acrylic sticker	Nos.	0	0	3400	0	3400
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	15	15
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	15	15

2. Bill of Quantity for Kajora Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	13	0	0	0	13
	(ii) Crowd detection	Nos.	8	0	0	0	8
	(iii) Automatic Number Plate Recognition	Nos.	0	0	0	0	0
	(iv) Object counting or people/ vehicle counting	Nos.	0	0	0	0	0
	(v) Classification of objects	Nos.	0	0	0	0	0
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	0	0	0	0	0
	(vii) Perimeter Violation	Nos.	10	0	0	0	10
	(viii) Loitering Detection	Nos.	0	0	0	0	0
	(ix) Fire & Smoke Detection	Nos.	7	0	0	0	7
	(x) Traffic Congestion detection	Nos.	0	0	0	0	0
	(xi) PPE KIT detection	Nos.	5	0	0	0	5
	(xii) Safety hazards detection	Nos.	0	0	0	0	0
	(xiii) Insufficient illumination	Nos.	0	0	0	0	0
	(xiv) Accidents Alerts	Nos.	0	0	0	0	0
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	44	0	44
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	21	11	0	32
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	14	0	0	14
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	11	0	11
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	3	11	0	14
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	11	0	11
25	Printer	Nos.	0	0	11	0	11
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	11	10	21
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	10	0	0	10
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	23	8	0	31
34	16 port POE managed Switch	Nos.	1	0	11	0	12
35	Media Converter	Pairs	0	0	15	0	15
36	CAT 6 Armoured STP cable	Mtrs	0	0	2300	500	2800
37	CAT 6 UTP cable	Mtrs	0	630	550	0	1180
38	Armoured Optical Fibre cable	Mtrs	0	0	11000	0	11000
39	Armoured Power cable	Mtrs	0	0	550	200	750
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	11	10	21
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	11	10	21
44	MCB	Nos.	0	0	11	10	21
45	Power Surge Protector	Nos.	0	0	11	10	21
46	Lightning arrestor with all accessories	Nos.	0	0	11	10	21
47	Earthing	Nos.	0	0	33	10	43
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	11	10	21
51	CPE radio	Nos.	0	0	2	0	2
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	13	0	13
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	0	0	0
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	22	0	22
57	Signal Light for Traffic control	Nos.	0	0	22	20	42

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	13	0	13
59	LED Flood Light along with all accessories	Nos.	0	0	13	0	13
60	Portable Cabins	Nos.	0	0	11	0	11
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	4	0	4
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	500	0	500
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	1000	0	1000
64	Reflective acrylic sticker	Nos.	0	0	2000	0	2000
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	10	10
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	10	10

3. Bill of Quantity for Kenda Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	19	0	0	0	19
	(ii) Crowd detection	Nos.	8	0	0	0	8
	(iii) Automatic Number Plate Recognition	Nos.	0	0	0	0	0
	(iv) Object counting or people/ vehicle counting	Nos.	0	0	0	0	0
	(v) Classification of objects	Nos.	0	0	0	0	0
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	0	0	0	0	0
	(vii) Perimeter Violation	Nos.	19	0	0	0	19
	(viii) Loitering Detection	Nos.	6	0	0	0	6
	(ix) Fire & Smoke Detection	Nos.	11	0	0	0	11
	(x) Traffic Congestion detection	Nos.	0	0	0	0	0
	(xi) PPE KIT detection	Nos.	4	0	0	0	4
	(xii) Safety hazards detection	Nos.	0	0	0	0	0
	(xiii) Insufficient illumination	Nos.	0	0	0	0	0
	(xiv) Accidents Alerts	Nos.	0	0	0	0	0
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	88	0	88
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	21	23	0	44
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	14	0	0	14
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	23	0	23
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	3	23	0	26
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	23	0	23
25	Printer	Nos.	0	0	23	0	23
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	23	11	34
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	10	0	0	10
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	22	8	0	30
34	16 port POE managed Switch	Nos.	1	0	23	0	24
35	Media Converter	Pairs	0	0	18	0	18
36	CAT 6 Armoured STP cable	Mtrs	0	0	4450	550	5000
37	CAT 6 UTP cable	Mtrs	0	630	1150	0	1780
38	Armoured Optical Fibre cable	Mtrs	0	0	13200	0	13200
39	Armoured Power cable	Mtrs	0	0	1150	220	1370
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	23	11	34
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	23	11	34
44	MCB	Nos.	0	0	23	11	34
45	Power Surge Protector	Nos.	0	0	23	11	34
46	Lightning arrestor with all accessories	Nos.	0	0	23	11	34
47	Earthing	Nos.	0	0	69	11	80
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	23	11	34
51	CPE radio	Nos.	0	0	4	0	4
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	24	0	24
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	2	0	2
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	44	0	44
57	Signal Light for Traffic control	Nos.	0	0	44	22	66

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	24	0	24
59	LED Flood Light along with all accessories	Nos.	0	0	24	0	24
60	Portable Cabins	Nos.	0	0	23	0	23
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	4	0	4
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	600	0	600
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	1200	0	1200
64	Reflective acrylic sticker	Nos.	0	0	2400	0	2400
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	11	11
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	11	11

4. Bill of Quantity for Kunustoria Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	18	0	0	0	18
	(ii) Crowd detection	Nos.	30	0	0	0	30
	(iii) Automatic Number Plate Recognition	Nos.	0	0	0	0	0
	(iv) Object counting or people/ vehicle counting	Nos.	0	0	0	0	0
	(v) Classification of objects	Nos.	0	0	0	0	0
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	0	0	0	0	0
	(vii) Perimeter Violation	Nos.	6	0	0	0	6
	(viii) Loitering Detection	Nos.	7	0	0	0	7
	(ix) Fire & Smoke Detection	Nos.	6	0	0	0	6
	(x) Traffic Congestion detection	Nos.	0	0	0	0	0
	(xi) PPE KIT detection	Nos.	8	0	0	0	8
	(xii) Safety hazards detection	Nos.	0	0	0	0	0
	(xiii) Insufficient illumination	Nos.	0	0	0	0	0
	(xiv) Accidents Alerts	Nos.	0	0	0	0	0
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	72	0	72
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	26	18	0	44
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	15	0	0	15
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	18	0	18
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	3	18	0	21
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	18	0	18
25	Printer	Nos.	0	0	18	0	18
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	18	11	29
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	10	0	0	10
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	28	5	0	33
34	16 port POE managed Switch	Nos.	1	0	18	0	19
35	Media Converter	Pairs	0	0	17	0	17
36	CAT 6 Armoured STP cable	Mtrs	0	0	3900	550	4450
37	CAT 6 UTP cable	Mtrs	0	780	900	0	1680
38	Armoured Optical Fibre cable	Mtrs	0	0	27800	0	27800
39	Armoured Power cable	Mtrs	0	0	900	220	1120
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	18	11	29
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	18	11	29
44	MCB	Nos.	0	0	18	11	29
45	Power Surge Protector	Nos.	0	0	18	11	29
46	Lightning arrestor with all accessories	Nos.	0	0	18	11	29
47	Earthing	Nos.	0	0	54	11	65
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	18	11	29
51	CPE radio	Nos.	0	0	4	0	4
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	24	0	24
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	0	0	0
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	36	0	36
57	Signal Light for Traffic control	Nos.	0	0	36	22	58

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	24	0	24
59	LED Flood Light along with all accessories	Nos.	0	0	24	0	24
60	Portable Cabins	Nos.	0	0	18	0	18
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	4	0	4
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	500	0	500
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	1000	0	1000
64	Reflective acrylic sticker	Nos.	0	0	2000	0	2000
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	11	11
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	11	11

5. Bill of Quantity for Sonepur Bazari Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	9	0	0	0	9
	(ii) Crowd detection	Nos.	31	0	0	0	31
	(iii) Automatic Number Plate Recognition	Nos.	1	0	0	0	1
	(iv) Object counting or people/ vehicle counting	Nos.	1	0	0	0	1
	(v) Classification of objects	Nos.	1	0	0	0	1
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	1	0	0	0	1
	(vii) Perimeter Violation	Nos.	0	0	0	0	0
	(viii) Loitering Detection	Nos.	0	0	0	0	0
	(ix) Fire & Smoke Detection	Nos.	0	0	0	0	0
	(x) Traffic Congestion detection	Nos.	1	0	0	0	1
	(xi) PPE KIT detection	Nos.	0	0	0	0	0
	(xii) Safety hazards detection	Nos.	1	0	0	0	1
	(xiii) Insufficient illumination	Nos.	1	0	0	0	1
	(xiv) Accidents Alerts	Nos.	1	0	0	0	1
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	24	0	24
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	134	0	0	134
13	5 MP IP Dome Camera	Nos.	0	17	7	0	24
14	2 MP IP PTZ Camera	Nos.	0	20	0	0	20
15	Enterprise CVMS (per Camera's license)	Nos.	0	171	0	0	171
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	12	0	0	12
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	9	7	0	16
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	5	0	0	5

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	4	0	0	4
20	Monitor 32 inches	Nos.	0	21	7	0	28
21	55" LED Display	Nos.	0	8	0	0	8
22	Server	Nos.	0	2	0	0	2
23	Video Storage	Nos.	0	2	0	0	2
24	Workstation PC	Nos.	0	2	7	0	9
25	Printer	Nos.	0	0	7	0	7
26	Network management Systems	Nos.	0	1	0	0	1
27	Video Management Software solution	Nos.	0	1	0	0	1
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	7	11	18
30	24 Port Layer 2, Managed switch	Nos.	0	1	0	0	1
31	8 port unmanaged gigabyte switch	Nos.	0	30	0	0	30
32	Industrial 8 Port POE Switch	Nos.	0	11	0	0	11
33	8 port managed Switch	Nos.	0	46	6	0	52
34	16 port POE managed Switch	Nos.	1	0	7	0	8
35	Media Converter	Pairs	0	25	6	0	31
36	CAT 6 Armoured STP cable	Mtrs	0	8250	1250	550	10050
37	CAT 6 UTP cable	Mtrs	0	5130	350	0	5480
38	Armoured Optical Fibre cable	Mtrs	0	28400	12500	0	40900
39	Armoured Power cable	Mtrs	0	4950	350	220	5520
40	Floor mounted Rack 24U enclosure	Nos.	1	1	0	0	2
41	9U rack	Nos.	0	39	7	11	57
42	Outdoor Vandal Proof box	Nos.	0	14	0	0	14
43	CVT/ Voltage Stabilizer	Nos.	0	26	7	11	44
44	MCB	Nos.	0	53	7	11	71
45	Power Surge Protector	Nos.	0	53	7	11	71
46	Lightning arrestor with all accessories	Nos.	0	53	7	11	71
47	Earthing	Nos.	0	168	21	11	200
48	1 KVA UPS with 60 min backup	Nos.	0	31	0	0	31
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	22	0	0	23
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	4	7	11	22
51	CPE radio	Nos.	0	18	4	0	22
52	Point to Multipoint Radios	Nos.	0	4	0	0	4
53	PTP radio	Nos.	0	12	0	0	12
54	10 m Tubular Swaged pole	Nos.	0	90	8	0	98
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	13	0	0	13
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	12	0	12
57	Signal Light for Traffic control	Nos.	0	0	12	22	34

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	8	0	8
59	LED Flood Light along with all accessories	Nos.	0	0	8	0	8
60	Portable Cabins	Nos.	0	0	7	0	7
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	6	0	6
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	300	0	300
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	600	0	600
64	Reflective acrylic sticker	Nos.	0	0	1200	0	1200
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	11	11
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	11	11

6. Bill of Quantity for Rajmahal Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	26	0	0	0	26
	(ii) Crowd detection	Nos.	10	0	0	0	10
	(iii) Automatic Number Plate Recognition	Nos.	1	0	0	0	1
	(iv) Object counting or people/ vehicle counting	Nos.	1	0	0	0	1
	(v) Classification of objects	Nos.	1	0	0	0	1
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	1	0	0	0	1
	(vii) Perimeter Violation	Nos.	0	0	0	0	0
	(viii) Loitering Detection	Nos.	2	0	0	0	2
	(ix) Fire & Smoke Detection	Nos.	12	0	0	0	12
	(x) Traffic Congestion detection	Nos.	1	0	0	0	1
	(xi) PPE KIT detection	Nos.	6	0	0	0	6
	(xii) Safety hazards detection	Nos.	1	0	0	0	1
	(xiii) Insufficient illumination	Nos.	1	0	0	0	1
	(xiv) Accidents Alerts	Nos.	1	0	0	0	1
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	12	0	12
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	24	0	0	24
13	5 MP IP Dome Camera	Nos.	0	17	6	0	23
14	2 MP IP PTZ Camera	Nos.	0	14	0	0	14
15	Enterprise CVMS (per Camera's license)	Nos.	0	172	0	0	172
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	7	0	0	7
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	10	6	0	16
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	2	0	0	2

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	19	6	0	25
21	55" LED Display	Nos.	0	1	0	0	1
22	Server	Nos.	0	1	0	0	1
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	1	6	0	7
25	Printer	Nos.	0	0	6	0	6
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	1	0	0	1
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	6	7	13
30	24 Port Layer 2, Managed switch	Nos.	0	1	0	0	1
31	8 port unmanaged gigabyte switch	Nos.	0	41	0	0	41
32	Industrial 8 Port POE Switch	Nos.	0	12	0	0	12
33	8 port managed Switch	Nos.	0	45	3	0	48
34	16 port POE managed Switch	Nos.	1	0	6	0	7
35	Media Converter	Pairs	0	45	3	0	48
36	CAT 6 Armoured STP cable	Mtrs	0	4000	600	350	4950
37	CAT 6 UTP cable	Mtrs	0	4420	300	0	4720
38	Armoured Optical Fibre cable	Mtrs	0	28000	3000	0	31000
39	Armoured Power cable	Mtrs	0	5000	300	140	5440
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	30	6	7	43
42	Outdoor Vandal Proof box	Nos.	0	12	0	0	12
43	CVT/ Voltage Stabilizer	Nos.	0	13	6	7	26
44	MCB	Nos.	0	45	6	7	58
45	Power Surge Protector	Nos.	0	45	6	7	58
46	Lightning arrestor with all accessories	Nos.	0	45	6	7	58
47	Earthing	Nos.	0	80	18	7	105
48	1 KVA UPS with 60 min backup	Nos.	0	30	0	0	30
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	12	0	0	13
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	1	6	7	14
51	CPE radio	Nos.	0	15	4	0	19
52	Point to Multipoint Radios	Nos.	0	3	0	0	3
53	PTP radio	Nos.	0	4	0	0	4
54	10 m Tubular Swaged pole	Nos.	0	40	6	0	46
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	15	2	0	17
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	6	0	6
57	Signal Light for Traffic control	Nos.	0	0	6	14	20

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	6	0	6
59	LED Flood Light along with all accessories	Nos.	0	0	6	0	6
60	Portable Cabins	Nos.	0	0	6	0	6
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	2	0	2
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	100	0	100
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	200	0	200
64	Reflective acrylic sticker	Nos.	0	0	400	0	400
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	7	7
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	7	7

7. Bill of Quantity for Satgram-Sripur Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	30	0	0	0	30
	(ii) Crowd detection	Nos.	31	0	0	0	31
	(iii) Automatic Number Plate Recognition	Nos.	0	0	0	0	0
	(iv) Object counting or people/ vehicle counting	Nos.	0	0	0	0	0
	(v) Classification of objects	Nos.	0	0	0	0	0
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	0	0	0	0	0
	(vii) Perimeter Violation	Nos.	11	0	0	0	11
	(viii) Loitering Detection	Nos.	6	0	0	0	6
	(ix) Fire & Smoke Detection	Nos.	7	0	0	0	7
	(x) Traffic Congestion detection	Nos.	0	0	0	0	0
	(xi) PPE KIT detection	Nos.	5	0	0	0	5
	(xii) Safety hazards detection	Nos.	0	0	0	0	0
	(xiii) Insufficient illumination	Nos.	0	0	0	0	0
	(xiv) Accidents Alerts	Nos.	0	0	0	0	0
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	60	0	60
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	41	16	0	57
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	32	0	0	32
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	16	0	16
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	12	16	0	28
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	16	0	16
25	Printer	Nos.	0	0	16	0	16
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	16	20	36
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	20	0	0	20
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	43	10	0	53
34	16 port POE managed Switch	Nos.	1	0	16	0	17
35	Media Converter	Pairs	0	0	15	0	15
36	CAT 6 Armoured STP cable	Mtrs	0	0	3000	1000	4000
37	CAT 6 UTP cable	Mtrs	0	1230	800	0	2030
38	Armoured Optical Fibre cable	Mtrs	0	0	13150	0	13150
39	Armoured Power cable	Mtrs	0	0	800	400	1200
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	16	20	36
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	16	20	36
44	MCB	Nos.	0	0	16	20	36
45	Power Surge Protector	Nos.	0	0	16	20	36
46	Lightning arrestor with all accessories	Nos.	0	0	16	20	36
47	Earthing	Nos.	0	0	48	20	68
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	16	20	36
51	CPE radio	Nos.	0	0	8	0	8
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	16	0	16
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	2	0	2
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	30	0	30

57	Signal Light for Traffic control	Nos.	0	0	30	40	70
58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	16	0	16
59	LED Flood Light along with all accessories	Nos.	0	0	16	0	16
60	Portable Cabins	Nos.	0	0	16	0	16
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	0	0	0
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	750	0	750
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	1500	0	1500
64	Reflective acrylic sticker	Nos.	0	0	3000	0	3000
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	20	20
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	20	20

8. Bill of Quantity for Pandaveswar Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	10	0	0	0	10
	(ii) Crowd detection	Nos.	3	0	0	0	3
	(iii) Automatic Number Plate Recognition	Nos.	0	0	0	0	0
	(iv) Object counting or people/ vehicle counting	Nos.	0	0	0	0	0
	(v) Classification of objects	Nos.	0	0	0	0	0
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	0	0	0	0	0
	(vii) Perimeter Violation	Nos.	2	0	0	0	2
	(viii) Loitering Detection	Nos.	4	0	0	0	4
	(ix) Fire & Smoke Detection	Nos.	3	0	0	0	3
	(x) Traffic Congestion detection	Nos.	0	0	0	0	0
	(xi) PPE KIT detection	Nos.	0	0	0	0	0
	(xii) Safety hazards detection	Nos.	0	0	0	0	0
	(xiii) Insufficient illumination	Nos.	0	0	0	0	0
	(xiv) Accidents Alerts	Nos.	0	0	0	0	0
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	34	0	34
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	23	9	0	32
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	22	0	0	22
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	9	0	9
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0
19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0

20	Monitor 32 inches	Nos.	0	5	9	0	14
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	9	0	9
25	Printer	Nos.	0	0	9	0	9
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	9	10	19
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	15	0	0	15
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	24	5	0	29
34	16 port POE managed Switch	Nos.	1	0	9	0	10
35	Media Converter	Pairs	0	0	19	0	19
36	CAT 6 Armoured STP cable	Mtrs	0	0	2100	500	2600
37	CAT 6 UTP cable	Mtrs	0	690	450	0	1140
38	Armoured Optical Fibre cable	Mtrs	0	0	14500	0	14500
39	Armoured Power cable	Mtrs	0	0	450	200	650
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	9	10	19
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	9	10	19
44	MCB	Nos.	0	0	9	10	19
45	Power Surge Protector	Nos.	0	0	9	10	19
46	Lightning arrestor with all accessories	Nos.	0	0	9	10	19
47	Earthing	Nos.	0	0	27	10	37
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	9	10	19
51	CPE radio	Nos.	0	0	2	0	2
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	17	0	17
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	0	0	0
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	17	0	17
57	Signal Light for Traffic control	Nos.	0	0	17	20	37

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	17	0	17
59	LED Flood Light along with all accessories	Nos.	0	0	17	0	17
60	Portable Cabins	Nos.	0	0	9	0	9
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	4	0	4
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	300	0	300
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	600	0	600
64	Reflective acrylic sticker	Nos.	0	0	1200	0	1200
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	10	10
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	10	10

9. Bill of Quantity for SP Mines Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	15	0	0	0	15
	(ii) Crowd detection	Nos.	12	0	0	0	12
	(iii) Automatic Number Plate Recognition	Nos.	1	0	0	0	1
	(iv) Object counting or people/ vehicle counting	Nos.	1	0	0	0	1
	(v) Classification of objects	Nos.	1	0	0	0	1
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	1	0	0	0	1
	(vii) Perimeter Violation	Nos.	4	0	0	0	4
	(viii) Loitering Detection	Nos.	2	0	0	0	2
	(ix) Fire & Smoke Detection	Nos.	5	0	0	0	5
	(x) Traffic Congestion detection	Nos.	1	0	0	0	1
	(xi) PPE KIT detection	Nos.	1	0	0	0	1
	(xii) Safety hazards detection	Nos.	1	0	0	0	1
	(xiii) Insufficient illumination	Nos.	1	0	0	0	1
	(xiv) Accidents Alerts	Nos.	1	0	0	0	1
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	32	0	32
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	55	0	0	55
13	5 MP IP Dome Camera	Nos.	0	11	11	0	22
14	2 MP IP PTZ Camera	Nos.	0	26	0	0	26
15	Enterprise CVMS (per Camera's license)	Nos.	0	92	0	0	92
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	8	0	0	8
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	3	11	0	14
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	4	0	0	4
20	Monitor 32 inches	Nos.	0	15	11	0	26
21	55" LED Display	Nos.	0	2	0	0	2
22	Server	Nos.	0	2	0	0	2
23	Video Storage	Nos.	0	1	0	0	1
24	Workstation PC	Nos.	0	1	11	0	12
25	Printer	Nos.	0	0	11	0	11
26	Network management Systems	Nos.	0	1	0	0	1
27	Video Management Software solution	Nos.	0	1	0	0	1
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	11	8	19
30	24 Port Layer 2, Managed switch	Nos.	0	4	0	0	4
31	8 port unmanaged gigabyte switch	Nos.	0	20	0	0	20
32	Industrial 8 Port POE Switch	Nos.	0	45	0	0	45
33	8 port managed Switch	Nos.	0	22	0	0	22
34	16 port POE managed Switch	Nos.	1	0	11	0	12
35	Media Converter	Pairs	0	34	31	0	65
36	CAT 6 Armoured STP cable	Mtrs	0	5200	1850	400	7450
37	CAT 6 UTP cable	Mtrs	0	3410	550	0	3960
38	Armoured Optical Fibre cable	Mtrs	0	20300	13900	0	34200
39	Armoured Power cable	Mtrs	0	4250	550	160	4960
40	Floor mounted Rack 24U enclosure	Nos.	1	1	0	0	2
41	9U rack	Nos.	0	13	11	8	32
42	Outdoor Vandal Proof box	Nos.	0	43	0	0	43
43	CVT/ Voltage Stabilizer	Nos.	0	15	11	8	34
44	MCB	Nos.	0	58	11	8	77
45	Power Surge Protector	Nos.	0	58	11	8	77
46	Lightning arrestor with all accessories	Nos.	0	87	11	8	106
47	Earthing	Nos.	0	145	33	8	186
48	1 KVA UPS with 60 min backup	Nos.	0	43	0	0	43
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	13	0	0	14
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	2	11	8	21
51	CPE radio	Nos.	0	19	10	0	29
52	Point to Multipoint Radios	Nos.	0	6	0	0	6
53	PTP radio	Nos.	0	8	0	0	8
54	10 m Tubular Swaged pole	Nos.	0	62	16	0	78
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	23	8	0	31
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	16	0	16

57	Signal Light for Traffic control	Nos.	0	0	16	16	32
58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	16	0	16
59	LED Flood Light along with all accessories	Nos.	0	0	16	0	16
60	Portable Cabins	Nos.	0	0	11	0	11
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	8	0	8
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	400	0	400
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	800	0	800
64	Reflective acrylic sticker	Nos.	0	0	1600	0	1600
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	8	8
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	8	8

10. Bill of Quantity for Jhanjra Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	30	0	0	0	30
	(ii) Crowd detection	Nos.	10	0	0	0	10
	(iii) Automatic Number Plate Recognition	Nos.	1	0	0	0	1
	(iv) Object counting or people/ vehicle counting	Nos.	1	0	0	0	1
	(v) Classification of objects	Nos.	1	0	0	0	1
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	1	0	0	0	1
	(vii) Perimeter Violation	Nos.	7	0	0	0	7
	(viii) Loitering Detection	Nos.	3	0	0	0	3
	(ix) Fire & Smoke Detection	Nos.	9	0	0	0	9
	(x) Traffic Congestion detection	Nos.	1	0	0	0	1
	(xi) PPE KIT detection	Nos.	5	0	0	0	5
	(xii) Safety hazards detection	Nos.	1	0	0	0	1
	(xiii) Insufficient illumination	Nos.	1	0	0	0	1
	(xiv) Accidents Alerts	Nos.	1	0	0	0	1
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	12	0	12
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	12	5	0	17
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	5	0	0	5
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	5	0	5
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	5	5	0	10
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	5	0	5
25	Printer	Nos.	0	0	5	0	5
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	5	7	12
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	10	0	0	10
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	14	0	0	14
34	16 port POE managed Switch	Nos.	1	0	5	0	6
35	Media Converter	Pairs	0	0	13	0	13
36	CAT 6 Armoured STP cable	Mtrs	0	0	600	350	950
37	CAT 6 UTP cable	Mtrs	0	360	250	0	610
38	Armoured Optical Fibre cable	Mtrs	0	0	16000	0	16000
39	Armoured Power cable	Mtrs	0	0	250	140	390
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	5	7	12
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	5	7	12
44	MCB	Nos.	0	0	5	7	12
45	Power Surge Protector	Nos.	0	0	5	7	12
46	Lightning arrestor with all accessories	Nos.	0	0	5	7	12
47	Earthing	Nos.	0	0	15	7	22
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	5	7	12
51	CPE radio	Nos.	0	0	6	0	6
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	5	0	5
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	3	0	3
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	6	0	6
57	Signal Light for Traffic control	Nos.	0	0	6	14	20

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	5	0	5
59	LED Flood Light along with all accessories	Nos.	0	0	5	0	5
60	Portable Cabins	Nos.	0	0	5	0	5
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	3	0	3
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	150	0	150
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	300	0	300
64	Reflective acrylic sticker	Nos.	0	0	600	0	600
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	7	7
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	7	7

11. Bill of Quantity for Sodepur Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	30	0	0	0	30
	(ii) Crowd detection	Nos.	10	0	0	0	10
	(iii) Automatic Number Plate Recognition	Nos.	1	0	0	0	1
	(iv) Object counting or people/ vehicle counting	Nos.	1	0	0	0	1
	(v) Classification of objects	Nos.	1	0	0	0	1
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	1	0	0	0	1
	(vii) Perimeter Violation	Nos.	7	0	0	0	7
	(viii) Loitering Detection	Nos.	3	0	0	0	3
	(ix) Fire & Smoke Detection	Nos.	9	0	0	0	9
	(x) Traffic Congestion detection	Nos.	1	0	0	0	1
	(xi) PPE KIT detection	Nos.	5	0	0	0	5
	(xii) Safety hazards detection	Nos.	1	0	0	0	1
	(xiii) Insufficient illumination	Nos.	1	0	0	0	1
	(xiv) Accidents Alerts	Nos.	1	0	0	0	1
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	12	0	12
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	12	5	0	17
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	5	0	0	5
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	5	0	5
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	5	5	0	10
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	5	0	5
25	Printer	Nos.	0	0	5	0	5
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	5	7	12
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	10	0	0	10
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	14	0	0	14
34	16 port POE managed Switch	Nos.	1	0	5	0	6
35	Media Converter	Pairs	0	0	13	0	13
36	CAT 6 Armoured STP cable	Mtrs	0	0	600	350	950
37	CAT 6 UTP cable	Mtrs	0	360	250	0	610
38	Armoured Optical Fibre cable	Mtrs	0	0	16000	0	16000
39	Armoured Power cable	Mtrs	0	0	250	140	390
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	5	7	12
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	5	7	12
44	MCB	Nos.	0	0	5	7	12
45	Power Surge Protector	Nos.	0	0	5	7	12
46	Lightning arrestor with all accessories	Nos.	0	0	5	7	12
47	Earthing	Nos.	0	0	15	7	22
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	5	7	12
51	CPE radio	Nos.	0	0	6	0	6
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	5	0	5
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	3	0	3
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	6	0	6
57	Signal Light for Traffic control	Nos.	0	0	6	14	20

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	5	0	5
59	LED Flood Light along with all accessories	Nos.	0	0	5	0	5
60	Portable Cabins	Nos.	0	0	5	0	5
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	3	0	3
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	150	0	150
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	300	0	300
64	Reflective acrylic sticker	Nos.	0	0	600	0	600
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	7	7
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	7	7

12. Bill of Quantity for Mugma Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	38	0	0	0	38
	(ii) Crowd detection	Nos.	34	0	0	0	34
	(iii) Automatic Number Plate Recognition	Nos.	0	0	0	0	0
	(iv) Object counting or people/ vehicle counting	Nos.	0	0	0	0	0
	(v) Classification of objects	Nos.	0	0	0	0	0
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	0	0	0	0	0
	(vii) Perimeter Violation	Nos.	0	0	0	0	0
	(viii) Loitering Detection	Nos.	8	0	0	0	8
	(ix) Fire & Smoke Detection	Nos.	18	0	0	0	18
	(x) Traffic Congestion detection	Nos.	0	0	0	0	0
	(xi) PPE KIT detection	Nos.	7	0	0	0	7
	(xii) Safety hazards detection	Nos.	0	0	0	0	0
	(xiii) Insufficient illumination	Nos.	0	0	0	0	0
	(xiv) Accidents Alerts	Nos.	0	0	0	0	0
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	58	0	58
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	40	15	0	55
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	17	0	0	17
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	15	0	15
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	4	15	0	19
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	15	0	15
25	Printer	Nos.	0	0	15	0	15
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	15	16	31
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	15	0	0	15
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	32	11	0	43
34	16 port POE managed Switch	Nos.	1	0	15	0	16
35	Media Converter	Pairs	0	0	4	0	4
36	CAT 6 Armoured STP cable	Mtrs	0	0	3400	800	4200
37	CAT 6 UTP cable	Mtrs	0	1200	750	0	1950
38	Armoured Optical Fibre cable	Mtrs	0	0	10050	0	10050
39	Armoured Power cable	Mtrs	0	0	750	320	1070
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	15	16	31
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	15	16	31
44	MCB	Nos.	0	0	15	16	31
45	Power Surge Protector	Nos.	0	0	15	16	31
46	Lightning arrestor with all accessories	Nos.	0	0	15	16	31
47	Earthing	Nos.	0	0	45	16	61
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	15	16	31
51	CPE radio	Nos.	0	0	4	0	4
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	25	0	25
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	0	0	0
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	29	0	29
57	Signal Light for Traffic control	Nos.	0	0	29	32	61

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	25	0	25
59	LED Flood Light along with all accessories	Nos.	0	0	25	0	25
60	Portable Cabins	Nos.	0	0	15	0	15
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	5	0	5
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	650	0	650
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	1300	0	1300
64	Reflective acrylic sticker	Nos.	0	0	2600	0	2600
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	16	16
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	16	16

13. Bill of Quantity for Salanpur Area

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	34	0	0	0	34
	(ii) Crowd detection	Nos.	14	0	0	0	14
	(iii) Automatic Number Plate Recognition	Nos.	1	0	0	0	1
	(iv) Object counting or people/ vehicle counting	Nos.	1	0	0	0	1
	(v) Classification of objects	Nos.	1	0	0	0	1
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	1	0	0	0	1
	(vii) Perimeter Violation	Nos.	7	0	0	0	7
	(viii) Loitering Detection	Nos.	4	0	0	0	4
	(ix) Fire & Smoke Detection	Nos.	2	0	0	0	2
	(x) Traffic Congestion detection	Nos.	1	0	0	0	1
	(xi) PPE KIT detection	Nos.	0	0	0	0	0
	(xii) Safety hazards detection	Nos.	1	0	0	0	1
	(xiii) Insufficient illumination	Nos.	1	0	0	0	1
	(xiv) Accidents Alerts	Nos.	1	0	0	0	1
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	48	0	48
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	19	12	0	31
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	12	0	0	12
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	12	0	12
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	3	12	0	15
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	12	0	12
25	Printer	Nos.	0	0	12	0	12
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	12	19	31
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	15	0	0	15
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	20	3	0	23
34	16 port POE managed Switch	Nos.	1	0	12	0	13
35	Media Converter	Pairs	0	0	11	0	11
36	CAT 6 Armoured STP cable	Mtrs	0	0	2400	950	3350
37	CAT 6 UTP cable	Mtrs	0	570	600	0	1170
38	Armoured Optical Fibre cable	Mtrs	0	0	5000	0	5000
39	Armoured Power cable	Mtrs	0	0	600	380	980
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	12	19	31
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	12	19	31
44	MCB	Nos.	0	0	12	19	31
45	Power Surge Protector	Nos.	0	0	12	19	31
46	Lightning arrestor with all accessories	Nos.	0	0	12	19	31
47	Earthing	Nos.	0	0	36	19	55
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	12	19	31
51	CPE radio	Nos.	0	0	2	0	2
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	12	0	12
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	0	0	0
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	24	0	24
57	Signal Light for Traffic control	Nos.	0	0	24	38	62

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	12	0	12
59	LED Flood Light along with all accessories	Nos.	0	0	12	0	12
60	Portable Cabins	Nos.	0	0	12	0	12
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	3	0	3
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	300	0	300
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	600	0	600
64	Reflective acrylic sticker	Nos.	0	0	1200	0	1200
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	19	19
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	19	19

14. Bill of Quantity for ECL HQ

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	0	0	0	0	0
2	Video Wall along with all hardware & software accessories	No.	0	0	0	0	0
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	34	0	0	0	34
	(ii) Crowd detection	Nos.	14	0	0	0	14
	(iii) Automatic Number Plate Recognition	Nos.	1	0	0	0	1
	(iv) Object counting or people/ vehicle counting	Nos.	1	0	0	0	1
	(v) Classification of objects	Nos.	1	0	0	0	1
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	1	0	0	0	1
	(vii) Perimeter Violation	Nos.	7	0	0	0	7
	(viii) Loitering Detection	Nos.	4	0	0	0	4
	(ix) Fire & Smoke Detection	Nos.	2	0	0	0	2
	(x) Traffic Congestion detection	Nos.	1	0	0	0	1
	(xi) PPE KIT detection	Nos.	0	0	0	0	0
	(xii) Safety hazards detection	Nos.	1	0	0	0	1
	(xiii) Insufficient illumination	Nos.	1	0	0	0	1
	(xiv) Accidents Alerts	Nos.	1	0	0	0	1
4	Servers for ICCC and Video Analytics	Nos.	1	0	0	0	1
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	0	0	0	0	0
6	42 U Rack for Server	No.	0	0	0	0	0
7	Furniture Set	Set	0	0	0	0	0
8	Air Conditioner, 2 Ton, 5 Star	Nos.	0	0	0	0	0
9	Unit Rate for future integration of different verticals with ICCC	No.	0	0	0	0	0
10	Technical Manpower for 01 Year	Set	0	0	0	0	0
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	48	0	48
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	0	0	0	0
13	5 MP IP Dome Camera	Nos.	0	19	12	0	31
14	2 MP IP PTZ Camera	Nos.	0	0	0	0	0
15	Enterprise CVMS (per Camera's license)	Nos.	0	0	0	0	0
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	12	0	0	12
17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	0	12	0	12
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	0	0	0	0

19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	0	0	0	0
20	Monitor 32 inches	Nos.	0	3	12	0	15
21	55" LED Display	Nos.	0	0	0	0	0
22	Server	Nos.	0	0	0	0	0
23	Video Storage	Nos.	0	0	0	0	0
24	Workstation PC	Nos.	0	0	12	0	12
25	Printer	Nos.	0	0	12	0	12
26	Network management Systems	Nos.	0	0	0	0	0
27	Video Management Software solution	Nos.	0	0	0	0	0
28	Central Server application	No.	0	0	0	0	0
29	Client Software	Nos.	0	0	12	19	31
30	24 Port Layer 2, Managed switch	Nos.	0	0	0	0	0
31	8 port unmanaged gigabyte switch	Nos.	0	15	0	0	15
32	Industrial 8 Port POE Switch	Nos.	0	0	0	0	0
33	8 port managed Switch	Nos.	0	20	3	0	23
34	16 port POE managed Switch	Nos.	1	0	12	0	13
35	Media Converter	Pairs	0	0	11	0	11
36	CAT 6 Armoured STP cable	Mtrs	0	0	2400	950	3350
37	CAT 6 UTP cable	Mtrs	0	570	600	0	1170
38	Armoured Optical Fibre cable	Mtrs	0	0	5000	0	5000
39	Armoured Power cable	Mtrs	0	0	600	380	980
40	Floor mounted Rack 24U enclosure	Nos.	1	0	0	0	1
41	9U rack	Nos.	0	0	12	19	31
42	Outdoor Vandal Proof box	Nos.	0	0	0	0	0
43	CVT/ Voltage Stabilizer	Nos.	0	0	12	19	31
44	MCB	Nos.	0	0	12	19	31
45	Power Surge Protector	Nos.	0	0	12	19	31
46	Lightning arrestor with all accessories	Nos.	0	0	12	19	31
47	Earthing	Nos.	0	0	36	19	55
48	1 KVA UPS with 60 min backup	Nos.	0	0	0	0	0
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	1	0	0	0	1
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	0	12	19	31
51	CPE radio	Nos.	0	0	2	0	2
52	Point to Multipoint Radios	Nos.	0	0	0	0	0
53	PTP radio	Nos.	0	0	0	0	0
54	10 m Tubular Swaged pole	Nos.	0	0	12	0	12
55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	0	0	0	0
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	24	0	24
57	Signal Light for Traffic control	Nos.	0	0	24	38	62

58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	12	0	12
59	LED Flood Light along with all accessories	Nos.	0	0	12	0	12
60	Portable Cabins	Nos.	0	0	12	0	12
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	3	0	3
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	300	0	300
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	600	0	600
64	Reflective acrylic sticker	Nos.	0	0	1200	0	1200
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	19	19
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	19	19

15. Bill of Quantity for Total Module wise

SN	Item Description	Unit	ICCC	CCTV	ACS	TCS	Total
1	Integrated Command and Control Centre (ICCC) software platform	Set	1	0	0	0	1
2	Video Wall along with all hardware & software accessories	No.	1	0	0	0	1
3	AI based Video Analytics Application Software having following analytics (i) Camera Tampering Detection	Nos.	310	0	0	0	310
	(ii) Crowd detection	Nos.	260	0	0	0	260
	(iii) Automatic Number Plate Recognition	Nos.	6	0	0	0	6
	(iv) Object counting or people/ vehicle counting	Nos.	5	0	0	0	5
	(v) Classification of objects	Nos.	5	0	0	0	5
	(vi) Loaded/Unloaded condition of the vehicles	Nos.	5	0	0	0	5
	(vii) Perimeter Violation	Nos.	95	0	0	0	95
	(viii) Loitering Detection	Nos.	60	0	0	0	60
	(ix) Fire & Smoke Detection	Nos.	105	0	0	0	105
	(x) Traffic Congestion detection	Nos.	5	0	0	0	5
	(xi) PPE KIT detection	Nos.	57	0	0	0	57
	(xii) Safety hazards detection	Nos.	6	0	0	0	6
	(xiii) Insufficient illumination	Nos.	6	0	0	0	6
	(xiv) Accidents Alerts	Nos.	5	0	0	0	5
4	Servers for ICCC and Video Analytics	Nos.	16	0	0	0	16
5	Online UPS of 30 KVA with 2-hour backup at Full Load	No.	1	0	0	0	1
6	42 U Rack for Server	No.	1	0	0	0	1
7	Furniture Set	Set	1	0	0	0	1
8	Air Conditioner, 2 Ton, 5 Star	Nos.	3	0	0	0	3
9	Unit Rate for future integration of different verticals with ICCC	No.	1	0	0	0	1
10	Technical Manpower for 01 Year	Set	3	0	0	0	3
11	5 MP IP Bullet Camera (Motorized vari-focal lens) for ANPR with license cost	Nos.	0	0	580	0	580
12	5 MP IP Bullet Camera (Motorized vari-focal lens)	Nos.	0	355	0	0	355
13	5 MP IP Dome Camera	Nos.	0	424	159	0	583
14	2 MP IP PTZ Camera	Nos.	0	64	0	0	64
15	Enterprise CVMS (per Camera's license)	Nos.	0	705	0	0	705
16	NVR (4 channel) loaded with 1 TB Storage	Nos.	0	260	0	0	260

17	NVR (8 channel) loaded with 4 TB Storage	Nos.	0	42	159	0	201
18	NVR (16 channel) loaded with 8 TB Storage	Nos.	0	10	0	0	10
19	NVR (32 channel) loaded with 16TB Storage	Nos.	0	14	0	0	14
20	Monitor 32 inches	Nos.	0	182	159	0	341
21	55" LED Display	Nos.	0	19	0	0	19
22	Server	Nos.	0	9	2	0	11
23	Video Storage	Nos.	1	5	0	0	6
24	Workstation PC	Nos.	3	6	160	0	169
25	Printer	Nos.	0	0	160	0	160
26	Network management Systems	Nos.	0	2	0	0	2
27	Video Management Software solution	Nos.	0	4	0	0	4
28	Central Server application	No.	0	0	1	0	1
29	Client Software	Nos.	0	0	159	153	312
30	24 Port Layer 2, Managed switch	Nos.	2	12	0	0	14
31	8 port unmanaged gigabyte switch	Nos.	0	276	0	0	276
32	Industrial 8 Port POE Switch	Nos.	0	68	0	0	68
33	8 port managed Switch	Nos.	0	452	69	0	521
34	16 port POE managed Switch	Nos.	13	0	161	0	174
35	Media Converter	Pairs	0	1902	172	0	2074
36	CAT 6 Armoured STP cable	Mtrs	0	23245	30800	7650	61695
37	CAT 6 UTP cable	Mtrs	0	28525	8560	0	37085
38	Armoured Optical Fibre cable	Mtrs	0	87470	151500	0	238970
39	Armoured Power cable	Mtrs	0	16040	8150	3060	27250
40	Floor mounted Rack 24U enclosure	Nos.	13	4	2	0	19
41	9U rack	Nos.	0	135	159	153	447
42	Outdoor Vandal Proof box	Nos.	0	69	0	0	69
43	CVT/ Voltage Stabilizer	Nos.	0	66	161	153	380
44	MCB	Nos.	0	214	161	153	528
45	Power Surge Protector	Nos.	0	214	161	153	528
46	Lightning arrestor with all accessories	Nos.	0	271	161	153	585
47	Earthing	Nos.	0	523	483	153	1159
48	1 KVA UPS with 60 min backup	Nos.	0	210	0	0	210
49	1 KVA online UPS with 120 min backup with isolation transformer	Nos.	13	53	0	0	66
50	3 KVA (or more) online UPS with power backup of minimum 2 hours with isolation transformer	Nos.	0	13	161	153	327
51	CPE radio	Nos.	0	56	58	0	114
52	Point to Multipoint Radios	Nos.	0	17	0	0	17
53	PTP radio	Nos.	0	26	0	0	26
54	10 m Tubular Swaged pole	Nos.	0	232	195	0	427

55	27 Meters 3 Leg Self Supported Galvanized Towers	Nos.	0	56	22	0	78
56	Long Range RFID Reader/Writer with pole and mounting	Nos.	0	0	290	0	290
57	Signal Light for Traffic control	Nos.	0	0	290	306	596
58	5 Mtr Long Boom Barrier complete with Trigger switch, Operating & Safety Sensor	Nos.	0	0	195	0	195
59	LED Flood Light along with all accessories	Nos.	0	0	195	0	195
60	Portable Cabins	Nos.	0	0	159	0	159
61	Handheld RFID reader along with all hardware & software accessories	Nos.	0	0	48	0	48
62	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Temporary)	Nos.	0	0	5500	0	5500
63	RFID tags - Tamperproof windshield Passive UHF RFID Tags with one sided printing (Permanent)	Nos.	0	0	11000	0	11000
64	Reflective acrylic sticker	Nos.	0	0	22000	0	22000
65	Truck Centering System with controller and all accessories	Nos.	0	0	0	153	153
66	Audio System along with all hardware & software accessories	Nos.	0	0	0	153	153